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Review

Catalysis of entanglement and other quantum resources

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Abstract

In chemistry, a catalyst is a substance which enables a chemical reaction or increases its rate, while remaining unchanged in the process. Instead of chemical reactions, *quantum catalysis* enhances our ability to convert quantum states into each other under physical constraints. The nature of the constraints depends on the problem under study and can arise, e.g. from energy preservation. This article reviews the most recent developments in quantum catalysis and gives a historical overview of this research direction. We focus on the catalysis of quantum entanglement and coherence, and also discuss this phenomenon in quantum thermodynamics and general quantum resource theories. We review applications of quantum catalysis and also discuss the recent efforts on universal catalysis, where the quantum state of the catalyst does not depend on the states to be transformed. Catalytic embezzling is also considered, a phenomenon that occurs if the catalyst's state can change in the transition.

Keywords: quantum catalysis, resource theories, entanglement, quantum coherence, quantum thermodynamics

(Some figures may appear in colour only in the online journal)

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1. Introduction

In 1835, Jacob Berzelius observed a common feature among several chemical reactions: they all required an additional substance, which remained unchanged in the process. He described this feature with the word 'catalysis', which goes back to Greek, and can be translated as 'down' or 'loosen' (Berzelius 1835, van Santen *et al* 1999). A systematic study of catalytic processes in chemistry has been performed later by Wilhelm Ostwald, who gave a precise definition which is also used today: 'Catalysis is the acceleration of a slow chemical process by the presence of a foreign substance' (Ostwald 1894). First known examples of chemical catalysis used by humans are fermentation processes, which have been used for thousands of years (van Santen *et al* 1999). Today, chemical catalysis is broadly used in the chemical industry and is essential for many industrial processes. In addition, the importance of catalytic reactions in biochemistry cannot be overstated: to the point that the ability of living organisms to 'catalyse chemical reactions efficiently and selectively' via metabolic pathways could be called one of the 'fundamental conditions for life' (Nelson *et al* 2008).

Quantum catalysis is conceptually similar to chemical catalysis but differs from it in several important details. A simple analogy between quantum and chemical catalysis can be established by replacing 'chemical reaction' with 'quantum state transition'. With this, a quantum catalyst is a quantum system which enables otherwise impossible transitions between quantum states. First investigations of quantum catalysis go back to the work of Jonathan and Plenio (1999), who studied transitions between entangled states shared by two remote parties (Alice and Bob). It is commonly assumed that the remote parties have the complete freedom to manipulate their quantum systems locally, including all kinds of transformations and measurements which are compatible with the laws of quantum mechanics. Additionally, Alice and Bob can exchange the outcomes of their measurements via a classical communication channel. This allows each party to adjust their local setups to the outcomes of the previous measurements obtained by the other party. This setup describes everything Alice and Bob can do to their shared quantum state if they cannot exchange other quantum particles (Bennett *et al* 1996c, Horodecki *et al* 2009).

In the first example of entanglement catalysis given by Jonathan and Plenio (1999), Alice and Bob aim to convert their quantum state $|\psi\rangle^{AB}$ into another state $|\phi\rangle^{AB}$ under the limitations described above. A few months earlier, Nielsen (1999) presented conditions for such state transformations, which allow to check whether $|\psi\rangle^{AB}$ can be converted into $|\phi\rangle^{AB}$ for any pair of states. Jonathan and Plenio (1999) found that for some states an otherwise impossible transition can be enabled by using an additional pair of particles A' and B' held by Alice and Bob, respectively. More precisely, by choosing an appropriate entangled state $|\eta\rangle^{A'B'}$, they showed that $|\psi\rangle^{AB}\otimes|\eta\rangle^{A'B'}$ can be converted into $|\phi\rangle^{AB}\otimes|\eta\rangle^{A'B'}$, even though a direct conversion from $|\psi\rangle^{AB}$ into $|\phi\rangle^{AB}$ is not possible. As a result the state $|\eta\rangle^{A'B'}$ remains unchanged and the two-particle system $A'B'$ can be seen as a quantum catalyst.

In recent years, it became clear that entanglement is not the only quantum feature to exhibit catalytic behaviour, mostly thanks to the development of quantum resource theories (Chitambar and Gour 2019), which goes back to the systematic study of quantum entanglement (Horodecki *et al* 2009) as a resource for the emerging quantum technology. Every quantum resource theory is based on a set of free states and free operations, corresponding to states and transformations which are easy to prepare and implement within well-defined physical constraints. Apart from quantum entanglement, the framework of quantum resource theories gave significant insight into the thermodynamical properties of quantum systems (Brandão *et al* 2013, Gour *et al* 2015, Goold *et al* 2016), and the role of purity (Horodecki *et al* 2003b, Gour *et al* 2015) and quantum coherence (Baumgratz *et al* 2014, Streltsov *et al* 2017a) for technological applications. Catalytic effects have been found in most resource theories studied in recent literature, allowing to overcome certain constraints of the theory by adding a particle which remains unchanged in the procedure.

The original formulation of quantum catalysis proposed by Jonathan and Plenio (1999) did not allow for any correlations between the catalyst and the primary system as a result of a catalytic process. This constraint has been lifted in recent literature, allowing the procedure to build up and preserve correlations. This fruitful approach has significantly improved our understanding of quantum catalysis, with applications in the resource theories of quantum coherence (Åberg 2014), quantum thermodynamics (Wilming *et al* 2017, Müller 2018, Shiraishi and Sagawa 2021, 2022), purity (Boes *et al* 2019, Wilming 2021), and entanglement (Kondra *et al* 2021a, Datta *et al* 2022). In many cases, it has significantly simplified the analysis of catalytic state transformations within the corresponding resource theory (Müller 2018, Boes *et al* 2019, Shiraishi and Sagawa 2021, 2022, Wilming 2021, Kondra *et al* 2021a, Datta *et al* 2022, 2023).

In this article, we review recent developments in the catalysis of quantum entanglement and other quantum resource theories. The article is structured as follows: in section 2 we recall the main features of quantum resource theories. Exact and approximate catalysis are discussed in detail in sections 3

and 4, respectively. In section 5 we review applications of catalysis for quantum information processing. Section 6 presents recent results on universal catalysis. Catalytic embezzling phenomena are discussed in section 7, and perspectives and open problems are presented in section 8.

2. Quantum resource theories

The study of quantum resource theories (Chitambar and Gour 2019) goes back to the early research in quantum entanglement (Horodecki *et al* 2009). After the first observation of quantum entanglement in nature (Aspect *et al* 1981, 1982), it was suggested that entanglement might eventually find interesting technological applications. Among others, Bennett *et al* (1993) established that maximally entangled singlet states could be regarded as a vehicle for quantum teleportation. And even before that, a cryptographic scheme with singlets was described in Ekert (1991). Thus, from the point of view of various technological applications, an important problem emerged, how to find an optimal method of transforming arbitrary quantum states into maximally entangled singlets:

$$|\phi_2^+\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right). \tag{1}$$

This question is highly relevant also in the context of mixed, or noisy, quantum states, which are the right model of a quantum system in any real experiment. One of the goals of quantum resource theories is to develop optimal strategies for transforming arbitrary quantum states, states perhaps not so useful for some applications into ones that are deemed more 'resourceful'.

Any quantum resource theory specifies a context in which certain states and transformations are considered *free*. Free states are always available at no cost—an additional subsystem as a free state can be added or discarded at will. For example, the resource theory of entanglement regards all separable states (Werner 1989) as free:

$$\rho_{\text{sep}} = \sum_i p_i \rho_i^A \otimes \sigma_i^B, \tag{2}$$

where ρ_i^A and σ_i^B are local quantum states. A composite quantum system can be brought to a separable state without the requirement of having shared quantum entanglement in advance, using only local operations and classical communication (LOCC). For a general resource theory, free operations are those transformations of quantum systems that can be implemented without the need of having access to resourceful, non-free states. In the resource theory of entanglement, free operations are precisely the class of LOCC.

It was the resource theory of quantum entanglement that was historically the first to undergo a systematic study (Horodecki *et al* 2009). Intuitively, entanglement is a valuable resource in systems that allow for clear spatial separation into several discrete subsystems. Not all prominent applications of quantum technologies, however, emphasise the role of such spatial separation. For example, an instance of quantum computation is typically assumed to be a local process. The role of entanglement in quantum computation has been the subject of numerous publications, and it has been shown that under certain assumptions, a quantum computer requires entanglement in order to exhibit an exponential speed-up compared to classical computation (Jozsa and Linden 2003). As of today, it remains unclear, however, whether it is quantum entanglement that is a crucial component of quantum computation, especially when noisy quantum states are considered. An interesting example to the contrary is the one clean qubit model (Knill and Laflamme 1998), which performs exponentially faster than the best-known classical algorithm, with vanishing entanglement (Datta *et al* 2005, Naseri *et al* 2022). Moreover, given certain quantum computational tasks, large amounts of entanglement can have a detrimental effect on the performance of a quantum processor (Gross *et al* 2009, Naseri *et al* 2022). One should also note that universal quantum computation is proven to be possible with arbitrary little entanglement (Van den Nest 2013).

The framework of the general quantum resource theory allows for investigating quantum systems beyond the constraints of particular resource manipulation scenarios. Since not all quantum information processing tasks require entanglement, a considerable amount of effort has been put in recent years into the research of other resource theories (Chitambar and Gour 2019), among others the resource theories of purity (Horodecki *et al* 2003b, Gour *et al* 2015, Streltsov *et al* 2018), quantum thermodynamics (Brandão *et al* 2013, Gour *et al* 2015, Goold *et al* 2016), coherence (Aberg 2006, Baumgratz *et al* 2014, Streltsov *et al* 2017a, Wu *et al* 2021c), asymmetry (Gour and Spekkens 2008, Vaccaro *et al* 2008, Gour *et al* 2009, Marvian and Spekkens 2014a, 2014b), and imaginarity (Hickey and Gour 2018, Wu *et al* 2021a, 2021b). As it has been shown that for every quantum resource theory there exists a channel discrimination task showing a quantitative advantage of utilising that particular resource (Takagi *et al* 2019, Takagi and Regula 2019), the newly developed theories may prove to be of considerable importance to future applications of processing quantum information.

3. Exact catalysis

Due to the rapid growth of quantum information science in recent years, we can identify and quantify the quantum resources required for different quantum information protocols. One of the important studies in this direction is to learn how quantum states can be transformed into each other under some physical constraints. More precisely, we are interested in knowing whether a state σ can be achieved from a given resourceful state ρ as a result of some physical operation allowed by the theory. Sometimes the transformation is possible deterministically, i.e. there exists a free operation Λ_f such that $\sigma = \Lambda_f[\rho]$. When the transformation from ρ to σ is not possible deterministically, we consider the possibility of probabilistic conversion. Nonetheless, there exist states which cannot be converted into each other even probabilistically (Dür *et al* 2000, Fang *et al* 2018, Sauerwein *et al* 2018, Wu *et al* 2020,

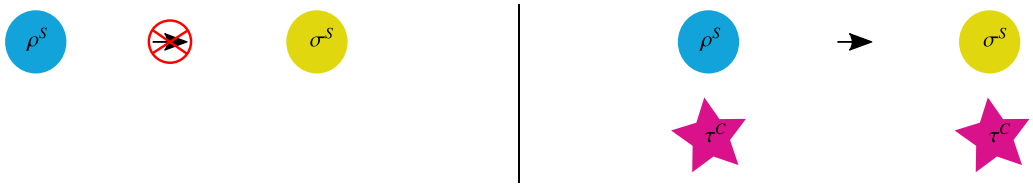


Figure 1. Quantum catalysis: consider two system states ρ^S and σ^S such that a conversion $\rho^S \to \sigma^S$ cannot be achieved with free operations (left part of the figure). In many resource theories, a conversion might still be enabled for some states by a catalyst in the state τ^C, allowing for the transformation $\rho^S \otimes \tau^C \to \sigma^S \otimes \tau^C$ (right part of the figure).

Regula 2022). In that case, there may exist an additional state that works as a catalyst, enabling the transformation while remaining unchanged in the process, see also figure 1.

In this section, we will discuss exact catalysis, where no error is allowed in the transformation. Historically, exact catalysis was first introduced for quantum entanglement (Jonathan and Plenio 1999), and extended to other quantum resource theories more recently. In the following, we will review recent results on quantum catalysis for different resource theories. Before presenting the results, we introduce the concept of exact catalysis. An exact catalytic transformation between two states ρ^S and σ^S of a system S is possible when there exists a catalyst C in a state τ^C and a free operation Λ_f acting on the system S and the catalyst C such that (Jonathan and Plenio 1999)

$$\Lambda_f\left(\rho^S \otimes \tau^C\right) = \sigma^S \otimes \tau^C. \tag{3}$$

Exact catalytic transformation is very stringent, and over time relaxations of it have been considered in the literature, allowing the catalyst to build up correlations with the primary system (Åberg 2014, Boes *et al* 2018, Lostaglio and Müller 2019, Rethinasamy and Wilde 2020, Yadin *et al* 2022). In particular, an exact correlated catalytic transformation between two states ρ^S and σ^S is possible if and only if there exists a catalyst state τ^C, a correlated state σ^{SC} and a free operation Λ_f such that

$$\Lambda_f\left(\rho^S \otimes \tau^C\right) = \sigma^{SC} \text{ with } \mathrm{Tr}_C\,\sigma^{SC} = \sigma^S \text{ and } \mathrm{Tr}_S\,\sigma^{SC} = \tau^C. \tag{4}$$

Having defined exact catalysis for general quantum resource theories, we will focus on the catalysis of entanglement in the following.

3.1. Entanglement

Historically, Jonathan and Plenio (1999) introduced the notion of exact catalysis of quantum states first in the context of the resource theory of entanglement. The idea is based on the following fundamental result, which characterises fully the necessary and sufficient conditions to be met by all admissible transformations of pure bipartite entangled states. Consider two states $|\psi\rangle^{AB} = \sum_i \sqrt{\alpha_i}|ii\rangle$ and $|\phi\rangle^{AB} = \sum_i \sqrt{\beta_i}|ii\rangle$, with their respective Schmidt coefficients $\{\alpha_i\}$ and $\{\beta_i\}$ sorted in non-increasing order.

Theorem 1 (Nielsen's theorem, see Nielsen (1999)). *The transformation $|\psi\rangle^{AB} \to |\phi\rangle^{AB}$ is possible via LOCC if and only if the reduced states ψ^A and ϕ^A fulfill the majorisation relation $\psi^A \prec \phi^A$, i.e. their Schmidt coefficients satisfy*

$$\sum_{i=0}^{n} \alpha_i \leqslant \sum_{i=0}^{n} \beta_i \;\text{ for all }\; 0 \leqslant n \leqslant d-1, \tag{5}$$

with $d = \min\{d_A, d_B\}$, where d_A and d_B are the dimensions of the Hilbert space of Alice and Bob, respectively.

The same result as in theorem 1 has been derived independently by Hardy (1999).

It is not difficult to find such states which do not satisfy the above relations, and hence an LOCC transition between them is not possible. As an example, consider the following two states (Jonathan and Plenio 1999):

$$|\psi\rangle = \sqrt{0.4}|00\rangle + \sqrt{0.4}|11\rangle + \sqrt{0.1}|22\rangle + \sqrt{0.1}|33\rangle, \tag{6}$$

$$|\phi\rangle = \sqrt{0.5}|00\rangle + \sqrt{0.25}|11\rangle + \sqrt{0.25}|22\rangle. \tag{7}$$

It can be easily verified that the above states do not satisfy Nielsen's theorem, and as a result, a deterministic conversion is not possible. When a deterministic transformation is not possible, one may consider probabilistic transformation with maximum success probability given by Vidal (1999):

$$P_{\max}\left(|\psi\rangle \to |\phi\rangle\right) = \min_{n\in[0,d-1]} \frac{\sum_{i=n}^{d-1} \alpha_i}{\sum_{i=n}^{d-1} \beta_i}. \tag{8}$$

For the states given in equations (6) and (7), the optimal probability of transforming $|\psi\rangle$ into $|\phi\rangle$ is 0.8. Surprisingly, if the two parties have access to an additional entangled state $|\eta\rangle = \sqrt{0.6}|00\rangle + \sqrt{0.4}|11\rangle$, then they can perform the transformation $|\psi\rangle \otimes |\eta\rangle \to |\phi\rangle \otimes |\eta\rangle$ with certainty (Jonathan and Plenio 1999, Nielsen 1999). After the transformation the state $|\eta\rangle$ remains unchanged and can be used again for the same purpose. We can say that the state $|\eta\rangle$ acts as a catalyst and enables a transformation that otherwise would not be possible. Note that in the context of quantum entanglement, what is nowadays predominantly called 'catalysis', used to be described as 'entanglement-assisted LOCC operation' (Jonathan and Plenio 1999, Eisert and Wilkens 2000).

A maximally entangled state of any dimension cannot catalyse a transformation between two arbitrary incomparable

bipartite pure states (Jonathan and Plenio 1999). A surprising result that already suggests a rich structure of catalytic transformations of quantum resources. Moreover, if for a pair of states $|\psi\rangle$ and $|\phi\rangle$, there exist catalytic transformations $|\phi\rangle \to |\psi\rangle$ and $|\psi\rangle \to |\phi\rangle$, i.e. if two states are interconvertible under catalytic LOCC, then there is a *local* unitary U such that $|\phi\rangle = U|\psi\rangle$ (Jonathan and Plenio 1999). As a consequence, if a direct, non-catalytic transformation from $|\psi\rangle$ to $|\phi\rangle$ is not possible, whereas a catalytic transformation exists in that case, then the reverse transition $|\phi\rangle \to |\psi\rangle$ cannot be realised even as a catalytic one. Hence, catalysis can have an advantage over the direct transformation of quantum states only in the case when the two states are incomparable: $|\phi\rangle \nleftrightarrow |\psi\rangle$, i.e. there is neither a direct transformation $|\psi\rangle \to |\phi\rangle$ nor $|\phi\rangle \to |\psi\rangle$. However, note that the above result is based on assuming pure state catalysts, and how the result behaves with mixed catalyst states is not known. The result suggests that, as far as transformations of two-qubit pure entangled states are concerned, there is no advantage with a catalyst in a pure state, since it is always true that either $|\psi\rangle \to |\phi\rangle$ or $|\phi\rangle \to |\psi\rangle$ in this case.

A catalytic LOCC transformation with catalysts in pure states has no advantage in the case of two-qutrit pure states as well. In order to see this, Jonathan and Plenio (1999) prove that a bipartite state $|\psi\rangle$ of local dimension d can be converted catalytically into another bipartite state $|\phi\rangle$, only if the following inequalities hold

$$\alpha_0 \leqslant \beta_0 \text{ and } \alpha_{d-1} \geqslant \beta_{d-1}, \tag{9}$$

where the Schmidt coefficients $\{\alpha_i\}$ and $\{\beta_i\}, i = 0, 1, \ldots, d-1$, of respectively $|\psi\rangle$ and $|\phi\rangle$ are sorted in non-increasing order. In the case of $d = 3$, the above necessary condition for catalysis can be stated as

$$\alpha_0 \leqslant \beta_0 \text{ and } \alpha_0 + \alpha_1 \leqslant \beta_0 + \beta_1, \tag{10}$$

since the numbers $\{\alpha_i\}$ and $\{\beta_i\}$ sum up to identity. But this is exactly the condition (5) of Nielsen's majorisation theorem! Hence, as in the two-qubit case, there is no advantage in catalysis for bipartite systems with $d = 3$: the existence of a catalytic conversion of entangled states implies that these states can be converted directly via LOCC. A genuine catalytic conversion of incomparable states occurs first in systems of dimension $d = 4$, as we have already seen in equations (6) and (7).

When a catalytic transition from $|\psi\rangle$ to $|\phi\rangle$ is not possible with certainty, it can happen that by adding a catalyst state we will increase the probability of the conversion. As an example, consider two incomparable states (Jonathan and Plenio 1999):

$$|\psi\rangle = \sqrt{0.6}|00\rangle + \sqrt{0.2}|11\rangle + \sqrt{0.2}|22\rangle, \tag{11}$$

$$|\phi\rangle = \sqrt{0.5}|00\rangle + \sqrt{0.4}|11\rangle + \sqrt{0.1}|22\rangle. \tag{12}$$

The optimal probability to convert $|\psi\rangle$ into $|\phi\rangle$ is $P_{\max}(|\psi\rangle \to |\phi\rangle) = 0.8$ (Vidal 1999), whereas with the help of another state $|\eta\rangle = \sqrt{0.65}|00\rangle + \sqrt{0.35}|11\rangle$, we have $P_{\max}(|\psi\rangle \otimes |\eta\rangle \to |\phi\rangle \otimes |\eta\rangle) = 0.904$. Of course, there are situations when even adding a catalyst in an arbitrary pure state cannot increase the optimal probability of transformation. This is the case, e.g. when the optimal probability of converting a bipartite state $|\psi\rangle$ into $|\phi\rangle$ is $P_{\max}(|\psi\rangle \to |\phi\rangle) = \alpha_{d-1}/\beta_{d-1}$. Indeed, let $\{\gamma_j\}_{j=0}^{d'-1}$ be the ordered, non-zero Schmidt coefficients of $|\eta\rangle$. Since the Schmidt coefficients, not necessarily ordered, of the state $|\psi\rangle \otimes |\eta\rangle$ are given by $\{\alpha_i\gamma_j\}_{i=0,j=0}^{d-1,d'-1}$, and similarly: $\{\beta_i\gamma_j\}_{i=0,j=0}^{d-1,d'-1}$ for the state $|\phi\rangle \otimes |\eta\rangle$, from equation (8) we obtain (Jonathan and Plenio 1999)

$$\begin{aligned} P_{\max}(|\psi\rangle \otimes |\eta\rangle \to |\phi\rangle \otimes |\eta\rangle) &= \min_{\substack{l=0,1,\ldots,d-1 \\ k=0,1,\ldots,d'-1}} \frac{\sum_{i\geqslant l, j\geqslant k} \alpha_i\gamma_j}{\sum_{i\geqslant l, j\geqslant k} \beta_i\gamma_j} \\ &\leqslant \frac{\alpha_{d-1}}{\beta_{d-1}}, \end{aligned} \tag{13}$$

where the inequality is obtained by setting $l = d-1, k = d'-1$, instead of minimising over all possible pairs l, k.

Of course, a pure catalyst state can also increase the transformation probability. Feng *et al* (2004, 2005) have studied the advantage of pure catalysts in probabilistic entangled state transformations. Here, the goal is to find a pure catalyst state $|\eta\rangle$ increasing the optimal conversion probability, i.e.

$$P_{\max}(|\psi\rangle \otimes |\eta\rangle \to |\phi\rangle \otimes |\eta\rangle) > P_{\max}(|\psi\rangle \to |\phi\rangle). \tag{14}$$

The following necessary and sufficient condition has been derived for the existence of a catalyst $|\eta\rangle$ for such an enhancement (Feng *et al* 2004):

$$P_{\max}(|\psi\rangle \to |\phi\rangle) < \min\left\{\frac{\alpha_{d-1}}{\beta_{d-1}}, 1\right\}, \tag{15}$$

where α_{d-1} and β_{d-1} are the smallest Schmidt coefficients of $|\psi\rangle$ and $|\phi\rangle$, respectively. Furthermore, they also discussed the scenario where the probability can be increased with the help of a two-qubit pure catalyst. In addition, an efficient algorithm has been proposed to find the most economical catalyst in such cases (Feng *et al* 2005).

As mentioned earlier, if the local dimension of the given state is smaller than or equal to three, a catalyst does not help in accomplishing a forbidden transformation (Jonathan and Plenio 1999). Now one may wonder that for a given final state $|\phi\rangle$ of dimension at least four there exists an initial state $|\psi\rangle$ with $\psi^A \not\prec \phi^A$, such that $|\psi\rangle$ can be converted catalytically into $|\phi\rangle$. It turns out that almost for all final states $|\phi\rangle$ there exists a catalytic conversion from $|\psi\rangle$ (Daftuar and Klimesh 2001). To be precise, the following has been shown: for a given final state $|\phi\rangle$, there exists a pure state catalyst and an initial state $|\psi\rangle$ with $\psi^A \not\prec \phi^A$, such that the transformation $|\psi\rangle \to |\phi\rangle$ is possible catalytically if and only if $\beta_0 \neq \beta_l$ and $\beta_m \neq \beta_{d-1}$ for some l, m with $0 < l < m < d-1$ (Daftuar and Klimesh 2001). Here, β_i are Schmidt coefficients of the state $|\phi\rangle$. This result emphasises that for almost all final states of dimension 4 or more, the sets of initial states to achieve a final state $|\phi\rangle$ using a pure state catalyst and without a catalyst are not the same, i.e. $T(\phi) \neq S(\phi)$. Here $T(\phi)$ and $S(\phi)$ represent the sets

of possible initial states $|\psi\rangle$ that can be converted into $|\phi\rangle$ with a catalyst in an arbitrary pure state and without a catalyst, respectively. Note that no restriction has been imposed on the dimension of the catalyst. For a restriction on the dimension of the catalyst, the set of initial states is represented by $T_k(\phi)$, where k is the maximum dimension of the catalyst. Then it has been shown that there exists no k such that $T_k(\phi) = T(\phi)$ when $T(\phi) \neq S(\phi)$ (Daftuar and Klimesh 2001). Although the set $T_k(\phi)$ is closed, the set $T(\phi)$ is not closed in general (Daftuar and Klimesh 2001). Therefore, the dimension of the catalyst is unbounded in general to determine which states can be catalytically converted into a given final state. Furthermore, as we have previously discussed maximally entangled state cannot be a catalyst. Naturally, one may ask which states are useful to be catalysts. As shown by Daftuar and Klimesh (2001), it turns out that almost all bipartite pure states are potential catalysts. Precisely the following has been shown: for a bipartite state $|\eta\rangle = \sum_{i=0}^{d-1} \sqrt{\xi_i}|ii\rangle$ of local dimension d with a nonuniform distribution of Schmidt coefficients there exist states $|\psi\rangle$ and $|\phi\rangle$ in dimension 4 with $\psi^A \not\prec \phi^A$, such that $\psi^A \otimes \eta^A \prec \phi^A \otimes \eta^A$ (Daftuar and Klimesh 2001).

Since the discovery of entanglement catalysis by Jonathan and Plenio (1999), an important problem was to characterise all pairs of states $|\psi\rangle^{AB}$ and $|\phi\rangle^{AB}$ allowing for a catalytic LOCC transformation. Results in this direction have been presented in Sun *et al* (2005), Aubrun and Nechita (2007, 2008), Klimesh (2007), Turgut (2007), focusing on pure catalyst states. When a catalyst enables a forbidden transformation between two bipartite pure states, the respective Schmidt coefficients satisfy a trumping condition. More precisely, a state $|\psi\rangle$ is trumped by another state $|\phi\rangle$, i.e. $\psi^A \prec_T \phi^A$ if there exists a catalyst in the state $|\eta\rangle$ such that $\psi^A \otimes \eta^A \prec \phi^A \otimes \eta^A$ holds (Daftuar and Klimesh 2001). The trumping condition is an extension of Nielsen's majorisation relation presented earlier in theorem 1.

A detailed characterisation of the trumping condition is mathematically very complex (Daftuar and Klimesh 2001). Sun *et al* (2005) provided a necessary and sufficient condition for the existence of a pure two-qubit catalyst state for $4 \otimes 4$ incomparable pure states. Furthermore, they have given an efficient algorithm to decide whether a $k \otimes k$ pure catalyst state exists for a pair of $n \otimes n$ incomparable pure states. Aubrun and Nechita (2007, 2008) provided some more partial results in this direction by determining the closure of the set $T(\phi)$ consisting of initial states $|\psi\rangle$ for a given final state $|\phi\rangle$. Precisely they showed that the closure of $T(\phi)$ in l_1 norm can be completely characterised by the following inequalities: $||\psi^A||_p \leqslant ||\phi^A||_p$ for all $p \geqslant 1$, where $||\cdot||_p$ represents l_p norm. Despite these efforts, the characterisation was not complete and a full characterisation has been provided by Turgut (2007) and Klimesh (2007). To be precise, it has been shown that the trumping condition is equivalent to (Klimesh 2007, Turgut 2007)

$$f_\alpha(s(\phi)) > f_\alpha(s(\psi)) \;\forall\, \alpha \in (-\infty, \infty), \tag{16}$$

where $s(\psi)$ and $s(\phi)$ represent the Schmidt coefficients of the states $|\psi\rangle$ and $|\phi\rangle$, respectively. The function $f_\alpha(x)$ for a d-dimensional probability vector x is expressed as

$$f_\alpha(x) = \begin{cases} \ln \sum_{i=1}^d x_i^\alpha & (\alpha > 1); \\ \sum_{i=1}^d x_i \ln x_i & (\alpha = 1); \\ -\ln \sum_{i=1}^d x_i^\alpha & (0 < \alpha < 1); \\ -\sum_{i=1}^d \ln x_i & (\alpha = 0); \\ \ln \sum_{i=1}^d x_i^\alpha & (\alpha < 0). \end{cases} \tag{17}$$

If one allows an infinitesimally small error in the initial state, i.e. $||\psi^A - \psi_\varepsilon^A||_1 \leqslant \varepsilon$ with the trace norm $||M||_1 = \mathrm{Tr}\sqrt{M^\dagger M}$, then the trumping condition $\psi_\varepsilon^A \prec_T \phi^A$ is equivalent to the monotonicity of the Rényi entropies (Brandão *et al* 2015).

Theorem 2 (see proposition 4 of the supplementary information in Brandão *et al* (2015)). *Consider two states $|\psi\rangle$ and $|\phi\rangle$ with Schmidt coefficients $s(\psi)$ and $s(\phi)$, respectively. Then for arbitrary $\varepsilon > 0$, there exists a state ψ_ε^A with $||\psi^A - \psi_\varepsilon^A||_1 \leqslant \varepsilon$ and $\psi_\varepsilon^A \prec_T \phi^A$ if and only if*

$$H_\alpha(s(\psi)) \geqslant H_\alpha(s(\phi)) \tag{18}$$

is true for all $\alpha \in (-\infty, \infty)$.

Here, H_α represent the Rényi entropies which can be expressed as

$$H_\alpha(x) = \frac{\mathrm{sgn}(\alpha)}{1-\alpha} \ln\left(\sum_{i=1}^d x_i^\alpha\right) \text{ for } \alpha \in \mathbb{R} \setminus \{0, 1\}. \tag{19}$$

$H_\alpha(x)$ for $\alpha \in \{-\infty, 0, 1, \infty\}$ are defined by taking limits and expressed as

$$H_0(x) = \ln \mathrm{rank}(x),\; H_1(x) = -\sum_{i=1}^d x_i \ln x_i \tag{20}$$

$$H_{-\infty}(x) = \ln x_{\min},\; H_\infty(x) = -\ln x_{\max}, \tag{21}$$

where $x_{\max}$ and $x_{\min}$ are the maximum and minimum elements of the vector x, respectively.

In real life, noise is inevitable, and as a result, we are forced to deal with mixed states rather than pure states. Mixed state transformations are much more complicated (Terhal and Horodecki 2000) as majorisation relations described in theorem 1 are not applicable in general. Consequently, the conversion of mixed states with the help of a catalyst gets convoluted. The reason is that we need to first check whether a mixed state transformation $\rho \to \sigma$ is possible. In general, it is a formidable task. Nevertheless, the problem has been solved for some classes of mixed states. Eisert and Wilkens (2000) explored transformations between a special class of rank two mixed states of the form

$$\rho = \nu|\psi\rangle\langle\psi| + (1-\nu)|\omega\rangle\langle\omega| \text{ and} \tag{22}$$

$$\sigma = \gamma|\phi\rangle\langle\phi| + (1-\gamma)\,|\omega\rangle\langle\omega|, \tag{23}$$

where $\gamma = \nu \mathrm{Tr}[\Omega]$, $\Omega = \Pi|\psi\rangle\langle\psi|\Pi$, $\Pi = \mathbb{1} - |\omega\rangle\langle\omega|$ and $|\omega\rangle$ is a product state. Moreover, we take $|\langle\phi|\omega\rangle|^2 = 0$. With this, an LOCC transformation from ρ to σ implies that (Eisert and Wilkens 2000)

$$\frac{\mathrm{Tr}_A\Omega}{\mathrm{Tr}\Omega} \prec \phi^A. \tag{24}$$

Now let us choose

$$|\psi\rangle = \sqrt{0.38}|00\rangle + \sqrt{0.38}|11\rangle + \sqrt{0.095}|22\rangle + \sqrt{0.095}|33\rangle + \sqrt{0.05}|44\rangle, \tag{25}$$

$$|\phi\rangle = \sqrt{0.5}|00\rangle + \sqrt{0.25}|11\rangle + \sqrt{0.25}|22\rangle, \tag{26}$$

$$|\omega\rangle = |44\rangle, \tag{27}$$

and $\gamma = 0.95\nu$, where $1 > \nu > 0$ (Eisert and Wilkens 2000). With this one can easily verify that equation (24) is not satisfied, and hence the transformation $\rho \to \sigma$ is not possible.

One may ask whether the forbidden transformation can be accomplished with the help of a catalyst. The answer is yes and the protocol is as follows (Eisert and Wilkens 2000). Alice performs a von Neumann measurement on her subsystem with Kraus operators $K_1 = \sum_{i=0}^{3}|i\rangle\langle i|$ and $K_2 = |4\rangle\langle 4|$. When she gets an outcome associated with K_2, no further operations are needed as we have state $|\omega\rangle$ with probability $(1-\gamma)$. In the case of other outcomes, the final state is

$$|\varphi\rangle = \sqrt{0.4}|00\rangle + \sqrt{0.4}|11\rangle + \sqrt{0.1}|22\rangle + \sqrt{0.1}|33\rangle. \tag{28}$$

Note that, the reduced state of $|\varphi\rangle$ is not majorised by the reduced state of $|\phi\rangle$. Therefore, a direct transformation is not possible and a catalyst could help in the transformation $|\varphi\rangle \to |\phi\rangle$. In fact, by taking a catalyst in state $|\zeta\rangle = \sqrt{0.4}|00\rangle + \sqrt{0.6}|11\rangle$, we obtain

$$\varphi^A \otimes \zeta^A \prec \phi^A \otimes \zeta^A. \tag{29}$$

Finally, discarding the classical information, we achieve the desired state σ.

In Li and Shi (2011a), the authors analyse the catalytic transformation of mixed states of two qubits. Building on the results from Gour (2005b), Li and Shi (2011b), their work sheds some light on the question of whether a mixed catalyst state could provide an advantage for entanglement catalysis. For a given mixed state ρ, the authors define a family of entanglement monotones:

$$\hat{E}_{k,l}(\rho) = \min_{\{p_j,|\psi_j\rangle\}} \sum_j p_j \hat{E}_{k,l}(|\psi_j\rangle), \tag{30}$$

with $l = 2,3,\ldots,d$ and $k \in (0,1]$, where the minimum is taken over all ensembles $\{p_j,|\psi_j\rangle\}$ of probability distributions p_j and bipartite pure states $|\psi_j\rangle$ such that $\rho = \sum_j p_j|\psi_j\rangle$, and

$$\hat{E}_{k,l}(|\psi\rangle) = \min\left\{1, \sum_{i=l}^{d}\frac{\alpha_i}{k}\right\}, \tag{31}$$

for any bipartite pure state $|\psi\rangle = \sum_i^d \sqrt{\alpha_i}|ii\rangle$ with ordered Schmidt coefficients. The ensemble that minimises equation (30) is called *optimal*.

In another paper, Li and Shi (2011b), the same authors give a non-standard definition of an LOCC transformation between mixed states. According to definition 2.6 in Li and Shi (2011b), a mixed state ρ can be converted into another mixed state σ via LOCC if there is an LOCC operation between optimal ensembles of ρ and σ. For a precise definition of a transformation between ensembles of pure states, see Li and Shi (2011b), p 97. For our discussion here, it is enough to notice that if there is an LOCC map Λ, such that $\Lambda(\rho) = \sigma$ in the ordinary sense, then ρ can be converted to σ via LOCC according to definition 2.6 in Li and Shi (2011b)—but not necessarily the other way round.

The main result from Li and Shi (2011a), namely proposition 2.2 therein, can be now stated as follows: Suppose ρ, σ, and ω are rank 2 mixed bipartite states of a two-qubit system. Then the state $\rho\otimes\omega$ can be converted into $\sigma\otimes\omega$ via LOCC according to definition 2.6 in Li and Shi (2011b) if and only if

$$\hat{E}_{k,l}(\rho\otimes\omega) \geqslant \hat{E}_{k,l}(\sigma\otimes\omega), \tag{32}$$

for all $k \in (0,1]$ and $l = 2,3,4$. In other words, as long as we consider rank 2 mixed states, a catalytic transformation between two-qubit systems is governed by the family of entanglement monotones defined in equation (30). The result is a generalisation of the trumping condition for pure states, see the discussion above, page 6.

Another important result in Li and Shi (2011a), proposition 2.5, follows quickly from the proof of the proposition 2.2, and gives us a partial answer to the question of whether mixed catalyst states could provide an advantage over pure states: if ρ, σ are rank 2 mixed bipartite states of a two-qubit system that *cannot* be converted into each other via LOCC, but for which a catalyst's state ω exists as in equation (32), then ω *cannot* be pure, even if we allowed for ω to have an arbitrary Schmidt rank. Note that this statement assumes that all LOCC transformations between mixed states are according to definition 2.6 in Li and Shi (2011b), but as we have already discussed above, that implies that there is no catalytic LOCC transformation in the ordinary sense as well.

Lastly, the authors give an example of a mixed catalyst state that facilitates a transformation between rank 2 mixed two-qubit states as above, see section 3 in Li and Shi (2011a). However, again, since the authors' definition of an LOCC transformation between mixed states describes a larger family of maps than standard LOCC on the convex set of mixed states, the question of whether an ordinary map and a mixed catalyst state exist remains open. Given that we know that no pure two-qubit state can be a catalyst in this case, a positive answer to that question would be a strong argument for the advantages of mixed-state catalysis.

Next, we will discuss the advantage of a catalyst for small transformations where we want to change the state slightly. In particular, the goal is to convert a state ρ into another state σ with $F(\rho,\sigma) > 1-\varepsilon$, where $F(\rho,\sigma) = \left(\mathrm{Tr}[\sqrt{\sigma}\rho\sqrt{\sigma}]^{1/2}\right)^2$

is the fidelity (Uhlmann 1976). In such a scenario, there exist states ρ and σ such that the transformation $\rho \to \sigma$ is impossible without a catalyst and can be accomplished with a suitable catalyst. Indeed, such states can be designed using equations (22) and (23) (Eisert and Wilkens 2000). On the contrary, a surprising result exists when we want to transform a pure state slightly. For all pure states $|\psi\rangle$ there exists an $\varepsilon > 0$, such that the following statement holds for all states $|\phi\rangle$ with $|\langle\psi|\phi\rangle|^2 > 1-\varepsilon$: the catalytic transformation $|\psi\rangle \to |\phi\rangle$ is impossible with a pure catalyst if it is impossible without the catalyst (Eisert and Wilkens 2000).

The structure of a catalyst for a successful transformation between two incomparable pure states was studied by Zhou and Guo (2000). To be precise, the authors showed that successful transformation of $|\psi\rangle = \sum_{i=0}^{d-1} \sqrt{\alpha_i}|ii\rangle$ into $|\phi\rangle = \sum_{i=0}^{d-1} \sqrt{\beta_i}|ii\rangle$ can be accomplished with the help of a pure catalyst state having m non-zero Schmidt coefficients, i.e. $|\eta\rangle = \sum_{i=0}^{m-1} \sqrt{\gamma_i}|ii\rangle$, only if (Zhou and Guo 2000)

$$m\gamma_{m-1} \leqslant P_{\max}\left(|\psi\rangle \to |\phi\rangle\right), \tag{33}$$

where $P_{\max}(|\psi\rangle \to |\phi\rangle)$ is described in equation (8), represents the optimal probability of transforming $|\psi\rangle$ into $|\phi\rangle$ (Vidal 1999). Furthermore, they extended the above result to stochastic scenarios. As a matter of fact, they showed that the probability of the transformation $P_{\max}(|\psi\rangle \to |\phi\rangle)$ can be increased with the help of a pure state catalyst $|\eta\rangle$, only if (Zhou and Guo 2000)

$$m\gamma_{m-1} \leqslant P_{\max}\left(|\psi\rangle \to |\phi\rangle\right)/P', \tag{34}$$

where $P' \geqslant P_{\max}(|\psi\rangle \to |\phi\rangle)$ represents the enhancement in the probability for catalytic transformation. Their results provide some insight into the structure of the catalyst.

In Feng *et al* (2002), the concept of mutual catalysis has been introduced. For this, suppose there are two pairs of states $(|\psi\rangle, |\phi\rangle)$ and $(|\eta\rangle, |\zeta\rangle)$. The transformations $|\psi\rangle \to |\phi\rangle$ and $|\eta\rangle \to |\zeta\rangle$ are not possible deterministically. However, sometimes it is possible that joint LOCC can transform $|\psi\rangle \otimes |\eta\rangle$ to $|\phi\rangle \otimes |\zeta\rangle$. Therefore, $|\psi\rangle$ and $|\eta\rangle$ help each other to achieve the desired target states. As an example, consider the following states (Feng *et al* 2002)

$$|\psi\rangle = \sqrt{0.4}|00\rangle + \sqrt{0.36}|11\rangle + \sqrt{0.14}|22\rangle + \sqrt{0.1}|33\rangle, \tag{35}$$

$$|\phi\rangle = \sqrt{0.5}|00\rangle + \sqrt{0.25}|11\rangle + \sqrt{0.25}|22\rangle, \tag{36}$$

$$|\eta\rangle = \sqrt{0.6}|00\rangle + \sqrt{0.4}|11\rangle, \tag{37}$$

$$|\zeta\rangle = \sqrt{0.55}|00\rangle + \sqrt{0.45}|11\rangle. \tag{38}$$

It is easy to check that $|\psi\rangle \nleftrightarrow |\phi\rangle$ and $|\eta\rangle \nrightarrow |\zeta\rangle$ (Nielsen 1999). However, they can be transformed jointly into the desired target states. Necessary conditions for such transformations have been derived in Feng *et al* (2002). Furthermore, the authors have discussed the scenario where a conversion from $|\psi\rangle$ to $|\phi\rangle$ is not possible even with the help of a catalyst but can be realised with mutual catalysis (Feng *et al* 2002).

According to theorem 1, there exist states $|\psi\rangle$ and $|\phi\rangle$, such that $|\psi\rangle \to |\phi\rangle$ cannot be realised with certainty. However, in some cases, if we possess multiple copies of $|\psi\rangle$, they can be transformed to the same number of $|\phi\rangle$ (Bandyopadhyay *et al* 2002). Along the same line, a concept of multicopy catalysis has been introduced by Duan *et al* (2005b). A state $|\eta\rangle$ is called a multiple-copy catalyst when it cannot catalyse the transformation $|\psi\rangle \to |\phi\rangle$, however the same can be accomplished with the help of $|\eta\rangle^{\otimes k}$ for some $k > 1$. As an example Duan *et al* (2005b) consider the following case. Consider the initial state $|\psi\rangle$ as given in equation (6) and the target state is

$$|\phi\rangle = \sqrt{0.5}|00\rangle + \sqrt{0.25}|11\rangle + \sqrt{0.22}|22\rangle + \sqrt{0.03}|33\rangle. \tag{39}$$

The transformation $|\psi\rangle \to |\phi\rangle$ cannot be realised with a pure two-qubit catalyst state (Sun *et al* 2005). However, note that the above transformation can be realised with a pure two-qutrit catalyst state. Now consider a situation where Alice and Bob only have access to some copies of a two-qubit pure state. One may wonder whether the above transformation can be realised with multiple copies of a two-qubit catalyst. Indeed there exists a two-qubit state $|\eta\rangle = \sqrt{0.6}|00\rangle + \sqrt{0.4}|11\rangle$, such that $\psi^A \otimes (\eta^A)^{\otimes 5} \prec \phi \otimes (\eta^A)^{\otimes 5}$ (Duan *et al* 2005b). Furthermore, they have derived some necessary conditions for an arbitrary pure state to be a catalyst for a given transformation $|\psi\rangle \to |\phi\rangle$. In addition, an advantage in probabilistic transformation with multiple copy catalysis has also been shown by Duan *et al* (2005b). Instead of a single copy of the initial state, if we have multiple copies of it and want to transform it into the same number of copies of the final state with the help of a multicopy catalyst, then a study has been done on the tradeoff between the number of copies of the source state and that of the catalyst state. To be precise, they have shown that the number of copies required for a catalyst state decreases with the number of copies of the source state (Duan *et al* 2005b). In Duan *et al* (2004, 2005a) it has been shown that any multiple copy entangled state transformation (Bandyopadhyay *et al* 2002) can be realised in single copy level with the help of a suitable pure catalyst. It is natural to wonder whether the opposite is also true. But the answer is negative. As shown by Feng *et al* (2006), the catalytic assisted transformation is strictly more powerful than multiple copy transformations. Now, it is natural to ask whether they are equivalent in the asymptotic scenario. Indeed it has been shown that catalytically assisted transformation is asymptotically equivalent to the multiple-copy transformation when the dimension of the catalyst and number of copies tend to infinity (Duan *et al* 2005c). More precisely, these two kinds of transformations are equivalent in the sense that they can simulate each other's ability to transform a given initial state to a given target state with the same optimal success probability. Optimal probability for multiple-copy state transformation from $|\psi\rangle$ to $|\phi\rangle$ can be defined as (Duan *et al* 2005c)

$$P_{\mathrm{M}}\left(|\psi\rangle \to |\phi\rangle\right) = \sup_m \left[P_{\max}\left(|\psi\rangle^{\otimes m} \to |\phi\rangle^{\otimes m}\right)\right]^{1/m}. \tag{40}$$

On the other hand, the optimal conversion probability for catalytically assisted transformation is given by Duan *et al* (2005c)

$$P_{\mathrm{C}}\left(|\psi\rangle \to |\phi\rangle\right) = \sup_{|\eta\rangle} P_{\max}\left(|\psi\rangle \otimes |\eta\rangle \to |\phi\rangle \otimes |\eta\rangle\right). \quad (41)$$

In Duan *et al* (2005c), it has been shown that $P_{\mathrm{M}}(|\psi\rangle \to |\phi\rangle) = P_{\mathrm{C}}(|\psi\rangle \to |\phi\rangle)$. Furthermore, Duan *et al* (2005a) have shown that multiple copy transformation, with the aid of a pure catalyst, is equivalent to single copy catalytic transformation with an arbitrary pure catalyst state. Let $V(\phi)$ be the set of all initial states $\{|\psi\rangle\}$ such that $|\psi\rangle \otimes |\eta\rangle \to |\phi\rangle \otimes |\eta\rangle$ for some $|\eta\rangle$, and $V_{\mathrm{M}}(\phi)$ is the set of all initial states $\{|\psi\rangle\}$ such that $|\psi\rangle^{\otimes m} \otimes |\eta\rangle \to |\phi\rangle^{\otimes m} \otimes |\eta\rangle$ for some $m \geqslant 1$ and $|\eta\rangle$. Obviously $V(\phi) \subseteq V_{\mathrm{M}}(\phi)$. In fact, it has been shown that they are equal $V(\phi) = V_{\mathrm{M}}(\phi)$ (Duan *et al* 2005a). Therefore, the power of catalytic assisted transformation (assuming pure catalyst states) cannot be elevated by increasing the number of copies of the original state.

While the previous discussion showed when a catalytic transformation is possible, it does not provide any information about the catalyst. To be precise, it does not give information about the entanglement content or the dimension of the catalyst required for a transformation. Sanders and Gour (2009) provided a lower bound on the dimension of a possible catalyst state $|\eta\rangle$ for a transformation $|\psi\rangle \otimes |\eta\rangle \to |\phi\rangle \otimes |\eta\rangle$. Suppose $|\psi\rangle, |\phi\rangle$ are two incomparable pure states with d non-zero Schmidt coefficients and the transformation $|\psi\rangle \otimes |\eta\rangle \to |\phi\rangle \otimes |\eta\rangle$ is possible with the help of a pure catalyst state $|\eta\rangle$ that has r non-zero Schmidt coefficients. Then it has been shown that the minimum number of non-zero Schmidt coefficients for the catalyst should be (Sanders and Gour 2009)

$$r \geqslant 1 + \frac{d-1}{d} \frac{\log_2 C_{d-1}(\phi) - \log_2 C_{d-1}(\psi)}{\log_2 C_d(\psi) - \log_2 C_d(\phi)}, \quad (42)$$

where C_k represents generalised kth concurrence and can be expressed as

$$C_k(\psi) = \left(\frac{f_k[s(\psi)]}{f_k[s(\psi^+)]}\right)^{1/k}, \quad (43)$$

where ψ^+ is a maximally entangled state of dimension d and $f_k(x)$ represents kth elementary symmetric function of d variables: $f_k(x) = \sum_{i_1<\cdots<i_k} x_{i_1} \cdots x_{i_k}$ (Gour 2005a).

So far, we have not discussed the amount of entanglement required for a catalyst to enable a transformation between a pair of incomparable states. However, the answer is not fully known and only some partial results are available in the literature. Duan *et al* (2005b) derived some necessary conditions for a bipartite pure entangled state to be a catalyst. In fact, the following has been proved: Suppose $|\psi\rangle$ and $|\phi\rangle$ are two incomparable states of Schmidt rank d with Schmidt coefficients $\{p\}$ and $\{q\}$, respectively, and a catalyst state $|\eta\rangle$ that has k number of non-zero Schmidt coefficients represented by $\{t\}$ enables the transformation from $|\psi\rangle$ to $|\phi\rangle$. Then for any $l \in L_{\psi,\phi}$ and $0 \leqslant i \leqslant k-2$, the Schmidt coefficients of the catalyst state satisfy the following three conditions:

$$\frac{t_0}{t_{k-1}} > \frac{q_l}{q_{l+1}}, \quad (44a)$$

$$\frac{t_0}{t_i} > \frac{q_l}{q_{l+1}} \text{ or } \frac{t_i}{t_{i+1}} < \frac{q_0}{q_l}, \quad (44b)$$

$$\frac{t_{i+1}}{t_{d-1}} > \frac{q_l}{q_{l+1}} \text{ or } \frac{t_i}{t_{i+1}} < \frac{q_{l+1}}{q_{d-1}}, \quad (44c)$$

where $L_{\psi,\phi}$ represents the set of all l such that

$$L_{\psi,\phi} = \left\{ l : 0 \leqslant l \leqslant d-1 \text{ and } \sum_{j=0}^{l} p_j > \sum_{j=0}^{l} q_j \right\}. \quad (45)$$

These restrictions on the Schmidt coefficients represent the deviation of state $|\eta\rangle$ from a product and a maximally entangled state.

Very recently, Grabowecky and Gour (2019) presented some stronger conditions than equation (44). They showed that for any two incomparable states $|\psi\rangle$, $|\phi\rangle$ with d non-zero Schmidt coefficients and a catalyst state $|\eta\rangle$ having k non-zero Schmidt coefficients, if $\psi^A \otimes \eta^A \prec \phi^A \otimes \eta^A$, then (Grabowecky and Gour 2019)

$$\max_{i\in\{0,\ldots,k-2\}} \frac{t_i}{t_{i+1}} < \min\left\{\frac{q_0}{q_m}, \frac{q_{n+1}}{q_{d-1}}\right\} \equiv a, \quad (46a)$$

$$\frac{t_0}{t_{k-1}} > \max_{l\in L_{\psi,\phi}} \frac{q_l}{q_{l+1}} \equiv b, \quad (46b)$$

where $m = \min L_{\psi,\phi}$ and $n = \max L_{\psi,\phi}$. These two conditions show how much a catalyst state can differ from both a product state and a maximally entangled state. Previously, in equation (42), we have seen a lower bound on the dimension of a catalyst in terms of the generalised kth concurrence (Sanders and Gour 2009). However, calculating kth concurrence for a higher-dimensional state is not easy in general. In that context, Grabowecky and Gour (2019) provided the following lower bound on the dimension of a catalyst, which can be easily calculated:

$$k > \frac{\ln b}{\ln a} + 1. \quad (47)$$

The constraints on the Schmidt coefficients of the catalyst state presented in equations (44) and (46) depend only on the Schmidt coefficients of the target state. Recently, Guo *et al* (2021) studied a similar scenario and showed that for $\psi^A \otimes \eta^A \prec \phi^A \otimes \eta^A$ the following result holds:

$$\frac{t_0}{t_{k-1}} > \max_{l\in L_{\psi,\phi}} \left(\min\left\{\frac{p_l}{p_{l+1}}, \frac{q_l}{q_{l+1}}\right\}\right) \equiv c, \quad (48)$$

where k is the number of non-zero Schmidt coefficients of state $|\eta\rangle$ and also represents the dimension of the catalyst.

Furthermore, they put a bound on the minimum dimension of the catalyst state that can be expressed as (Guo *et al* 2021)

$$k > \frac{\ln c}{\ln\left(a\sqrt{b}\right)} + 1. \tag{49}$$

Unlike equation (47), the above bound depends on both $\{p\}$ and $\{q\}$. One can compare and check which bound provides a good lower bound.

Duarte *et al* (2016) explored an interesting question about self-catalysis. Given two incomparable states $|\psi\rangle$ and $|\phi\rangle$, they studied whether $|\psi\rangle$ can catalyse the transformation $|\psi\rangle \otimes |\psi\rangle \to |\phi\rangle \otimes |\psi\rangle$. As an example, consider the following incomparable states (Duarte *et al* 2016):

$$|\psi\rangle = \sqrt{0.9}|00\rangle + \sqrt{0.081}|11\rangle + \sqrt{0.01}|22\rangle + \sqrt{0.009}|33\rangle, \tag{50}$$

$$|\phi\rangle = \sqrt{0.95}|00\rangle + \sqrt{0.03}|11\rangle + \sqrt{0.02}|22\rangle. \tag{51}$$

The transformation of $|\psi\rangle$ into $|\phi\rangle$ can be accomplished with the help of $|\psi\rangle$ as a catalyst, i.e. $|\psi\rangle \otimes |\psi\rangle \to |\phi\rangle \otimes |\psi\rangle$. One can modify it further by considering multi-copy self catalysis, where $|\psi\rangle \otimes |\psi\rangle \nrightarrow |\phi\rangle \otimes |\psi\rangle$ is not possible but $|\psi\rangle \otimes |\psi\rangle^{\otimes N} \to |\phi\rangle \otimes |\psi\rangle^{\otimes N}$ is possible for some $N > 1$ (Duarte *et al* 2016). For instance, consider the initial state to be

$$|\psi_1\rangle = \sqrt{0.918}|00\rangle + \sqrt{0.07}|11\rangle + \sqrt{0.006}|22\rangle + \sqrt{0.006}|33\rangle \tag{52}$$

and the target state to be $|\phi\rangle$ in equation (51). Then the transformation of $|\psi_1\rangle$ into $|\phi\rangle$ can be achieved using $|\psi_1\rangle^{\otimes 4}$ as a catalyst. Duarte *et al* (2016) tried to answer the question of how typical the self-catalysis phenomenon is using numerical methods. Although they found that self-catalysis occurs in systems of any dimension, the phenomenon is relatively rare (Duarte *et al* 2016). They also explored the self-catalysis of stochastic transformations and concluded that self-catalysis in this case occurs more often than in the deterministic one (Duarte *et al* 2016). In a recent paper, Acácio and Duarte (2022) discuss in detail whether neural network models can detect catalysis or self-catalysis for bipartite pure entangled state conversions.

The practical implementation of majorisation relations was first studied by Gagatsos *et al* (2013). In fact, they showed that beam splitter can be used to achieve several classes of majorisation relations. Furthermore, they construct incomparable states and showed that these states can be catalytically converted. Such catalyst states can be easily implemented by using single-photon states or two-mode squeezed vacuum states (Gagatsos *et al* 2013).

Recently, Santra and Malinovsky (2021a) studied the catalytic entanglement concentration of two-qubit pure states. They derived a formula for the maximum probability of converting a finite number of copies of a general two-qubit pure state into a singlet with the help of a two-qubit pure catalyst state. Furthermore, they have shown that any pure two-qubit entangled state can be a potential catalyst. Based on the approach given in Santra and Malinovsky (2021a), the authors further discussed the advantage of catalysis in the long-range entanglement distribution over a quantum network in Santra and Malinovsky (2021b).

In a recent numerical study, the entanglement properties of pure catalyst states have been explored (Gupta *et al* 2022). The average amount of entanglement required to catalyse a forbidden transformation decreases with the increment in entanglement difference between the pure initial and target states. Furthermore, for multiple copy state transformations, the amount of entanglement required to catalyse the transformation is lower compared to a single copy catalytic transformation.

An intriguing role of entangled catalysts in the context of implementing some nonlocal operations has been discussed by Vidal and Cirac (2002). In fact, Vidal and Cirac (2002) have shown that certain nonlocal tasks cannot be implemented without using additional entanglement. Interestingly, this additional entanglement does not need to be consumed and can be recovered exactly, thus serving as a catalyst in the process. More precisely, there exists a certain two-qubit unitary gate $\tilde{U}_{\mathrm{AB}}$ that cannot be implemented from the available two-qubit unitary gate U_{AB} by means of LOCC only (Vidal and Cirac 2002). However, the same can be implemented by means of local unitary operations and using a maximally entangled state as a catalyst (Vidal and Cirac 2002).

Kay and Ericsson (2006) explored the cloning of non-maximally entangled states with LOCC and a copy of a maximally entangled state. The authors completely characterised the set of clonable states. Furthermore, it has been shown that catalysts do not provide any advantage in the cloning process (Kay and Ericsson 2006).

In general, multipartite entangled state transformation is more complex in nature (Walter *et al* 2016) and that makes catalytic transformation even more complex. Unlike catalytic transformations in bipartite scenarios, very few results are known for catalytic state transformations in multipartite scenarios. One of the first results in this direction was provided by Chen *et al* (2010). More precisely, they studied the advantage of catalysis in stochastic LOCC (SLOCC) transformation of multipartite pure states. In fact, it has been shown that there exist state transformations that cannot be accomplished by SLOCC but can be realised by both catalytic SLOCC and multicopy SLOCC. Catalytic SLOCC represents the probabilistic transformation of $|\psi\rangle$ to $|\phi\rangle$ with the help of a catalyst state $|\eta\rangle$, such that the transformation $|\psi\rangle \otimes |\eta\rangle \to |\phi\rangle \otimes |\eta\rangle$ has non-vanishing probability. On the other hand, a transformation is possible via multicopy SLOCC if there exists a k, such that $|\psi\rangle^{\otimes k}$ can be converted into $|\phi\rangle^{\otimes k}$ with non-vanishing probability. As specific examples, consider the following transformations (Chen *et al* 2010):

(i) $|\psi\rangle_{ABC} \to |\phi_2^+\rangle_{AB} \otimes |\phi_2^+\rangle_{BC} \otimes |\phi_2^+\rangle_{CA}$, where $|\psi\rangle_{ABC}$ is any generalised GHZ-type state of tensor rank 6.

(ii) $|\mathrm{GHZ}_n^{n-1}\rangle \to |W_n\rangle$, where $|\mathrm{GHZ}_n^d\rangle = \sum_{i=0}^{d-1} |i\rangle^{\otimes n}/\sqrt{d}$ and $|W_n\rangle$ is n-partite generalisation of $|W_3\rangle = (|001\rangle + |010\rangle + |100\rangle)/\sqrt{3}$.

Both of the above state transformations are not possible even stochastically but can be accomplished via both catalytic SLOCC and multicopy SLOCC. Furthermore, it has been shown that if $|\psi\rangle$ can be converted to $|\phi\rangle$ via multicopy SLOCC, then the same can be realised with catalytic SLOCC (Chen *et al* 2010).

Recently Neven *et al* (2021) found a majorisation condition for a class of GHZ-like states, which completely characterises LOCC transformations. For this, suppose g and h are two invertible, complex and diagonal matrices satisfy $\mathrm{Tr}(g^\dagger g) = \mathrm{Tr}(h^\dagger h)$. Then the transformation

$$\mathbb{1}\otimes\cdots\otimes\mathbb{1}\otimes g|\mathrm{GHZ}_n^d\rangle \to \mathbb{1}\otimes\cdots\otimes\mathbb{1}\otimes h|\mathrm{GHZ}_n^d\rangle \tag{53}$$

is possible by LOCC if and only if

$$\left(|g_1|^2,\ldots,|g_d|^2\right)^{\mathrm{T}} \prec \left(|h_1|^2,\ldots,|h_d|^2\right)^{\mathrm{T}} \tag{54}$$

holds. Here, $(g_1,\ldots,g_d)$ and $(h_1,\ldots,h_d)$ are diagonal elements of the matrices g and h, respectively. Note that g and h correspond to GHZ-like states of the form $\sum_{i=0}^{d-1}\sqrt{a_i}|i\rangle^{\otimes n}$. As a direct consequence of this we see that $|\psi\rangle$ and $|\phi\rangle$ characterised by

$$g = \mathrm{diag}\left(\sqrt{0.45},\sqrt{0.35},\sqrt{0.12},\sqrt{0.08}\right), \tag{55}$$

$$h = \mathrm{diag}\left(\sqrt{0.56},\sqrt{0.21},\sqrt{0.17},\sqrt{0.06}\right), \tag{56}$$

respectively, are incomparable as they fail to satisfy equation (54). However, the transformation can be accomplished with another GHZ-like state $|\varphi\rangle$ characterised by

$$f = \mathrm{diag}\left(\sqrt{0.63},\sqrt{0.27},\sqrt{0.07},\sqrt{0.03}\right), \tag{57}$$

that acts as a catalyst in the process. In fact, the transformation $|\psi\rangle\otimes|\varphi\rangle \to |\phi\rangle\otimes|\varphi\rangle$ is equivalent to the following transformation:

$$\mathbb{1}\otimes\cdots\otimes\mathbb{1}\otimes(g\otimes f)\,|\mathrm{GHZ}_n^{16}\rangle \to \mathbb{1}\otimes\cdots\otimes\mathbb{1}\otimes(h\otimes f)\,|\mathrm{GHZ}_n^{16}\rangle. \tag{58}$$

This completes our discussion of exact entanglement catalysis, and we will consider the exact catalysis of coherence in the following.

3.2. Quantum coherence

Recent studies suggest that quantum coherence might be another useful resource for many information processing tasks that are otherwise impossible without it (Streltsov *et al* 2017a). There are two different notions of coherence in the literature: *unspeakable* and *speakable* (Marvian and Spekkens 2016). In the case of unspeakable coherence, the levels of eigenspaces are important for the quantification of coherence, for example, energy. Whereas for the latter one, it is irrelevant which eigenspaces are responsible for the coherence. While the former type of coherence is characterised using the resource theory of asymmetry (Marvian and Spekkens 2013), the latter is characterised using the usual notion of resource theory of coherence introduced by Baumgratz *et al* (2014). In the following, we will discuss catalysis in both cases.

Quantum catalysis in the resource theory of coherence was introduced in Åberg (2014). In this paper, the author studied the question of how to use a catalyst to create superposition between different energy levels under energy-preserving unitary operations. As energy must remain conserved, a system in a definite energy level cannot be changed to a superposition of different energy levels. However, an additional system or reservoir with a high degree of coherence can help to achieve this. Due to this, the amount of coherence in the reservoir can decrease. In contrast, Åberg (2014) showed that the coherence in the reservoir can be used to turn the state of the system into a superposition of energy levels without consuming it in the process. Hence, coherence in this reservoir acts like a catalyst in the process. The advantage has been shown by constructing two examples: the doubly-infinite ladder and the half-infinite ladder (AAberg 2014). Furthermore, as a direct consequence, it has been shown that this catalytic procedure has application in work extraction (Åberg 2014). Note that in Åberg's protocol the state of the reservoir does not remain the same. Nonetheless, the argument for catalysis is that the coherence resource in the reservoir does not degrade in the process and can be reused on another system to apply the same transformation.

The change in the reservoir's state is unusual to the definition of catalysis. Hence, an alternative description to Åberg (2014) has been provided by Korzekwa *et al* (2016), who preferred the term 'repeatability' over catalysis as the state of the reservoir changes. They studied work extraction from the coherence of the given system while the reservoir can be used repeatedly. The amount of work that can be extracted from a system in state ρ is equal to the difference in free energy $\Delta F(\rho)$ when the allowed operations conserve the energy only on average (Skrzypczyk *et al* 2014). Here $\Delta F(\rho) = F(\rho) - F(\gamma)$ with the Helmholtz free energy

$$F(\rho) = \mathrm{Tr}\left[\rho H\right] - kTS(\rho), \tag{59}$$

where H and γ correspond to the system Hamiltonian and thermal state, respectively, and $S(\rho) = -\mathrm{Tr}[\rho\log\rho]$ is the von Neumann entropy[3]. However, in Korzekwa *et al* (2016), the authors are interested in the work extraction only from the coherence of the system, and in that case,

$$F(\rho) - F(D[\rho]) = kTS(\rho||D[\rho]) \tag{60}$$

is the amount of free energy available for the extraction from the coherence of state ρ, where D is a dephasing operation in the energy eigenbasis and $S(\rho||D[\rho])$ represents a measure of coherence (Gour *et al* 2009) with the quantum relative entropy $S(\rho||\sigma) = \mathrm{Tr}[\rho\log\rho] - \mathrm{Tr}[\rho\log\sigma]$. Korzekwa *et al* (2016) showed that the amount of work extracted from the coherence of state ρ is arbitrarily close to the free energy extraction limit, i.e. $kTS(\rho||D[\rho])$. However, all the coherence

[3] If not stated otherwise, we define the von Neumann entropy with the natural logarithm.

cannot be converted to work, as some of the work will be needed to repump the reservoir so that it can be reused. Nevertheless, they showed that the amount of extracted work can be arbitrarily close to the coherence provided when the coherence of the reference system is sufficiently large.

Recently, Vaccaro *et al* (2018) analysed the proposal of Åberg (2014) to check whether coherence can be really used repeatedly without any degradation in its performance. In particular, Vaccaro *et al* (2018) showed that if the reservoir is used repeatedly then the final systems can have correlations that have not been considered by Åberg (2014). The correlations between the final systems actually affect the efficiency of repeated operations. To be precise, the coherence drawn from the reservoir by uncorrelated systems is completely different compared to the coherence drawn by correlated systems. Furthermore, they showed that neglecting these correlations among the final systems could lead to unphysical phenomena, for example, two non-orthogonal reservoir states can be discriminated almost perfectly (Vaccaro *et al* 2018).

In the context of whether coherence is catalytic, an important study was done by Lostaglio and Müller (2019). In fact, it has been shown that coherence can neither be cloned nor broadcasted for finite dimensional systems. The coherence of a finite-dimensional system ρ^E can be broadcast if there exists a time-translation covariant channel Λ and an incoherent state ρ^S, such that $[\rho^S, H^S] = 0$ with

$$\Lambda\left(\rho^E \otimes \rho^S\right) = \sigma^{ES}, \tag{61}$$

where $\rho^E = \mathrm{Tr}_S(\sigma^{ES})$, $\sigma^S = \mathrm{Tr}_E(\sigma^{ES})$ and $[\sigma^S, H^S] \neq 0$. The action of a time-translation covariant channel Λ on a state ρ with Hamiltonian H is expressed as $\Lambda(\rho) = \mathrm{Tr}_A[U(\rho \otimes \sigma^A)U^\dagger]$, where σ^A is some ancillary state with Hamiltonian H_A and U represents some global unitary with $[U, H + H_A] = 0$. On the other hand, the coherence of the system ρ^S can be cloned when there is no correlation between the sub-systems, i.e.

$$\Lambda\left(\rho^E \otimes \rho^S\right) = \sigma^{ES} = \rho^E \otimes \sigma^S. \tag{62}$$

Here, the system E acts like a catalyst. In such a scenario, correlated catalysis under covariant operations cannot broadcast or clone coherence in an initially incoherent state ρ^S (Lostaglio and Müller 2019). However, the questions of coherence cloning and broadcasting remain open for infinite-dimensional systems. Note that the results of Åberg (2014) suggest that a weaker version of coherence broadcasting, where the state ρ^E is allowed to change in the procedure, is possible for infinite dimensional systems. Hence, it seems like broadcasting as defined earlier in equation (61), where the state ρ^E remains unchanged may not be possible.

In the above result, we see that broadcasting of coherence to an initially incoherent state is not possible from a finite dimensional system (Lostaglio and Müller 2019). On the contrary, Åberg's result suggests that it could be possible with infinite-dimensional systems (AAberg 2014). Recently, Ding *et al* (2021) tried to create a connection between the two results by studying the role of a finite-dimensional catalyst in enlarging the set of accessible states under time-translational covariant operations. In their protocol they allow correlation between the system and the catalyst after the transformation, however, the reduced state of the catalyst remains unchanged. As correlation is allowed, they first showed that pure state catalysts do not help in any state transformations under covariant operations. Precisely, if ρ cannot be transformed into ρ' under covariant operations, then there exists no pure catalyst state $|\phi\rangle$ that can catalyse the transformation $\rho \otimes \phi \to \rho' \otimes \phi$ under covariant catalytic transformations. This result is a generalisation of the previously studied result by Marvian and Spekkens (2013), where ρ and ρ' are considered to be pure. Now if one can allow correlation between the system and the catalyst then there exist coherent states such that the achievable set of states from it is strictly larger under covariant operations. This has been shown explicitly for qubit states (Ding *et al* 2021). Furthermore, with the dimension of the catalyst the accessible set of states increases (Ding *et al* 2021).

In the same spirit as Nielsen's theorem in entanglement theory (see theorem 1), a counterpart has been derived in coherence theory to address the conversions among pure states (Du *et al* 2015a, Winter and Yang 2016, Zhu *et al* 2017). Before presenting the result, we introduce incoherent states and incoherent operations. To introduce these, we first fix a basis $\{|i\rangle\}$ (called as an incoherent basis) in a finite-dimensional Hilbert space. The incoherent states are diagonal in the incoherent basis and can be represented as $\sum_i p_i |i\rangle\langle i|$, where $p_i \geqslant 0$ and $\sum_i p_i = 1$. The set of all such incoherent states is represented by $\mathcal{I}$. A quantum operation characterised by a set of Kraus operators $\{K_n\}$ satisfying $\sum_n K_n^\dagger K_n = \mathbb{1}$, is called incoherent operation if the Kraus operators satisfy $K_n \rho K_n^\dagger / \mathrm{Tr}\left(K_n \rho K_n^\dagger\right) \in \mathcal{I}$ for all n and $\rho \in \mathcal{I}$ (Baumgratz *et al* 2014). Let us consider a pair of states: $|\psi\rangle = \sum_{i=0}^{d-1} \alpha_i |i\rangle$ and $|\phi\rangle = \sum_{i=0}^{d-1} \beta_i |i\rangle$, where α_i and β_i are arranged in the following way $|\alpha_0|^2 \geqslant \cdots \geqslant |\alpha_{d-1}|^2$ and $|\beta_0|^2 \geqslant \cdots \geqslant |\beta_{d-1}|^2$. Then $|\psi\rangle$ can be converted to $|\phi\rangle$ via incoherent operations iff $\Delta[\psi] \prec \Delta[\phi]$, or equivalently the following holds (Du *et al* 2015a, Winter and Yang 2016, Zhu *et al* 2017)

$$\sum_{i=0}^{k} |\alpha_i|^2 \leqslant \sum_{i=0}^{k} |\beta_i|^2 \;\; \forall k = \{0, \ldots, d-1\}, \tag{63}$$

where $\psi = |\psi\rangle\langle\psi|$ and $\Delta[\rho] = \sum_i \langle i|\rho|i\rangle |i\rangle\langle i|$ represents complete dephasing in the incoherent basis. Note that the above result also holds true for other classes of free operations considered in coherence theory (Winter and Yang 2016).

Following similar ideas as for entanglement, it is not difficult to find examples of incomparable states where the above conditions are not satisfied. However, in some cases, an additional coherent state can help in accomplishing a forbidden transformation without being changed in the process (Du *et al* 2015a). The above result in equation (63) sheds some light on the problem of coherence catalysis, similar to the previously discussed cases for entanglement (Jonathan and Plenio 1999). In fact, maximally coherent states cannot catalyse a forbidden transformation between two arbitrary incomparable pure states (Du *et al* 2015a). Furthermore, if a forbidden transformation $|\psi\rangle \nrightarrow |\phi\rangle$ can be accomplished with the help of a pure catalyst state, then the reverse transformation $|\phi\rangle \nrightarrow$

$|\psi\rangle$ cannot be catalysed with a pure catalyst state (Du *et al* 2015a). Hence, pure catalysts can only catalyse transformations between incomparable pure states. Another important consequence of equation (63) is that conversion of $|\psi\rangle$ into $|\phi\rangle$ is possible with the help of a pure catalyst if $|\alpha_0| \leqslant |\beta_0|$ and $|\alpha_{d-1}| \geqslant |\beta_{d-1}|$ (Du *et al* 2015a).

Bu *et al* (2016) studied the necessary and sufficient conditions for catalytic coherence transformations considering pure catalyst states. They derived necessary and sufficient conditions for the existence of a catalyst for a pair of incomparable pure states. Suppose $|\psi\rangle$ and $|\phi\rangle$ are two incomparable states of dimension d. Then there exists a catalyst state $|\eta\rangle$ accomplishing the transformation $|\psi\rangle \otimes |\eta\rangle \to |\phi\rangle \otimes |\eta\rangle$, if and only if (Bu *et al* 2016)

$$H_\alpha(\Delta[\psi]) > H_\alpha(\Delta[\phi]) \ \forall \alpha \in (-\infty, \infty)/0 \text{ and} \tag{64}$$

$$\mathrm{Tr}(\log \Delta[\psi]) > \mathrm{Tr}(\log \Delta[\phi]). \tag{65}$$

Here

$$H_\alpha(\rho) = \frac{\mathrm{sgn}(\alpha)}{1-\alpha} \log \mathrm{Tr}[\rho^\alpha] \tag{66}$$

represents Rényi entropy. The strict inequalities in the above equations can be made less strict by allowing an infinitesimally small error in the initial or in the final state. In fact, for arbitrary $\varepsilon > 0$, there exists a state $|\psi_\varepsilon\rangle$ with $||\psi - \psi_\varepsilon||_1 \leqslant \varepsilon$ such that $|\psi_\varepsilon\rangle$ can be converted into $|\phi\rangle$ via catalytic incoherent operations (assuming pure catalyst states), if and only if (Bu *et al* 2016)

$$H_\alpha(\Delta[\psi]) \geqslant H_\alpha(\Delta[\phi]) \ \forall \alpha \in (-\infty, \infty)/0 \text{ and} \tag{67}$$

$$\mathrm{Tr}(\log \Delta[\psi]) \geqslant \mathrm{Tr}(\log \Delta[\phi]). \tag{68}$$

Note the analogy between this result and the case of entanglement as discussed in equation (18). Furthermore, Bu *et al* (2016) explored the structure of qubit catalyst states for four-dimensional state transformations. The necessary and sufficient conditions for the existence of a qubit catalyst $|\eta\rangle = \sqrt{a}|0\rangle + \sqrt{1-a}|1\rangle$ for two four dimensional incomparable states $|\psi\rangle = \sum_{i=0}^{3} \sqrt{\alpha_i}|i\rangle$ and $|\phi\rangle = \sum_{i=0}^{3} \sqrt{\beta_i}|i\rangle$ are

$$\alpha_0 \leqslant \beta_0, \tag{69}$$

$$\alpha_0 + \alpha_1 > \beta_0 + \beta_1, \tag{70}$$

$$\alpha_0 + \alpha_1 + \alpha_2 \leqslant \beta_0 + \beta_1 + \beta_2, \tag{71}$$

and additionally

$$\max\left\{\frac{\alpha_0+\alpha_1-\beta_0}{\beta_1+\beta_2}, 1-\frac{\alpha_3-\beta_3}{\beta_2-\alpha_2}\right\} \leqslant a \leqslant \min\left\{\frac{\beta_0}{\alpha_0+\alpha_1}, \frac{\beta_0-\alpha_0}{\alpha_1-\beta_1}, 1-\frac{\beta_3}{\alpha_2+\alpha_3}\right\}. \tag{72}$$

Here, without loss of generality we assume that $\{\alpha_i\}$ and $\{\beta_i\}$ are real and they are arranged in descending order. Hence, equation (72) provides the range of the catalyst for the above transformation. Similar to the result in entanglement theory (Jonathan and Plenio 1999), catalytic conversions (assuming pure catalysts) of incomparable pure states are not possible for two and three-dimensional systems, and the first nontrivial result can be found in dimension four as shown in equation (72), see Bu *et al* (2016).

So far we have discussed deterministic state transformations. If a pair of states cannot be transformed deterministically, i.e, they violate equation (63), they still can be transformed stochastically with some optimal probability $P_{\max}$ (Du *et al* 2015b). Bu *et al* (2016) studied the advantage of a catalyst in stochastic state transformations. In fact, they showed that for two incomparable pure states $|\psi\rangle = \sum_{i=0}^{d-1} \alpha_i|i\rangle$ and $|\phi\rangle = \sum_{i=0}^{d-1} \beta_i|i\rangle$, there exists a catalyst state $|\eta\rangle$ such that $P_{\max}(|\psi\rangle \to |\phi\rangle) < P_{\max}(|\psi\rangle \otimes |\eta\rangle \to |\phi\rangle \otimes |\eta\rangle)$, if and only if

$$P_{\max}(|\psi\rangle \to |\phi\rangle) < \min\left\{\frac{|\alpha_{d-1}|^2}{|\beta_{d-1}|^2}, 1\right\}. \tag{73}$$

Note that a similar result is also present in entanglement theory (Jonathan and Plenio 1999). By analogy to self-catalysis in entanglement (Duarte *et al* 2016), in Bu *et al* (2016) the authors also explored self-catalysis in the context of coherence theory. As an example, consider the states

$$|\psi\rangle = \sqrt{0.9}|0\rangle + \sqrt{0.081}|1\rangle + \sqrt{0.01}|2\rangle + \sqrt{0.009}|3\rangle, \tag{74}$$

$$|\phi\rangle = \sqrt{0.95}|0\rangle + \sqrt{0.03}|1\rangle + \sqrt{0.02}|2\rangle. \tag{75}$$

It is easy to check that $|\psi\rangle \nrightarrow |\phi\rangle$, while at the same time $|\psi\rangle \otimes |\psi\rangle \to |\phi\rangle \otimes |\psi\rangle$ (Bu *et al* 2016).

Earlier, we discussed the advantage of catalysis-assisted probabilistic transformation of pure states. Naturally, one may wonder whether a catalyst exists such that it can enhance the probability of transforming a mixed state into a pure state. Recently, Liu and Zhou (2020) explored this scenario for strictly incoherent operations and derived necessary and sufficient conditions for the existence of a pure catalyst such that the optimal transformation probability is enhanced. A quantum operation characterised by Kraus operators $\{K_n\}$ is called a strictly incoherent operation if all K_n and $K_n^\dagger$ are incoherent in the given incoherent basis (Winter and Yang 2016, Yadin *et al* 2016). For such operations Liu and Zhou (2020) showed that $P_{\max}(\rho \otimes \eta \to \psi \otimes \eta) > P_{\max}(\rho \to \psi)$ is possible, if and only if the following holds at least for one μ:

$$P_{\max}(\phi^\mu \to \psi) < \min\left\{\frac{\phi_n^\mu}{\psi_n}, 1\right\}, \tag{76}$$

where $\{|\phi^\mu\rangle\}$ correspond to the maximally dimensional pure coherent state subspaces of ρ, $|\phi^\mu\rangle = \sum_{i=0}^{d_1-1} \phi_i^\mu|i\rangle$, $|\psi\rangle = \sum_{i=0}^{d_2-1} \psi_i|i\rangle$, $n = \max\{d_1-1, d_2-1\}$, and d_1 and d_2 denote the dimensions of $|\phi\rangle^\mu$ and $|\psi\rangle$, respectively. The coefficients of the pure states are arranged in descending order. A state ρ has a pure coherent state subspace of dimension n if there exists an incoherent projector $\mathbb{P}$ such that $\frac{\mathbb{P}_\mu \rho \mathbb{P}_\mu}{\mathrm{Tr}\mathbb{P}_\mu \rho \mathbb{P}_\mu} = \phi^\mu$ and ϕ^μ has coherence rank n (Liu and Zhou 2020). Coherence rank corresponds to the number of nonzero terms ϕ_i^μ in $|\phi\rangle^\mu$. The subspace is maximally coherent when the rank of ϕ^μ cannot be increased.

Catalytic coherence transformation of some classes of mixed states has been studied by Xing and Yang (2020a). Consider the following rank-2 mixed states of the forms

$$\rho = \alpha|\psi\rangle\langle\psi| + (1-\alpha)\,|4\rangle\langle 4| \text{ and} \tag{77}$$
$$\sigma = \beta|\phi\rangle\langle\phi| + (1-\beta)\,|4\rangle\langle 4|, \tag{78}$$

where

$$|\psi\rangle = \sqrt{0.38}|0\rangle + \sqrt{0.38}|1\rangle + \sqrt{0.095}|2\rangle + \sqrt{0.095}|3\rangle + \sqrt{0.05}|4\rangle, \tag{79}$$
$$|\phi\rangle = \sqrt{0.5}|0\rangle + \sqrt{0.25}|1\rangle + \sqrt{0.25}|2\rangle. \tag{80}$$

The transformation of ρ into σ is not possible via incoherent transformation (Xing and Yang 2020a). However, as shown in Xing and Yang (2020a), the transformation can be achieved with the help of a catalyst in state $|\varphi\rangle = \sqrt{0.6}|0\rangle + \sqrt{0.4}|1\rangle$. Note that the above states and the catalytic conversion protocol are similar to the entanglement case discussed in Eisert and Wilkens (2000). The above transformation of rank-2 mixed states can be translated to some special class of higher-rank states as well. In addition, for general mixed states catalytic coherence transformation is also discussed in Xing and Yang (2020a).

Xing and Yang (2020b) also studied *supercatalytic* coherence transformation. Suppose we have two incomparable states $|\psi\rangle$ and $|\phi\rangle$, and we want to achieve the transformation $|\psi\rangle \to |\phi\rangle$. We already see that such transformations can be realised with the help of some auxiliary state if equations (64) and (65) are satisfied. In contrast to this, the transformation can be realised in another way where the auxiliary state changes in the process, but the coherence of the final auxiliary system is increased. More precisely, $|\psi\rangle \otimes |\varphi_1\rangle \to |\phi\rangle \otimes |\varphi_2\rangle$, where $|\varphi_1\rangle$ and $|\varphi_2\rangle$ represent auxiliary states with $C_{\mathrm{r}}(\varphi_2) \geqslant C_{\mathrm{r}}(\varphi_1)$. Here C_{r} represents the relative entropy of coherence (Baumgratz *et al* 2014). Examples and some interesting results of such supercatalytic coherence transformation are discussed in Xing and Yang (2020b).

3.3. Quantum thermodynamics

Catalysis has also been broadly explored in quantum thermodynamics, investigating transformations of quantum systems under thermal operations in the presence of catalysts. Thermal operations model transformations of a quantum system S with Hamiltonian H_S interacting with a heat reservoir B with Hamiltonian H_B and in a thermal state $\gamma^B = \mathrm{e}^{-\beta H_B}/Z_B$, where $\beta = 1/kT$ is the inverse temperature and $Z_B = \mathrm{Tr}\,\mathrm{e}^{-\beta H_B}$ is the partition function. The interaction is described by a unitary U_{SB} that conserves the total energy: $[U_{SB}, H_S + H_B] = 0$. A thermal operation Λ on a state ρ^S is defined as (Janzing *et al* 2000)

$$\Lambda\left(\rho^S\right) = \mathrm{Tr}_B\left(U_{SB}\left[\rho^S \otimes \gamma^B\right] U_{SB}^{\dagger}\right). \tag{81}$$

See also Horodecki and Oppenheim (2013) for a detailed discussion on thermal operations.

To describe *catalytic* thermal operations, Brandão *et al* (2015) allow for an additional system σ^C as a catalyst. Precisely, a transformation from ρ^S to μ^S is possible via catalytic thermal operations, if and only if there exists a thermal operation Λ and a state σ^C taking $\rho^S \otimes \sigma^C$ to $\mu^S \otimes \sigma^C$ (Brandão *et al* 2015), i.e.

$$\Lambda\left(\rho^S \otimes \sigma^C\right) = \mathrm{Tr}_B\left(U_{SCB}\left[\rho^S \otimes \sigma^C \otimes \gamma^B\right] U_{SCB}^{\dagger}\right) = \mu^S \otimes \sigma^C, \tag{82}$$

where the unitary U_{SCB} satisfies $[U_{SCB}, H_S + H_C + H_B] = 0$ and H_C represents the Hamiltonian of the catalyst system C. Moreover, the authors find necessary and sufficient conditions for such transformation to be possible when the initial and the final states are block-diagonal in energy eigenbasis (Brandão *et al* 2015).

Theorem 3 (see family of second laws in Brandão *et al* (2015)). *There exists a catalytic thermal operation mapping a diagonal state ρ to another diagonal state ρ', if and only if*

$$F_\alpha(\rho, \gamma_\beta) \geqslant F_\alpha(\rho', \gamma_\beta) \text{ for } \alpha \geqslant 0. \tag{83}$$

Here, the generalised free energies are defined as

$$F_\alpha(\rho, \gamma_\beta) = kTS_\alpha(\rho||\gamma_\beta) - kT\log Z, \tag{84}$$

where

$$S_\alpha(\rho||\sigma) = \begin{cases} \frac{1}{\alpha-1}\log \mathrm{Tr}\left(\rho^\alpha \sigma^{1-\alpha}\right), & \alpha \in [0,1) \\ \frac{1}{\alpha-1}\log \mathrm{Tr}\left[\left(\sigma^{\frac{1-\alpha}{2\alpha}}\rho\sigma^{\frac{1-\alpha}{2\alpha}}\right)^\alpha\right], & \alpha > 1 \end{cases} \tag{85}$$

is the Rényi relative entropy and $S_1(\rho||\sigma) = \mathrm{Tr}[\rho(\log\rho - \log\sigma)]$ is found by taking the limit $\alpha \to 1$. Also note that for $\lim_{\alpha\to 1} F_\alpha(\rho, \gamma_\beta)$ we obtain the ordinary Helmholtz free energy, see also equation (59). The family of inequalities in equation (83) is known as the *second laws of quantum thermodynamics* (Brandão *et al* 2015). In contrast, note that at the macroscopic scale or in the standard thermodynamic limit, transitions are governed by free energy only, and its monotonicity provides the standard second law. Furthermore, Brandão *et al* (2015) discussed a generalisation of the above result by considering transformations of arbitrary states and provided some necessary conditions. A state ρ can be transformed into ρ' with arbitrary accuracy only if (Brandão *et al* 2015)

$$F'_\alpha(\rho, \gamma_\beta) \geqslant F'_\alpha(\rho, \gamma_\beta)\ \forall \alpha \geqslant \frac{1}{2}, \tag{86a}$$
$$F'_\alpha(\gamma_\beta, \rho) \geqslant F'_\alpha(\gamma_\beta, \rho')\ \text{for}\ \frac{1}{2} \leqslant \alpha \leqslant 1, \text{ and,} \tag{86b}$$
$$F''_\alpha(\rho, \gamma_\beta) \geqslant F''_\alpha(\rho', \gamma_\beta)\ \text{for}\ 0 \leqslant \alpha \leqslant 2, \tag{86c}$$

where F' and F'' are defined as

$$F'(\rho, \gamma_\beta) = \frac{kT}{\alpha-1}\log \mathrm{Tr}\left[\left(\gamma_\beta^{\frac{1-\alpha}{2\alpha}}\rho\gamma_\beta^{\frac{1-\alpha}{2\alpha}}\right)^\alpha\right] - kT\log Z, \tag{87}$$

$$F''(\rho, \gamma_\beta) = \frac{kT\,\mathrm{sgn}(\alpha)}{\alpha-1}\log \mathrm{Tr}\left(\rho^\alpha \gamma_\beta^{1-\alpha}\right) - kT\log Z. \tag{88}$$

van der Meer *et al* (2017) studied how the above mentioned free energies behave in the asymptotic limit for block-diagonal states. As a result of their study, they found out that if $F(\rho) > F(\rho')$, then for any $\varepsilon > 0$ there exists a sufficiently large n and a *catalytic* thermal operation Λ, taking $\rho^{\otimes n}$ arbitrarily close to $\rho'^{\otimes n}$, i.e. $||\Lambda(\rho^{\otimes n}) - \rho'^{\otimes n}||_1 \leqslant \varepsilon$.

In contrast to the previous result by Brandão *et al* (2015), Lostaglio *et al* (2015b) showed that only the monotonicity of free energy is required for state transformations if one allows correlations among the particles of the catalyst and an arbitrarily small error in the final state ρ'. In their definition of catalytic state transformation, a block-diagonal state ρ is transformed to another block-diagonal state ρ' when for all $\varepsilon > 0$, there exists a thermal operation Λ, a state ρ'_ε with $||\rho' - \rho'_\varepsilon|| < \varepsilon$ and a correlated state $\sigma_{C_1\cdots C_N}$ with marginals $\sigma_{C_1}, \cdots, \sigma_{C_N}$ such that (Lostaglio *et al* 2015b)

$$\Lambda\left(\rho \otimes \sigma_{C_1} \otimes \cdots \otimes \sigma_{C_N}\right) = \rho'_\varepsilon \otimes \sigma_{C_1\cdots C_N}. \tag{89}$$

Such a transformation is possible, if and only if $F(\rho) \geqslant F(\rho')$ (Lostaglio *et al* 2015b). However, note that although the local states of the particles remain the same, the global state of the catalyst changes in this procedure. This may lead to embezzling, a phenomenon discussed in more detail in section 7.

The above result emphasises that if a transformation is possible in the thermodynamic limit then it is also possible in the single copy settings. Hence, it provides an operational meaning to free energy. Motivated by this result, Müller and Pastena (2016) derived a necessary and sufficient condition for the transformations of probability distributions. For this, they studied transformations of probability distributions under bistochastic maps, which are represented by bistochastic matrices that are composed of non-negative real numbers with every individual row and column summing up to one. In particular, Müller and Pastena (2016) considered transformations of probability distributions of the form

$$p \otimes r_1 \otimes \cdots \otimes r_N \to q \otimes r_{1,\ldots,N}, \tag{90}$$

where the probability distribution $p = (p_1, \ldots, p_m)$ is mapped to another probability distribution $q = (q_1, \ldots, q_m)$ with the help of auxiliary systems that get correlated in the procedure, but at the same time, the marginals remain unchanged. In such a scenario, if p and q are not identical up to permutation and both contain non-zero elements, then the transformation from p to q is possible if and only if $H(p) < H(q)$ (Müller and Pastena 2016), where $H(p) = -\sum_i p_i \log p_i$ represents the Shannon entropy.

In most of the previous studies (Brandão *et al* 2015, Lostaglio *et al* 2015b, van der Meer *et al* 2017), only block-diagonal states are considered, which contain no coherence in the eigenbasis of the Hamiltonian. Although the conditions in equations (86) hold for arbitrary states, the role of coherence in state transformations is not apparent from them. On the contrary, in Lostaglio *et al* (2015a) the authors considered arbitrary states and derived the necessary conditions for state transformations under catalytic thermal operation. They considered the exact catalytic thermal operation presented in equation (82). If a state ρ can be converted to another state ρ' via a catalytic thermal operation where the catalyst state is in the block-diagonal form, then we have (Lostaglio *et al* 2015a)

$$A_\alpha(\rho) \geqslant A_\alpha(\rho') \ \forall \alpha \geqslant 0, \tag{91}$$

where $A_\alpha(\rho) = S_\alpha(\rho||D[\rho])$ represents free coherence of the state ρ (Lostaglio *et al* 2015a) and D corresponds to the dephasing operation in the given energy eigenbasis.

In Wilming *et al* (2017), the authors considered marginal-catalytic transformation and correlated-catalytic transformation under Gibbs preserving operations and showed that free energy is the only monotonic function under such free operations. In their definition, the marginal-catalytic transformation has a similar structure as presented in equation (89), with the only difference that the final state is returned exactly and the maps are Gibbs preserving (Wilming *et al* 2017). On the other hand, in the case of correlated-catalytic transformation of ρ^S into μ^S, we have $\rho^S \otimes \sigma^C \to \mu^{SC}$, where the final state μ^{SC} is correlated with marginals μ^S and σ^C (Wilming *et al* 2017). Then it has been shown that any monotone under these two kinds of catalytic transformations, which is additive on the tensor product and continuous, is proportional to the free energy difference (Wilming *et al* 2017). Hence, free energy is the only monotone for such kind of catalytic transformations.

Recently, the role of exact uncorrelated catalysis for the Markovian thermal process has been investigated by Korzekwa and Lostaglio (2022). In fact, catalysis has been demonstrated to be advantageous in the cooling of systems. More specifically, the authors demonstrated that a qubit catalyst can cool a system initially attached to a bath to temperatures lower than the bath temperature, whereas without a catalyst, the system can only be cooled to the bath temperature (Korzekwa and Lostaglio 2022).

In a recent paper (Son and Ng 2022), the role of exact uncorrelated catalysis in elementary thermal operations has been discussed. Elementary thermal operations are a subset of thermal operations, and their practical realisations are apparent in comparison to thermal operations (Lostaglio *et al* 2018). In Son and Ng (2022), the authors provided a few examples to illustrate that for a given initial qutrit state, the set of final states under thermal operations can almost be achieved with elementary thermal operations helped by qubit catalysts. In addition, Son and Ng (2022) demonstrated an example in which the set of final states under thermal operations is strictly smaller compared to the set of final states that can be achieved under elementary thermal operations aided by qubit catalysts. Therefore, as elementary thermal operations can be experimentally realised, this method could provide an experimentally feasible way of achieving the set of final states under thermal operations using only elementary thermal operations and catalysts.

3.4. Purity

In the resource theory of purity, a key question is to find the minimum amount of noise required to perform a given

physical process on a quantum system (Gour *et al* 2015, Boes *et al* 2018). In this theory, the free operations are noisy operations. In general, a noisy operation can be implemented in the following steps: (i) first, a given system is attached to an ancillary system prepared in a completely mixed state; (ii) then, a unitary operation is performed to couple them; and (iii) finally, we discard the ancillary system by taking partial trace over the ancillary system. The overall transformation on a state ρ^S in the Hilbert space $\mathcal{H}_S$ can be written as

$$\Lambda\left[\rho^S\right] = \mathrm{Tr}_A\left[U\left(\rho^S \otimes \frac{\mathbb{1}_A}{d_A}\right)U^\dagger\right], \tag{92}$$

where U is in $\mathcal{H}_S \otimes \mathcal{H}_A$, $\mathcal{H}_A$ represents the Hilbert space of the ancillary system, and d_A corresponds to the dimension of the ancillary system. Note that this kind of operation was first reported in Horodecki *et al* (2003a, 2003b). All the states except the maximally mixed state are considered resources in this theory. Furthermore, note that without loss of generality, we can assume the states to be diagonal matrices as unitary operations are free in this resource theory. The necessary and sufficient conditions for exact state transformations are given by the majorisation relation. More precisely, a transformation from ρ to σ is possible under a noisy operation if and only if $\sigma \prec \rho$ (Gour *et al* 2015). In contrast to the entanglement theory, the majorisation relation is in the opposite direction.

An immediate consequence is that there exist transformations which are not possible only via noisy operations but could be accomplished with the help of a catalyst (Gour *et al* 2015). To analyse state transformations under catalytic noisy operations, one needs to consider a trumping relation $\sigma \prec_T \rho$, which states that there exists a catalyst η that helps in converting ρ into σ via noisy operations, i.e. $\sigma \otimes \eta \prec \rho \otimes \eta$ (Gour *et al* 2015). The noisy trumping relation is equivalent to (Klimesh 2007, Gour *et al* 2015)

$$f_\alpha(\rho) > f_\alpha(\sigma)\ \forall\ \alpha \in (-\infty, \infty), \tag{93}$$

where the functions f_α are the extensions of the functions in equation (17) to density matrices, i.e.

$$f_\alpha(\rho) = \begin{cases} \ln \mathrm{Tr}\left[\rho^\alpha\right] & (\alpha > 1); \\ \mathrm{Tr}\left[\rho \ln \rho\right] & (\alpha = 1); \\ -\ln \mathrm{Tr}\left[\rho^\alpha\right] & (0 < \alpha < 1); \\ -\mathrm{Tr}\left[\ln \rho\right] & (\alpha = 0); \\ \ln\left[\mathrm{Tr}\,\rho^\alpha\right] & (\alpha < 0). \end{cases} \tag{94}$$

Similar to the case of entanglement and coherence, an alternative definition of noisy trumping is introduced (Gour *et al* 2015), where an infinitesimally small error is allowed in the initial or final state. More formally a state ρ trumps noisily σ, if there exists a state η such that $\rho \otimes \nu_\delta \otimes \eta \to \sigma \otimes \eta$ for all $\delta > 0$. Here, $\nu_\delta = (1/m, \ldots, 1/m, 0, \ldots, 0)$ is a sharp state of dimension n with $m \leqslant n$ and satisfies $||\nu_\delta - \mathbb{1}_n/n||_1 \leqslant \delta$. With this, Gour *et al* (2015) provided necessary and sufficient conditions for noisy trumping as follows: a state ρ noisy trumps σ, if and only if

$$H_\alpha(\rho) \leqslant H_\alpha(\sigma)\ \forall\ \alpha \geqslant 0, \tag{95}$$

where H_α represent the Rényi entropies defined earlier in equation (66). Furthermore, they also explored approximate catalysis where a state ρ is transformed into σ approximately via noisy operations and a catalyst η. However, no further restriction is imposed on the catalyst except that the final state is ε close to $\sigma \otimes \eta$. Therefore, the catalyst might change infinitesimally in the procedure. The slight change in the catalyst leads to embezzling, which is discussed in more detail in section 7. In such a scenario, for a given $\varepsilon > 0$ the noisy operations are trivial, i.e. every state can be converted to every other state (Brandão *et al* 2015, Gour *et al* 2015). We also refer to section 4 for a more detailed analysis of approximate catalysis.

Previously, we discussed the necessary and sufficient conditions for state transformations under catalytic noisy operations. Now, one may wonder what is the minimum amount of randomness required to implement state transitions via noisy operations, and if there is any difference between classical and quantum randomness. Before answering these questions, an understanding of classical and quantum randomness is important. The nature of the source of randomness defines whether it is quantum or classical. A transformation from ρ to σ using m units of *classical randomness* is possible if there exists a set of unitaries $\{U_i\}$ such that (Boes *et al* 2018)

$$\sigma = \frac{1}{m}\sum_{i=1}^{m} U_i \rho U_i^\dagger \tag{96}$$

and the transformation is denoted by $\rho \xrightarrow[m]{C} \sigma$. On the other hand, a transformation from ρ to σ using m units of *quantum randomness* is possible if there exists a unitary U and an ancillary system A of dimension m in the maximally mixed state $\mathbb{1}/m$ such that (Boes *et al* 2018)

$$\sigma = \mathrm{Tr}_A\, U\left(\rho \otimes \frac{\mathbb{1}}{m}\right)U^\dagger, \tag{97}$$

and the transformation is denoted by $\rho \xrightarrow[m]{Q} \sigma$. When there is no bound on the amount of available randomness (quantified by m), the set of transitions that can be achieved with quantum randomness sources can also be achieved with classical randomness sources (Marshall *et al* 2011, Boes *et al* 2018). Equivalently we have,

$$\rho \xrightarrow[\infty]{C} \sigma \iff \rho \xrightarrow[\infty]{Q} \sigma \iff \rho \succ \sigma. \tag{98}$$

The transformations of the form (97) in the limit $m \to \infty$ have been previously studied in Gour *et al* (2015), where noise is treated as free or in other words we can use a maximally mixed state of arbitrary dimension to implement a noisy operation. On the contrary, Boes *et al* (2018) treated noise as a resource and investigated the minimum amount of noise required to accomplish a given state transformation. For a given Hilbert space $\mathcal{H}_S$ of dimension d, the minimum amount of noise or randomness can be quantified as

$$m_X^*(d) = \arg\min_m \left[\rho \xrightarrow[m]{X} \sigma\ \forall\ \rho, \sigma \in \mathcal{H}_S \text{ with } \rho \succ \sigma\right], \tag{99}$$

where X corresponds to either C or Q. It is important to note that for a specific transition the required randomness could be smaller than the above specified value in equation (99). Boes *et al* (2018) provided an analytical expression for these quantities as following

$$m_C^*(d) = d, \tag{100}$$

$$m_Q^*(d) = \lceil \sqrt{d} \rceil. \tag{101}$$

Here, we will focus mainly on quantum randomness. In proving the optimality of m_Q^*, the authors of Boes *et al* (2018), used the Schur–Horn lemma (Horn 1954) with the majorisation relation $\rho \succ \sigma$, which states that for every state transition under noisy operation there exists a map of the form

$$\sigma = \Lambda(\rho) = \mathcal{U}' \circ \Delta_B \circ \mathcal{U}(\rho) \tag{102}$$

where, $\mathcal{U}$, $\mathcal{U}'$ are unitary channels that are specific to the states ρ and σ. Here Δ_B represents a dephasing map in a fixed basis $B = \{|i\rangle\}_{i=1}^d$ of dimension d and defined as $\langle i|\Delta_B(\rho)|j\rangle = \langle i|\rho|j\rangle\delta_{ij}$. As the implementations of unitary channels do not require any noise, all the noise needed for a state transition is required to implement the dephasing map Δ_B in equation (102). Next Boes *et al* (2018) provided an explicit protocol implementing dephasing map Δ_B in any basis B, using a noise model and requiring $m_Q^*(d) = \lceil \sqrt{d} \rceil$. Specifically, for any dimension d and basis B, Boes *et al* (2018) showed that there exists a unitary U, such that

$$\mathrm{Tr}_A \left[U \left(\rho \otimes \frac{\mathbb{1}}{\lceil \sqrt{d} \rceil} \right) U^\dagger \right] = \Delta_B(\rho), \tag{103}$$

$$\mathrm{Tr}_S \left[U \left(\rho \otimes \frac{\mathbb{1}}{\lceil \sqrt{d} \rceil} \right) U^\dagger \right] = \frac{\mathbb{1}}{\lceil \sqrt{d} \rceil} \; \forall \, \rho. \tag{104}$$

Although the final state may have correlations between the system and the ancilla, the second condition implies that the ancillary system A is returned exactly in the same state as before and thus acts as a catalyst in the process. As mentioned previously, Boes *et al* (2018) considered noise as a resource, and interestingly, here the noise is not consumed in the process and can be reused again. Furthermore, note that equation (102) emphasises that every possible state transition utilising randomness can be accomplished if a dephasing map can be constructed and Boes *et al* (2018) provided an explicit protocol for that.

In Boes *et al* (2018), only maximally mixed states have been used as a source of randomness. In contrast, Lie *et al* (2021) considered any arbitrary randomness source η and derived a necessary and sufficient condition such that the randomness source η can be used to dephase an arbitrary input state ρ. In fact, it has been shown that any randomness source η can dephase a d^2 dimensional input states catalytically iff the following holds (Lie *et al* 2021)

$$S_{\min}(\eta) \geqslant \log_2 d, \tag{105}$$

where $S_{\min}(\eta) = -\log_2 ||\eta||$ represents min-entropy and the operator norm $||\eta||$ corresponds to the largest singular value of η. However, note that in their formulation, the state of the catalyst η can change, but the same catalyst can be used repeatedly to dephase arbitrary input states of the same dimension. In validation of this, the authors of Lie *et al* (2021) showed that randomness described by min-entropy is not consumed in the process and can be reused.

Based on the result in Boes *et al* (2018), Boes *et al* (2019) proved that von Neumann entropy completely characterises single-shot state transitions under unitary operations. To achieve this, the authors in Boes *et al* (2019) used an ancillary system as a catalyst and an environment which dephases a system in a preferred basis. In particular, the following has been shown (Boes *et al* 2019).

Theorem 4 (see theorem 1 in Boes *et al* (2019)). *For any two states ρ and σ of the same finite dimension and different spectra, the following statements are equivalent*

(i) $S(\sigma) > S(\rho)$ *and* $rank(\sigma) \geqslant rank(\rho)$

(ii) There exists a unitary U and a finite dimensional catalyst in the state η, which can be chosen to be diagonal in any basis B such that

$$\mathrm{Tr}_A \, U[\rho \otimes \eta] U^\dagger = \sigma, \tag{106}$$

$$\Delta_B \left(\mathrm{Tr}_S \, U[\rho \otimes \eta] U^\dagger \right) = \eta. \tag{107}$$

Note that the result corresponding to equations (103) and (104) is a special case of the above theorem. Naturally, one may wonder whether the dephasing map given in equation (107) is at all necessary to implement all state transitions such that the von Neumann entropy does not decrease. In support of this, Boes *et al* (2019) provided the following *catalytic entropy conjecture*.

Conjecture 1 (see conjecture 1 in Boes *et al* (2019)). *For any two states ρ and σ of the same finite dimension and different spectra, the following statements are equivalent:*

(i) $S(\sigma) > S(\rho)$ *and* $\mathrm{rank}(\sigma) \geqslant \mathrm{rank}(\rho)$,

(ii) There exists a finite dimensional state η and a unitary U such that,

$$\mathrm{Tr}_A \, U[\rho \otimes \eta] U^\dagger = \sigma, \tag{108}$$

$$\mathrm{Tr}_S \, U[\rho \otimes \eta] U^\dagger = \eta. \tag{109}$$

This conjecture has been proven very recently by Wilming (2022). In fact, Wilming (2022) showed the following: let ρ and ρ' be two states of dimension d in $\mathcal{H}_d$ that are not unitarily equivalent. Then there exists a natural number n and a state ρ_n' on $\mathcal{H}_d^{\otimes n}$ such that

$$\rho^{\otimes n} \succ \rho_n' \text{ and } \mathrm{Tr}_{\{1,\ldots,n\}\setminus i} \, \rho_n{}' = \rho' \; \forall i \in \{1,\ldots,n\} \tag{110}$$

holds true if and only if $S(\rho) < S(\rho')$ and $\mathrm{rank}(\rho) \leqslant \mathrm{rank}(\rho')$. The validity of the catalytic entropy conjecture is a consequence of this result (Wilming 2022).

Recently, in Lie and Jeong (2021a, 2021b), randomness to implement quantum channels has been explored. In particular, Lie and Jeong (2021b) used an ancillary system as a source of randomness, such that when this ancillary system is used to implement a quantum channel, no information is leaked into it.

They named this a *randomness-utilising process*, as no information is leaked into the ancillary system while implementing a quantum channel. As a consequence, the ancillary system after the implementation of the process should not depend on the input state of the channel. A randomness-utilising channel on $\mathcal{B}(\mathcal{H}_S)$ can be expressed as (Lie and Jeong 2021b)

$$\Lambda(\rho) = \mathrm{Tr}_A\left[U(\rho\otimes\sigma)U^\dagger\right], \tag{111}$$

with a randomness source σ on $\mathcal{H}_A$, some unitary U on $\mathcal{H}_S\otimes\mathcal{H}_A$, and the state $\mathrm{Tr}_S[U(\rho\otimes\sigma)U^\dagger]$ is independent of the input state. Lie and Jeong (2021b) called this state of the ancillary system after the implementation of the randomness-utilising process *residue randomness*. If the residue randomness has the same spectrum as σ, the process is called catalytic. Hence, the randomness source can be reused to implement the same process on another input state. Equipped with this Lie and Jeong (2021b) showed that any dimension-preserving randomness-utilising process is catalytic. By 'dimension-preserving' the authors of Lie and Jeong (2021b) meant that the input and output systems are of the same dimension. Naturally one may wonder whether all dimension-preserving channels can be implemented by randomness-utilising process. It turns out that only unital quantum channels can be implemented using randomness-utilising process (Lie and Jeong 2021b). A unital quantum channel Λ_U keeps the identity operator invariant, i.e. $\Lambda_U(\mathbb{1}) = \mathbb{1}$. However, whether all unital quantum channels are catalytic, or, in other words, all unital quantum channels can be implemented using randomness-utilising processes, remained open in Lie and Jeong (2021b). A negative answer to this problem is given in Lie and Jeong (2021a) by showing that not all unital quantum channels are catalytic. Furthermore, Lie and Jeong (2021a) defined catalytic entropy to quantify the maximum amount of randomness that can be extracted catalytically from a randomness source σ. The catalytic entropy of a randomness source σ with eigenspace decomposition $\sigma = \sum_i \lambda_i P_i$ is given in Lie and Jeong (2021a) by

$$S_C(\sigma) := -\sum_i \lambda_i r_i \log_2 \frac{\lambda_i}{r_i}. \tag{112}$$

Here, P_i are the orthogonal projectors associated with the eigenvalue λ_i and satisfy $P_iP_j = 0$ if $\lambda_i \neq \lambda_j$, and $r_i = \mathrm{Tr}P_i$ represents degeneracy of λ_i. $S_C(\sigma)$ is the maximum amount of entropy or randomness that can be produced from σ (Lie and Jeong 2021a). In another recent work, the authors extended the role of catalytic quantum randomness for dynamical maps (Lie and Jeong 2023).

Henao and Uzdin (2023) explored the role of catalysts in reducing the impact of information erasure on the environment. In any information erasing process, an initial state ρ_s of a system s with von Neumann entropy $S(\rho_s)$ is taken to another state ρ_s' that has less entropy than the initial state ρ_s, i.e. $\Delta S_s = S(\rho_s) - S(\rho_s') > 0$, and the change in entropy $-\Delta S_s$ is dumped as heat in the environment. For a finite-dimensional environment, the authors in Henao and Uzdin (2023) showed that the correlation between the system and the environment can be maneuvered catalytically to reduce the entropy or heat waste in the environment. To set the scene, we introduce an environment e in the state ρ_e that is attached with the state ρ_s to reduce the entropy of the system s. By applying a global unitary transformation U_{se} on the system and the environment, the uncorrelated state $\rho_s\otimes\rho_e$ is transformed to the final state $\sigma_{se} = U(\rho_s\otimes\rho_e)U^\dagger$, such that the entropy of the system decreases, i.e. $S(\sigma_s) - S(\rho_s) < 0$, where $\sigma_s = \mathrm{Tr}_e\,\sigma_{se}$. As a result, the heat dumped in the environment is $Q_e = \mathrm{Tr}[H_e(\sigma_e - \rho_e)]$, where H_e represents the Hamiltonian associated with the environment and $\sigma_e = \mathrm{Tr}_s\,\sigma_{se}$. Now with the help of a catalyst c in the state ρ_c, Henao and Uzdin (2023) aimed to reduce the heat dissipation in the environment to a value Q_e' such that $Q_e' < Q_e$. This leads to a two-step process where $\rho_s\otimes\rho_e$ is transformed to σ_{se} and then σ_{se} is transformed to ρ_{sec}' with the help of a suitable catalyst state ρ_c and a global unitary U_{sec} such that $Q_e' = \mathrm{Tr}[H_e(\rho_e' - \rho_e)] < Q_e$. Here, $\rho_{sec}' = U_{sec}(\sigma_{se}\otimes\rho_c)U_{sec}^\dagger$ and $\rho_e' = \mathrm{Tr}_{sc}\,\rho_{sec}'$. Apart from ensuring $\rho_s' = \mathrm{Tr}_{ec}\,\rho_{sec}' = \sigma_s$, we need to make sure that the transformation is catalytic, i.e. $\mathrm{Tr}_{se}\,\rho_{sec}' = \rho_c$. With this, Henao and Uzdin (2023) proved the following: there exists a unitary transformation U_{sec} and a catalyst state ρ_c such that $\rho_e' \succ \sigma_s$, if and only if σ_{se} is a correlated state of the system s and the environment e. Furthermore, they showed that if ρ_e' majorises σ_s, then $Q_e \geqslant Q_e'$ (Henao and Uzdin 2023). Note that as the unitary is free in this process, the catalyst is the only party responsible for the reduction in heat dissipation to the environment. Furthermore, they showed $S(\rho_e') \leqslant S(\sigma_e)$ which reflects that the entropy change of the environment can also be reduced. Therefore, in any information erasure process involving a correlated state σ_{se}, it is possible to reduce the heat and entropy dissipation to the environment by using a suitable catalyst state.

3.5. Other quantum resource theories

Symmetry arguments play a crucial part in understanding many important results in physics. For example, in many cases, when the evolution of a system under some dynamics gets complicated to analyse, one can look for symmetry arguments to derive some useful results. In quantum theory, Marvian and Spekkens (2013) explored the role that symmetry plays in shaping the dynamics of a quantum system. More precisely, they explored state transformations in the context of time evolution with a given symmetry. To analyse this feature systematically, they introduced the resource theory of asymmetry in Marvian and Spekkens (2013). Suppose a group G with elements g represents a set of symmetry transformations. The action of each group element g is described by a unitary transformation U_g. In this resource theory, the free states are those states that remain invariant under the action of all g, i.e. $\mathcal{U}_g(\rho) = \rho\ \forall g \in G$, where $\mathcal{U}_g(\rho) = U_g(\rho)U_g^\dagger$. In addition, a quantum operation Λ is free if it respects the symmetry, i.e.

$$\Lambda\left[U_g^{\mathrm{i}}\rho\left(U_g^{\mathrm{i}}\right)^\dagger\right] = U_g^{\mathrm{o}}\Lambda[\rho]\left(U_g^{\mathrm{o}}\right)^\dagger \quad \forall g \in G, \tag{113}$$

where U_g^{i} and U_g^{o} represent the input and output representation of G, respectively. This type of operations are named as G-covariant operations in the literature (Marvian and Spekkens 2013).

In such a resource theory, Marvian and Spekkens (2013) explored the role of catalysis for pure state transformations. If a state $|\psi\rangle$ cannot be transformed into $|\phi\rangle$ under any deterministic G-covariant operation, but the same transformation can be achieved deterministically with a G-covariant operation and a catalyst in a state $|\eta\rangle$ such that $|\phi\rangle\otimes|\eta\rangle\to|\psi\rangle\otimes|\eta\rangle$, then such a transformation from $|\psi\rangle$ to $|\phi\rangle$ has been called a nontrivial example of catalysis (Marvian and Spekkens 2013). With this Marvian and Spekkens (2013) showed that no nontrivial example of catalysis is possible using a finite-dimensional catalyst when the symmetries are associated with compact connected Lie groups. However, this does not work for finite groups and a nontrivial example of catalysis has been shown for finite groups (Marvian and Spekkens 2013).

Recently, Rethinasamy and Wilde (2020) studied the role of catalysis in relative majorisation for classical probability distributions. A pair (p',q') of probability distributions p' and q' is relatively majorised by another pair (p,q) of probability distributions p and q, i.e. $(p,q)\succ(p',q')$ if and only if there exists a classical channel or a doubly stochastic matrix that transforms p and q to p' and q', respectively (Dahl 1999, Renes 2016, Buscemi and Gour 2017). In Rethinasamy and Wilde (2020), the authors reformulated the above mentioned transformation protocol in the presence of a catalyst pair (r,s) of probability distributions r and s. Now the input pair (p,q) is used jointly with the catalyst pair (r,s) to achieve the desired output pair (p',q') by applying a classical channel Λ on the joint pair $(p\otimes r,q\otimes s)$. Although, they allowed correlation on $\Lambda(p\otimes r)=t'$, but restrict that the first and second marginals of t' should be p and r, respectively. Furthermore, for the second one, they impose no correlation, i.e. $\Lambda(q\otimes s)=q'\otimes s$. As the catalyst is returned exactly, it can be reused to transform another independent set of (p,q). With this, the authors in Rethinasamy and Wilde (2020) showed that relative entropy completely determines exact catalytic pair transformations. Suppose p,q,p',q' are probability distributions on the same alphabet with (p,q) and (p',q') having different relative spectra, and q,q' have only rational entries and full support. Then the following two statements are equivalent: (i) $D(p||q)>D(p'||q')$ and $D_0(p||q)\geqslant D_0(p'||q')$. (ii) For all $\delta>0$, there exists a probability distribution r, and a joint probability distribution t' with marginals p' and r such that $(p\otimes r,q\otimes\nu)\succ(t',q'\otimes\nu)$, and $D(t'||p'\otimes r)<\delta$ (Rethinasamy and Wilde 2020). Here $D(p||q)=\sum_x p(x)\ln(p(x)/q(x))$, $D_0(p||q)=-\ln\sum_{x:p(x)\neq0}q(x)$, and ν is a uniform distribution that is used in place of s. The relative spectrum of a pair (p,q) is defined as $\{p(x)/q(x)\}_x$ (Rethinasamy and Wilde 2020). Therefore, relative entropy completely characterises exact transformations under correlated catalysis.

Note that the correlation can be made arbitrarily small such that t' is almost the same as $p'\otimes r$. The constraints on different relative spectra, q,q' are rational, and $D_0(p||q)\geqslant D_0(p'||q')$ can be lifted by allowing an infinitesimally small error on the target probability distribution p' (Rethinasamy and Wilde 2020). Suppose p,q,p',q' are probability distributions on the same alphabet with q,q' have full support. Then the following two statements are equivalent: (i) $D(p||q)\geqslant D(p'||q')$. (ii) For all $\delta>0$ and $\varepsilon>0$, there exists a probability distribution r, and a joint probability distribution t'_ε with marginals p'_ε and r, such that $(p\otimes r,q\otimes\nu)\succ(t'_\varepsilon,q'\otimes\nu)$, $||p'_\varepsilon-p'||_1\leqslant2\varepsilon$, and $D(t'_\varepsilon||p'_\varepsilon\otimes r)<\delta$. Here, $||p-q||_1=\sum_i|p_i-q_i|$. Considering q and q' to be the uniform distributions, a special case arises. Suppose p,p' are probability distributions on the same alphabet with $p\neq\mathrm{Per}(p')$ for all permutation matrices Per. Then the following two statements are equivalent: (i) $H(p)\geqslant H(p')$. (ii) For all $\delta>0$ and $\varepsilon>0$, there exists a probability distribution r, and a joint probability distribution t'_ε with marginals p'_ε and r, such that $p\otimes r\succ t'_\varepsilon$, $||p'_\varepsilon-p'||\leqslant2\varepsilon$, and $D(t'_\varepsilon||p'_\varepsilon\otimes r)<\delta$ (Rethinasamy and Wilde 2020). Here H represents the Shannon entropy of a probability distribution. Therefore, a single entropic condition completely dictates the approximate catalytic relative majorisation of classical probabilities. Note that this result can be applied to other quantum resource theories, for example, purity, quasiclassical thermodynamics, and asymmetric distinguishability (Rethinasamy and Wilde 2020).

The role of catalysis has also been explored in the framework of Gaussian thermal resource theory (Yadin *et al* 2022). Let us consider an n-mode bosonic continuous variable quantum system on the Hilbert space $\mathcal{H}=\otimes_{k=1}^n\mathcal{H}_k$, with corresponding bosonic creation and annihilation field operators $\hat{a}_k$ and $\hat{a}_k^\dagger$, respectively. From the field operators, the quadrature operators (or position and momentum-like operators) are constructed as $\hat{x}_k=(\hat{a}_k+\hat{a}_k^\dagger)/\sqrt{2}$ and $\hat{p}_k=\mathrm{i}(\hat{a}_k^\dagger-\hat{a}_k)/\sqrt{2}$. For convenience, all the quadrature operators are grouped together to make a column vector $\hat{\mathbf{r}}=(\hat{x}_1,\hat{p}_1,\ldots,\hat{x}_n,\hat{p}_n)^{\mathrm{T}}$. For a given quantum state ρ, the corresponding first and second moments are defined as (Weedbrook *et al* 2012)

$$\mathbf{r}=(\langle\hat{x}_1\rangle_\rho,\langle\hat{p}_1\rangle_\rho,\ldots,\langle\hat{x}_n\rangle_\rho,\langle\hat{p}_n\rangle_\rho)^{\mathrm{T}}\ \text{and,} \tag{114}$$

$$\sigma_{ij}=\frac{\langle\{\hat{\mathbf{r}}_i,\hat{\mathbf{r}}_j\}\rangle_\rho}{2}-\langle\hat{\mathbf{r}}_i\rangle_\rho\langle\hat{\mathbf{r}}_j\rangle_\rho \tag{115}$$

respectively, where the second moments σ_{ij} are the elements of the covariance matrix σ. Gaussian states are completely characterised by its first moments and covariance matrix (Weedbrook *et al* 2012).

After defining Gaussian states, we introduce Gaussian operations which preserve Gaussianity, or, in other words, take Gaussian states to Gaussian states. Generally, a Gaussian operation can be realised by allowing an interaction between an environment in a Gaussian state and a given system through a global Gaussian unitary and then performing a projective measurement on the environment (Weedbrook *et al* 2012). In the case of Gaussian thermal operations, the global unitary and the state of the environment should be energy preserving and in a thermal state, respectively. As Gaussian states are completely characterised by $\mathbf{r}$ and σ, it is enough to study the transformations of these two under Gaussian thermal operations. In Yadin *et al* (2022), the authors considered Gaussian states with $\mathbf{r}=0$ and studied the transformation of σ under strong and weak catalytic Gaussian thermal operations. Note

that the covariance matrix can be characterised by two more matrices M and A with elements $M_{ij} = \langle a_j^\dagger a_i \rangle - \delta_{ij}\nu$ and $A_{ij} = \langle a_j a_i \rangle$, respectively (Simon *et al* 1994) and this has been used to study state transformations under Gaussian thermal operations (Serafini *et al* 2020, Yadin *et al* 2022). Here $\nu = (e^{\beta\omega}+1)/(e^{\beta\omega}-1)$, ω is the eigenfrequency of the modes and $\beta = 1/kT$ is the inverse temperature. A strong catalytic Gaussian transformation of ρ to ρ' is possible when there exists a catalyst state σ such that $\rho\otimes\sigma \to \rho'\otimes\sigma$ is achieved under Gaussian thermal operation. On the other hand, a weak catalytic Gaussian transformation of ρ to ρ' is possible when there exists a catalyst state σ and a joint final state ν^{SC} such that $\rho\otimes\sigma \to \nu^{SC}$ with $\mathrm{Tr}_S\,\nu = \sigma$ and $\mathrm{Tr}_C\,\nu = \rho'$ is accomplished under Gaussian thermal operation. Equipped with these tools, the authors in Yadin *et al* (2022) studied transformations of single-mode bosonic Gaussian states. The transition from σ to σ' under strong catalytic Gaussian thermal operation is possible if and only if there exists a $p \in [0,1]$ such that the conditions

$$\mu' = p\mu \text{ and } \alpha' = p\alpha \tag{116}$$

hold (Yadin *et al* 2022). Here σ and σ' are the covariance matrices of a single-mode bosonic Gaussian state. For an n-mode bosonic state, μ and α correspond to the eigenvalues and singular values of the matrix M and A, respectively. Note that the conditions in equation (116) are exactly the same as the non-catalytic transition presented in Serafini *et al* (2020). Hence, strong catalysis does not provide any advantage in this scenario. Now, if we allow correlations between the system and the catalyst, then in the weak catalytic scenario, the above transition is possible if and only if there exists $p, q \in [0,1]$ with $p \geqslant q$ such that the conditions

$$\mu' = p\mu \text{ and } \alpha' = q\alpha \tag{117}$$

are satisfied (Yadin *et al* 2022). Furthermore, in Yadin *et al* (2022) multi-mode state transformations have been discussed as well.

The role of quantum catalysts in the distillation of magic states has also been investigated recently (Campbell 2011). Magic states have been considered to be one of the key ingredients for achieving fault-tolerant quantum computation (Knill 2004, Bravyi and Kitaev 2005, Campbell *et al* 2017). Generally, prepared states are noisy, and one needs to distill pure magic states from those to achieve fault-tolerant quantum computation. In magic state distillation, a smaller number of more useful pure magic states have been distilled from multiple noisy quantum states (Knill 2004, Bravyi and Kitaev 2005, Veitch *et al* 2014, Campbell *et al* 2017). Quantum catalysis has been shown to be useful in magic state distillation as well (Campbell 2011). Consider a state $|\psi\rangle \propto |H_{0,0,0}\rangle + |H_{1,1,1}\rangle$, where $|H_{0,0,0}\rangle = |H_0\rangle\otimes|H_0\rangle\otimes|H_0\rangle$, $|H_{1,1,1}\rangle = |H_1\rangle\otimes|H_1\rangle\otimes|H_1\rangle$. $|H_0\rangle$ and $|H_1\rangle$ are qubit magic states and also eigenvectors of the Hadamard matrix. Note that $|\psi\rangle \to |H_0\rangle$ cannot be achieved deterministically under Clifford unitary operations (Campbell 2011). However, if we have an additional state $|H_0\rangle$ in possession, the following transformation $|\psi\rangle\otimes|H_0\rangle \to |H_0\rangle\otimes|H_0\rangle$ can be achieved deterministically with Clifford unitary operations (Campbell 2011).

Another important aspect in the field of quantum computation is quantum error correction. Recently, the role of catalysis in it has been discussed by Brun *et al* (2014). Consider the following time evolution channel $\mathcal{N} : \mathcal{L}^{\otimes n} \to \mathcal{L}^{\otimes n}$ between Alice and Bob, which takes quantum states on Alice's system as inputs and provides quantum states on Bob's system as outputs. Here $\mathcal{L}$ represents the space of linear operators acting on the qubit Hilbert space $\mathcal{H}$. Note that here Alice and Bob are separated in time. Suppose Alice and Bob also have access to a noiseless channel $\mathrm{id}^{\otimes c}$, and Alice has an initial quantum state of k qubits. Equipped with this a catalytic quantum error correcting code can be defined as the following: (i) Alice performs an encoding operation $\mathcal{E} : \mathcal{L}^{\otimes k}\otimes\mathcal{L}^{\otimes c} \to \mathcal{L}^{\otimes n}$, (ii) After the time evolution Bob performs a decoding operation $\mathcal{D} : \mathcal{L}^{\otimes n}\otimes\mathcal{L}^{\otimes c} \to \mathcal{L}^{\otimes k}$ such that

$$\mathcal{D}\circ\left(\mathcal{N}\otimes\mathrm{id}^{\otimes c}\right)\circ\mathcal{E} = \mathrm{id}^{\otimes k} = \mathrm{id}^{\otimes k-c}\otimes\mathrm{id}^{\otimes c}. \tag{118}$$

The net rate of this error correcting code is $(k-c)/n$ and the noiseless channel $\mathrm{id}^{\otimes c}$ served as a catalyst in the process. If we additionally have access to m copies of the channel $\mathcal{N}$, then the following result has been proved (Brun *et al* 2014): If equation (118) holds, then for any arbitrary integer $m \geqslant 0$ there exists an encoding operation $\mathcal{E}_m$ and a decoding operation $\mathcal{D}_m$ for the channel $\mathcal{N}^{\otimes m}$ such that

$$\mathcal{D}_m\circ\left(\mathcal{N}^{\otimes m}\otimes\mathrm{id}^{\otimes c}\right)\circ\mathcal{E}_m = \mathrm{id}^{\otimes m(k-c)}\otimes\mathrm{id}^{\otimes c}. \tag{119}$$

The above result shows that the size of the catalyst can be reduced from mc to c.

In a recent study (Karvonen 2021), the role of catalysis has been explored in the resource theory of contextuality (Kochen and Specker 1967, Budroni *et al* 2022) and non-locality (Bell 1966, Brunner *et al* 2014). Karvonen (2021) considered the following kind of catalytic transformation: $p\otimes c \rightsquigarrow q\otimes c$, where p, q, and c represent some correlations, and q can be simulated ($\rightsquigarrow$) from p with the help of a catalyst c that remains intact in the process. Based on this, Karvonen (2021) proved that catalysis does not help in the resource theory of contextuality and non-locality, or, in other words, there exists no correlation that can be used to catalyse an otherwise impossible transformation. More precisely, if $p\otimes c \rightsquigarrow q\otimes c$ is possible in the resource theory of contextuality and non-locality, then $p \rightsquigarrow q$ is also possible (Karvonen 2021).

4. Approximate catalysis

The definition of exact catalysis, initially introduced for the resource theory of entanglement (Jonathan and Plenio 1999), does not allow for any errors in the transformation. In recent years, this notion has been extended, allowing for an error in the initial or final state (Brandão *et al* 2015, Lostaglio *et al*

2015b) while at the same time requiring that the catalyst be completely decoupled from the primary system at the end of the process. In parallel, there has been a growing interest in correlated catalysis, allowing the catalyst to build up correlations with the system, which may also be preserved at the end of the procedure (AAberg 2014, Wilming *et al* 2017, Müller 2018, Boes *et al* 2019, Shiraishi and Sagawa 2021, 2022, Wilming 2021, Kondra *et al* 2021a).

Recently, the framework of *approximate catalysis* has been proposed in Kondra *et al* (2021a), Datta *et al* (2022). While this framework allows for correlations between the main system and the catalyst, it additionally requires that these correlations can be made arbitrarily small. In particular, an approximate catalytic transformation between two states ρ^S and σ^S of a system S is possible if and only if for any $\varepsilon > 0$ there exists a catalyst state τ^C and a free operation Λ_f acting on the system and the catalyst such that (Kondra *et al* 2021a, Datta *et al* 2022)

$$\left\|\Lambda_f\left(\rho^S \otimes \tau^C\right) - \sigma^S \otimes \tau^C\right\|_1 \leqslant \varepsilon, \tag{120}$$

$$\mathrm{Tr}_S\left[\Lambda_f\left(\rho^S \otimes \tau^C\right)\right] = \tau^C. \tag{121}$$

In contrast to the exact catalysis discussed in section 3, the above definition allows for correlations between the system and the catalyst in the final state $\mu^{SC} = \Lambda_f(\rho^S \otimes \tau^C)$. However, equation (120) ensures that these correlations can be made arbitrarily small by choosing an appropriate state of the catalyst τ^C and a free operation Λ_f. The condition in equation (121) guarantees that the state of the catalyst does not change in the procedure. This ensures that the catalyst can be reused for the same transformation in the future. Moreover, this requirement makes it impossible to 'embezzle' quantum resources by using small changes of the catalyst state, see also section 7.

An alternative framework for approximate catalysis has been proposed in Müller (2018), Rethinasamy and Wilde (2020), Shiraishi and Sagawa (2022). While equation (120) represents that the correlations between the system and the catalyst can be made arbitrarily small as quantified by the trace norm, it is possible to formulate a similar requirement in terms of mutual information $I^{A:B}(\rho^{AB}) = S(\rho^A) + S(\rho^B) - S(\rho^{AB})$. Then, an approximate catalytic transformation from ρ^S into σ^S is possible if and only if for any $\varepsilon > 0$ and any $\delta > 0$ there exists a free operation Λ_f and a catalyst state τ^C such that (Müller 2018, Rethinasamy and Wilde 2020, Shiraishi and Sagawa 2022)

$$\left\|\mathrm{Tr}_C\left[\Lambda_f\left(\rho^S \otimes \tau^C\right)\right] - \sigma^S\right\|_1 \leqslant \varepsilon, \tag{122}$$

$$\mathrm{Tr}_S\left[\Lambda_f\left(\rho^S \otimes \tau^C\right)\right] = \tau^C, \tag{123}$$

$$I^{S:C}\left[\Lambda_f\left(\rho^S \otimes \tau^C\right)\right] \leqslant \delta. \tag{124}$$

Recently, Rubboli and Tomamichel (2022) demonstrated equivalence between this definition and the definition in equations (120) and (121).

Note that the term 'approximate catalysis' has been previously used by Gour *et al* (2015) in the resource theory of purity, referring to transformations which fulfill equation (120) for any $\varepsilon > 0$. Since the condition in equation (121) is not imposed, this type of transformations leads to embezzling phenomena (Gour *et al* 2015), see also section 7. Throughout this article, whenever we refer to approximate catalysis we mean transformations which obey equations (120) and (121).

In the following, we will review recent results on approximate catalysis for concrete resource theories, including quantum thermodynamics, entanglement, and quantum coherence.

4.1. Quantum thermodynamics

Approximate catalysis has been first investigated in quantum thermodynamics by Müller (2018). For the case that the initial and the final states are both block diagonal in the energy eigenbasis, it has been shown that state transformations are completely characterised by the Helmholtz free energy. In particular, a block diagonal state ρ can be converted into another block diagonal state σ via thermal operations with approximate catalysis if and only if (Müller 2018)

$$F(\rho) \geqslant F(\sigma). \tag{125}$$

Shiraishi and Sagawa (2021, 2022) have investigated a closely related setting, replacing thermal operations with the more general class of Gibbs-preserving operations. In this case, equation (125) completely characterises transformations between all quantum states, leading to the following theorem.

Theorem 5 (see theorem 1 in Shiraishi and Sagawa (2021)). *A state ρ can be converted into another state σ via Gibbs-preserving operations with approximate catalysis if and only if $F(\rho) \geqslant F(\sigma)$.*

It is possible to express the Helmholtz free energy in terms of the quantum relative entropy as follows:

$$F(\rho) = kT[S(\rho||\gamma) - \log Z], \tag{126}$$

where Z is the partition function and γ is the Gibbs state. Using equation (126), we see that equation (125) is equivalent to

$$S(\rho||\gamma) \geqslant S(\sigma||\gamma). \tag{127}$$

The techniques developed by Shiraishi and Sagawa (2021, 2022) to prove the above theorem turned out to be applicable to other resource theories, and we will provide a sketch of the proof in the following. The major insight behind the proof is a connection between asymptotic transformations and approximate catalysis, which has been investigated earlier in Duan *et al* (2005a). Let us assume that ρ can be converted into σ via Gibbs-preserving operations in the asymptotic limit, i.e. for any $\delta > 0$ there exists an integer n and a Gibbs-preserving operation Λ acting on n copies of ρ such that

$$\left\|\Lambda\left(\rho^{\otimes n}\right) - \sigma^{\otimes n}\right\|_1 \leqslant \delta. \tag{128}$$

In the following S_i will denote copies of the system S, and we further define $\Gamma = \Lambda(\rho^{\otimes n})$, which is a quantum state on $S_1 \otimes S_2 \otimes \cdots \otimes S_n$.

Shiraishi and Sagawa (2021, 2022) showed that equation (128) implies an approximate catalytic transformation from ρ into σ. The catalyst achieving this transformation is given as

$$\tau^C = \frac{1}{n}\sum_{k=1}^{n} \rho^{\otimes k-1} \otimes \Gamma_{n-k} \otimes |k\rangle\langle k|. \tag{129}$$

Here, $\Gamma_0 = 1$, and for $i > 0$ the state Γ_i is the reduced state of Γ on $S_1 \otimes S_2 \otimes \cdots \otimes S_i$. The Hilbert space of the catalyst is in $S^{\otimes n-1} \otimes K$, where K represents an ancillary system of dimension n with a fully degenerated Hamiltonian and basis states $\{|k\rangle\}_{k=1}^n$.

The Gibbs-preserving operation achieving the desired conversion consists of the following three steps applied onto the total state $\rho^S \otimes \tau^C$ (Shiraishi and Sagawa 2021, 2022):

1. A controlled Gibbs-preserving operation is performed, with the ancillary system K as the control, and all other particles as the target. Conditioned on the ancilla being in the state $|n\rangle$, the Gibbs-preserving operation Λ is applied onto $S_1 \otimes S_2 \otimes \cdots \otimes S_n$ such that equation (128) is fulfilled.
2. A unitary is applied onto the ancillary system that converts $|n\rangle \to |1\rangle$ and $|i\rangle \to |i+1\rangle$.
3. The systems are shifted as $S_i \to S_{i+1}$ and $S_n \to S_1$.

The initial state of the system and the catalyst has the form

$$\rho^S \otimes \tau^C = \frac{1}{n}\sum_{k=1}^{n} \rho^{\otimes k} \otimes \Gamma_{n-k} \otimes |k\rangle\langle k|. \tag{130}$$

After applying 1. step of the procedure, we obtain the following state:

$$\mu_1 = \frac{1}{n}\sum_{k=1}^{n-1} \rho^{\otimes k} \otimes \Gamma_{n-k} \otimes |k\rangle\langle k| + \frac{1}{n}\Gamma \otimes |n\rangle\langle n|. \tag{131}$$

In 2. step, μ_1 is transformed into the state

$$\mu_2 = \frac{1}{n}\sum_{k=1}^{n} \rho^{\otimes k-1} \otimes \Gamma_{n+1-k} \otimes |k\rangle\langle k|. \tag{132}$$

Tracing out S_n from μ_2 gives the initial state of the catalyst τ, see equation (129). Therefore, the shift in 3. step transforms μ_2 to the final state μ^{SC} having the property $\mathrm{Tr}_S[\mu^{SC}] = \tau^C$, which implies that the condition (121) for approximate catalysis is fulfilled.

Using the protocol presented above, Shiraishi and Sagawa (2021) further showed that the final system state μ^S can be made arbitrarily close to the desired state σ^S, and also the mutual information between the system and the catalyst in the total final state μ^{SC} can be made arbitrarily small whenever ρ^S can be converted into σ^S via Gibbs-preserving operations in the asymptotic limit. Thus, the procedure fulfills the criteria for approximate catalysis in equations (122)–(124). Recalling that these criteria are equivalent to equations (120) and (121), we will now prove that the transformation presented above fulfills equation (120). For this, we will show that for any $\varepsilon > 0$ the protocol described above can be performed such that

$$\left\|\mu^{SC} - \sigma^S \otimes \tau^C\right\|_1 \leqslant \varepsilon. \tag{133}$$

In this part of the proof we will use the techniques presented for entanglement catalysis in Kondra *et al* (2021a), Lipka-Bartosik and Skrzypczyk (2021b), adjusting them to the problem under study. Note that μ^{SC} is equivalent to the state μ_2 in equation (132) up to a cyclic shift. Thus, our figure of merit $||\mu^{SC} - \sigma^S \otimes \tau^C||_1$ is equal to $||\mu_2 - \nu||_1$, where the state ν has the form

$$\nu = \frac{1}{n}\sum_{k=1}^{n} \rho^{\otimes k-1} \otimes \Gamma_{n-k} \otimes \sigma \otimes |k\rangle\langle k|. \tag{134}$$

Using the triangle inequality, equation (128), and the monotonicity of trace distance under partial trace we obtain

$$\begin{aligned}\left\|\mu^{SC} - \sigma^S \otimes \tau^C\right\|_1 &= \|\mu_2 - \nu\|_1 \\ &= \frac{1}{n}\sum_{k=1}^{n} \|\Gamma_{n+1-k} - \Gamma_{n-k} \otimes \sigma\|_1 \\ &\leqslant \frac{1}{n}\sum_{k=1}^{n} \left\|\Gamma_{n+1-k} - \sigma^{\otimes(n+1-k)}\right\|_1 \\ &\quad + \frac{1}{n}\sum_{k=1}^{n} \left\|\Gamma_{n-k} \otimes \sigma - \sigma^{\otimes(n+1-k)}\right\|_1 \leqslant 2\delta.\end{aligned} \tag{135}$$

Setting $\delta = \varepsilon/2$ we see that equation (133) is fulfilled. This proves that the Gibbs-preserving operation presented above together with the catalyst state in equation (129) achieves the desired transformation. This holds true whenever ρ can be converted into σ via Gibbs-preserving operations in the asymptotic limit.

Shiraishi and Sagawa (2021, 2022)further proved that a conversion from ρ into σ via asymptotic Gibbs-preserving operations is possible whenever $F(\rho) > F(\sigma)$ is fulfilled. For proving this statement, Shiraishi and Sagawa (2021, 2022) make use of the quantum Stein's lemma (Hiai and Petz 1991, Ogawa and Nagaoka 2000) together with the properties of quantum hypothesis testing relative entropy and relative Rényi entropies (Datta 2009, Wang and Renner 2012, Buscemi *et al* 2019). Note that this result has previously been proved in a more general setting, specifically for the resource theory of asymmetric distinguishability (Wang and Wilde 2019), and Gibbs-preserving operations can be considered as a special scenario of this resource theory. With the arguments presented above, it follows that for any states ρ and σ fulfilling $F(\rho) > F(\sigma)$, a conversion from ρ into σ is possible via Gibbs-preserving operations with approximate catalysis (Shiraishi and Sagawa 2021, 2022). By a simple continuity argument this result can be extended to the case $F(\rho) = F(\sigma)$ (Shiraishi and Sagawa 2021, 2022).

It is left to show that a conversion via Gibbs-preserving operations with approximate catalysis is not possible if

equation (125) is violated. For this, consider the following inequalities:

$$\begin{aligned} S\left(\rho^S||\gamma^S\right)+S\left(\tau^C||\gamma^C\right) &= S\left(\rho^S\otimes\tau^C||\gamma^S\otimes\gamma^C\right) \\ &\geqslant S\left(\Lambda\left[\rho^S\otimes\tau^C\right]||\Lambda\left[\gamma^S\otimes\gamma^C\right]\right) \\ &= S\left(\Lambda\left[\rho^S\otimes\tau^C\right]||\gamma^S\otimes\gamma^C\right) \\ &\geqslant S\left(\mu^S||\gamma^S\right)+S\left(\tau^C||\gamma^C\right). \end{aligned} \quad (136)$$

The above result follows from the properties of the quantum relative entropy (Wehrl 1978, Capel *et al* 2018): the second line follows from the data processing inequality, the third line from the fact that Λ is a Gibbs-preserving operation, and the last line follows from superadditivity. Moreover, in the last line of equation (136) we have used the state $\mu^{SC}=\Lambda[\rho^S\otimes\tau^C]$, noting that $\mu^C=\tau^C$. Noting that the Gibbs states γ^S and γ^C have full rank, the quantum relative entropies appearing in equation (136) are all finite and continuous (Audenaert and Eisert 2005). We arrive at $S(\rho^S||\gamma^S)\geqslant S(\mu^S||\gamma^S)$ for any state μ^S which is achievable from ρ^S via Gibbs-preserving operations with approximate catalysis. By continuity, it follows that the same inequality also holds for any state σ^S which can be achieved with an arbitrary small error as in equation (120). Recalling that equation (125) is equivalent to equation (127), the proof of theorem 5 is complete.

The results of Shiraishi and Sagawa (2021, 2022) have also been used by Wilming (2021) to provide a proof of the catalytic entropy conjecture (Boes *et al* 2019) in an approximate form, see also section 3.4. In particular, for any two system states ρ and σ with $S(\rho)\leqslant S(\sigma)$ and any $\varepsilon>0$ there exists a catalyst state τ and a unitary U acting on the system and the catalyst such that (Wilming 2021)

$$\mathrm{Tr}_S\left[U\rho\otimes\tau U^\dagger\right]=\tau, \quad (137)$$

$$\left\|\mathrm{Tr}_C\left[U\rho\otimes\tau U^\dagger\right]-\sigma\right\|_1\leqslant\varepsilon. \quad (138)$$

Applications of catalysis in quantum thermodynamics have also been reported by Lipka-Bartosik *et al* (2023), who proposed a notion of temperature of a non-equilibrium state, inspired by the 'zeroth law of thermodynamics'. They showed that for every quantum state there exist two effective temperatures—hot and cold temperatures—quantifying the ability of a quantum system to cool down or heat up via thermal operations. Precisely, Lipka-Bartosik *et al* (2023) showed that, if the temperature of the bath is less than the cold temperature, then the system can never gain energy via any thermal operation (and vice versa). Interestingly, Lipka-Bartosik *et al* (2023) showed that approximate catalysis leads to much colder or hotter effective temperatures.

4.2. Entanglement

Approximate catalysis has also been studied within the resource theory of entanglement (Kondra *et al* 2021a, Lipka-Bartosik and Skrzypczyk 2021b, Datta *et al* 2022). In this case, Kondra *et al* (2021a) provided a full characterisation of state transformations for all bipartite pure states.

Theorem 6 (see equation (4) in Kondra *et al* (2021a)). *Using LOCC and approximate catalysis, Alice and Bob can convert a state $|\psi\rangle^{AB}$ into $|\phi\rangle^{AB}$ if and only if*

$$S\left(\psi^A\right)\geqslant S\left(\phi^A\right). \quad (139)$$

Similar to the method of Shiraishi and Sagawa (2021, 2022) for quantum thermodynamics, the proof of theorem 6 presented by Kondra *et al* (2021a) uses an analogy between approximate catalysis and asymptotic state transformations via LOCC. For this, recall that the optimal rate for transforming $|\psi\rangle^{AB}$ into $|\phi\rangle^{AB}$ via LOCC in the asymptotic limit is given by $S(\psi^A)/S(\phi^A)$ (Bennett *et al* 1996a). If equation (139) is fulfilled, then asymptotic conversion from $|\psi\rangle^{AB}$ into $|\phi\rangle^{AB}$ is possible with rate at least one. It is then possible to construct a catalyst state and an LOCC protocol achieving the transformation with approximate catalysis, in analogy to the construction proposed by Shiraishi and Sagawa (2021, 2022) for quantum thermodynamics, see the previous section 4.1 for more details. We also note that the result in theorem 6 does not assume that the state of the catalyst is pure, and in fact a catalyst in a mixed state is sometimes required to achieve the desired transformation (Kondra *et al* 2021a, Datta *et al* 2022).

Kondra *et al* (2021a) further used the properties of squashed entanglement (Christandl and Winter 2004) to prove that an approximate catalytic transformation is not possible if equation (139) is violated. Squashed entanglement is defined as (Christandl and Winter 2004)

$$E_{\mathrm{sq}}\left(\rho^{AB}\right)=\inf\left\{\frac{1}{2}I(A;B|E):\rho^{ABE}\text{ extension of }\rho^{AB}\right\}, \quad (140)$$

with the quantum conditional mutual information $I(A;B|E)$. As has been shown by Christandl and Winter (2004), squashed entanglement does not increase under LOCC, and moreover, it fulfills

$$E_{\mathrm{sq}}\left(\rho^{AA'BB'}\right)\geqslant E_{\mathrm{sq}}\left(\rho^{AB}\right)+E_{\mathrm{sq}}\left(\rho^{A'B'}\right) \quad (141)$$

with equality if $\rho^{AA'BB'}=\rho^{AB}\otimes\rho^{A'B'}$. For pure states, squashed entanglement coincides with the von Neumann entropy (with logarithm to the base 2) of the reduced state (Christandl and Winter 2004). Using these properties, Kondra *et al* (2021a) proved the following:

Theorem 7 (see theorem 2 in Kondra *et al* (2021a)). *If a bipartite state ρ^{AB} can be transformed into another state ν^{AB} via LOCC with approximate catalysis, then*

$$E_{\mathrm{sq}}\left(\rho^{AB}\right)\geqslant E_{\mathrm{sq}}\left(\nu^{AB}\right). \quad (142)$$

We will present the proof of this theorem following the arguments of Kondra *et al* (2021a). If ρ^{AB} can be transformed into ν^{AB} with approximate catalysis, then for any $\varepsilon>0$ there

exists a catalyst state $\tau^{A'B'}$ and an LOCC protocol Λ such that the final state $\sigma^{AA'BB'} = \Lambda(\rho^{AB} \otimes \tau^{A'B'})$ fulfills

$$\left\| \mathrm{Tr}_{A'B'} \left[\sigma^{AA'BB'} \right] - \nu^{AB} \right\|_1 < \varepsilon, \tag{143}$$

$$\mathrm{Tr}_{AB} \left[\sigma^{AA'BB'} \right] = \tau^{A'B'}. \tag{144}$$

We further obtain

$$\begin{aligned} E_{\mathrm{sq}} \left(\sigma^{AA'BB'} \right) &\leqslant E_{\mathrm{sq}} \left(\rho^{AB} \otimes \tau^{A'B'} \right) \\ &= E_{\mathrm{sq}} \left(\rho^{AB} \right) + E_{\mathrm{sq}} \left(\tau^{A'B'} \right), \end{aligned} \tag{145}$$

where in the inequality we used the fact that squashed entanglement does not increase under LOCC, and the equality follows from the additivity of E_{sq} on tensor products. From equation (141), we further have

$$E_{\mathrm{sq}} \left(\sigma^{AA'BB'} \right) \geqslant E_{\mathrm{sq}} \left(\mathrm{Tr}_{A'B'} \left[\sigma^{AA'BB'} \right] \right) + E_{\mathrm{sq}} \left(\tau^{A'B'} \right). \tag{146}$$

Combining equations (145) and (146) we obtain

$$E_{\mathrm{sq}} \left(\rho^{AB} \right) \geqslant E_{\mathrm{sq}} \left(\mathrm{Tr}_{A'B'} \left[\sigma^{AA'BB'} \right] \right). \tag{147}$$

The continuity of squashed entanglement (Alicki and Fannes 2004) together with equation (143) implies $E_{\mathrm{sq}} \left(\rho^{AB} \right) \geqslant E_{\mathrm{sq}} \left(\nu^{AB} \right)$, which completes the proof of theorem 7.

Theorem 7 together with the faithfulness of squashed entanglement (Li and Winter 2014) implies that it is not possible to create entanglement from separable states by using LOCC and approximate catalysis.

The proof of theorem 6 presented in Kondra *et al* (2021a) uses the fact that for bipartite pure states $|\psi\rangle^{AB}$ and $|\phi\rangle^{AB}$ fulfilling $S(\psi^A) > S(\phi^A)$, asymptotic conversion from $|\psi\rangle^{AB}$ into $|\phi\rangle^{AB}$ is possible with unit rate (Bennett *et al* 1996a). States with such properties have also been studied by Bennett *et al* (2000), where this feature has been called *asymptotic reducibility*. A general relation between asymptotic reducibility and approximate catalysis has been established by Kondra *et al* (2021a) for multipartite systems and multipartite LOCC protocols.

Theorem 8 (see theorem 1 in Kondra *et al* (2021a)). *If ρ is asymptotically reducible to σ, then ρ can be transformed into σ via LOCC with approximate catalysis.*

A stronger form of asymptotic reducibility can be constructed where the parties can additionally use a sublinear amount of entanglement with the standard LOCC protocol to accomplish the same transformation, which is denoted by asymptotic $\mathrm{LOCC_q}$ reducibility (Bennett *et al* 2000). Similarly, catalytic asymptotic reducibility $\mathrm{LOCC_c}$ can be defined as follows: a state $|\psi\rangle$ is asymptotically $\mathrm{LOCC_c}$ reducible to a state $|\phi\rangle$ if there exists a state $|\eta\rangle$ and a n such that $(|\psi\rangle \otimes |\eta\rangle)^{\otimes n}$ can be converted arbitrarily close to $(|\phi\rangle \otimes |\eta\rangle)^{\otimes n}$. Bennett *et al* (2000) showed that $\mathrm{LOCC_c}$ is at least as powerful as $\mathrm{LOCC_q}$. However, the converse i.e. whether $\mathrm{LOCC_c}$ can be simulated by $\mathrm{LOCC_q}$, has been shown only for some special cases (Bennett *et al* 2000).

In the context of $\mathrm{LOCC_c}$ reducibility, a recent result on the catalysis of arbitrary entangled states compares single-copy catalysis with asymptotic catalysis (Datta *et al* 2022). A transformation $\rho \to \sigma$ on system S is possible via asymptotic catalysis if and only if for any $\varepsilon > 0$ and any $\delta > 0$ there exist integers n and m with $n > m$, a catalyst state τ^C and an LOCC protocol Λ such that[4]

$$\left\| \Lambda \left[\rho^{\otimes n} \otimes \tau^C \right] - \sigma^{\otimes m} \otimes \tau^C \right\|_1 \leqslant \varepsilon, \tag{148a}$$

$$\mathrm{Tr}_{S^{\otimes m}} \left[\Lambda \left(\rho^{\otimes n} \otimes \tau^C \right) \right] = \tau^C, \tag{148b}$$

$$\frac{m}{n} + \delta \geqslant 1. \tag{148c}$$

With this, Datta *et al* (2022) showed that state transformations under asymptotic catalysis do not provide any advantage over single-copy catalysis. To be precise, if a state transformation $\rho \to \sigma$ can be achieved via asymptotic catalysis with unit rate, then the same transformation can be realised with single-copy catalysis too.

Datta *et al* (2022) also considered asymptotic settings beyond the independent and identically distributed ('iid') case, where the quantum state to be transformed asymptotically is no longer assumed to be just the tensor product $|\psi\rangle^{\otimes n}$. In particular, Datta *et al* (2022) considered a pure-state distillation scenario, in which Alice and Bob share a more general total state $\otimes_i |\psi_i\rangle^{AB}$, where each $|\psi_i\rangle^{AB}$ are bipartite states that need not be identical. As we will see, approximate catalysis provides, in this setting, a significant advantage over non-catalytic LOCC transformations.

By definition, a singlet state $|\phi_2^+\rangle$ can be extracted from a sequence of states $\{|\psi_i\rangle^{AB}\}$ with fidelity f and probability p, if there exists an integer n and a probabilistic LOCC protocol Λ such that (Datta *et al* 2022)

$$\mathrm{Tr}\left(\Lambda\left[\otimes_{i=1}^n \psi_i\right]\right) = p, \quad \langle \phi_2^+ | \Lambda\left[\otimes_{i=1}^n \psi_i\right] | \phi_2^+ \rangle = pf, \tag{149}$$

where we set $\psi_i = |\psi_i\rangle\langle\psi_i|^{AB}$ for short. Compare also equations (34) and (35) in Datta *et al* (2022). Moreover, in the context of pure state entanglement as the relevant resource, we can say that the same sequence allows for catalytic extraction of m singlets if the conditions provided in equations (120) and (121) are satisfied, with $\sigma^S = |\phi_2^+\rangle\langle\phi_2^+|^{\otimes m}$ and Λ being an LOCC protocol. With this, Datta *et al* (2022) proved the following:

Theorem 9 (see theorem 3 in Datta *et al* (2022)). *Given fidelity $f > \frac{1}{2}$ and any $\varepsilon > 0$, there is a sequence of two-qubit states $\{|\psi_i\rangle^{AB}\}$ such that for any $n \geqslant 1$, the probability of converting the state $\otimes_{i=1}^n |\psi_i\rangle^{AB}$ into a singlet, via LOCC with fidelity f, is smaller than ε. At the same time, an unbounded number of singlets can be extracted from $\otimes_{i=1}^n |\psi_i\rangle^{AB}$ with the help of catalysis, as $n \to \infty$.*

[4] The LOCC protocol Λ acts on n copies of S and the catalyst C, and produces a state on m copies of S and the catalyst.

The proof is by explicit construction (Datta *et al* 2022). The sequence $|\psi_i\rangle^{AB}$ given in the proof is designed in such a way that although the state $\otimes_{i=1}^{n}|\psi_i\rangle^{AB}$ stays arbitrarily close to a product state with respect to systems A and B, its total local entropy grows indefinitely as $n\to\infty$, see Datta *et al* (2022) for more details. The above results demonstrate clearly the advantage of an approximate catalytic transformation over all non-catalytic LOCC protocols. Whereas for the two-qubit sequence specified in the proof of theorem 9, the amount of entanglement available via catalytic transformation is unbounded, and not a single maximally entangled state can be extracted with reasonable probability and fidelity without catalysis. It is worth mentioning that entanglement distillation from non-iid sequences of states has been investigated earlier in Bowen and Datta (2008), Buscemi and Datta (2010), Waeldchen *et al* (2016).

Eisert and Wilkens (2000) have investigated the role of catalysis for approximate transformations between quantum states via LOCC, requiring however that the catalyst is not correlated with the primary system at the end of the procedure. In particular, Eisert and Wilkens (2000) demonstrated that catalysis is useful for approximately converting a mixed entangled state into a pure state. For this, Eisert and Wilkens (2000) considered the approximate conversion fidelity

$$F_{\mathrm{LOCC}}(\rho\to\sigma)=\max_{\Lambda\in\mathrm{LOCC}}F(\Lambda[\rho],\sigma), \tag{150}$$

$$F_{\mathrm{CLOCC}}(\rho\to\sigma)=\max_{\Lambda\in\mathrm{CLOCC}}F(\Lambda[\rho],\sigma), \tag{151}$$

where CLOCC represents catalytic LOCC. Eisert and Wilkens (2000) proved that the conversion fidelity for CLOCC can exceed the same for LOCC when some initial mixed states ρ are being converted into a pure target state $|\zeta\rangle$:

$$F_{\mathrm{CLOCC}}\left(\rho\to|\zeta\rangle\langle\zeta|\right)>F_{\mathrm{LOCC}}\left(\rho\to|\zeta\rangle\langle\zeta|\right). \tag{152}$$

To prove the statement, Eisert and Wilkens (2000) used the state ρ given in equation (22) with $|\omega\rangle$ given in equation (27) and

$$|\psi\rangle=\varepsilon\left(\sqrt{0.4}|00\rangle+\sqrt{0.4}|11\rangle+\sqrt{0.1}|22\rangle+\sqrt{0.1}|33\rangle\right)+\sqrt{1-\varepsilon^2}|44\rangle. \tag{153}$$

Setting $|\zeta\rangle=|\phi\rangle$ given in equation (26), Eisert and Wilkens (2000) proved that the inequality given in equation (152) holds. The above equation holds for all $\varepsilon\in(\varepsilon',1)$ with some appropriate $\varepsilon'\in(0,1)$ and independent of the value of $\nu\in(0,1)$.

Early results on the role of entanglement catalysis in the presence of noise have also been presented by Vidal *et al* (2000). Suppose a catalyst in state $|\eta\rangle$ improves the performance of the approximate transformation $|\psi\rangle\to|\phi\rangle$, i.e.

$$D\left(\psi\otimes\eta\to\phi\otimes\eta\right)<D\left(\psi\to\phi\right), \tag{154}$$

where $D(\rho\to\sigma)=\min_{\Lambda}D(\Lambda(\rho),\sigma)$, Λ represents the set of LOCC operations, and $D(\rho,\sigma)=||\rho-\sigma||_1/2$ is the trace distance. The quantity

$$\delta=D\left(\psi\to\phi\right)-D\left(\psi\otimes\eta\to\phi\otimes\eta\right) \tag{155}$$

can thus be regarded as a quantifier of the catalytic advantage in this procedure. Vidal *et al* (2000) have investigated how much noise is allowed in the initial state so that the catalytic advantage remains, i.e. $\delta>0$. Suppose the initial state $|\psi\rangle\otimes|\eta\rangle$ is distorted to ρ such that $D(\rho,\psi\otimes\eta)=\varepsilon$. If $\varepsilon<\delta$, it has been shown that the catalytic advantage is preserved as $D(\rho\to\phi\otimes\eta)<D(\psi\to\phi)$ (Vidal *et al* 2000).

The role of catalysis in the approximate transformation of bipartite quantum channels has been discussed recently in Kim *et al* (2021). In fact, Kim *et al* (2021) explored the dynamic resource theory of entanglement using the superchannel theory and also discussed the one-shot catalytic dynamic entanglement cost of bipartite quantum channels. More precisely, the authors provided a lower and an upper bound for the one-shot catalytic dynamic entanglement cost of an arbitrary bipartite channel. Before presenting the bounds, in the following, we introduce the one-shot catalytic dynamic entanglement cost of a bipartite channel $\mathcal{N}_{AB}$ for a given $\delta>0$ and $\varepsilon\geqslant 0$ (Kim *et al* 2021):

$$E^{\varepsilon}_{\mathrm{C},\delta\text{-SEPPSC}}\left(\mathcal{N}_{AB}\right):=\min\Big\{\log_2 m^2:\Theta_{A'B'CD\to ABCD}\left(\mathcal{F}^{m}_{A'B'}\otimes\mathcal{F}^{n}_{CD}\right)=\mathcal{N}'_{AB}\otimes\mathcal{F}^{n}_{CD},\Theta_{A'B'CD\to ABCD}\in\delta\text{-SEPPSC},\ ||\mathcal{N}_{AB}-\mathcal{N}'_{AB}||_{\diamond}\leqslant 2\varepsilon\Big\}, \tag{156}$$

where m,n represent natural numbers and $\Theta_{A'B'CD\to ABCD}$ is a δ-separability-preserving superchannel (δ-SEPPSC) from $A'B'CD$ to $ABCD$. For a given $\delta>0$, a superchannel $\Theta_{AB\to A'B'}$ is δ-SEPPSC when $R(\Theta_{AB\to A'B'}[\mathcal{S}_{AB}])\leqslant\delta$ holds for all separable channels $\mathcal{S}_{AB}$, where R is the generalised robustness with respect to the separable channels (Kim *et al* 2021). Here $\mathcal{F}^{m}_{AB}$ denotes m-SWAP channel which can be represented by the application of m-SWAP gate $F^{m}_{AB}=\sum_{i,j=0}^{m-1}|ij\rangle\langle ji|_{A_0B_0\to A_1B_1}$. With this, we are ready to introduce the following bounds: For $\delta>0$, $\varepsilon\geqslant 0$, there exists a natural number $n>0$ with $n^2\geqslant(\delta+1)/\delta$ such that the one-shot catalytic dynamic entanglement cost of an arbitrary bipartite channel $\mathcal{N}_{AB}$ is bounded from below and above as (Kim *et al* 2021)

$$\begin{aligned}&R^{\varepsilon}_{\mathrm{L}}\left(\mathcal{N}_{AB}\otimes\mathcal{F}^{n}_{CD}\right)-\log_2 n^2-\log_2\left(1+\delta\right)\\&\quad\leqslant E^{\varepsilon}_{\mathrm{C},\delta\text{-SEPPSC}}\left(\mathcal{N}_{AB}\right)\\&\quad\leqslant R^{\varepsilon'}_{\mathrm{L}}\left(\mathcal{N}_{AB}\otimes\mathcal{F}^{n}_{CD}\right)-\log_2 n^2-\log_2\left(1-2\varepsilon'\right)+2,\end{aligned} \tag{157}$$

where $\varepsilon'=\varepsilon^2/(2|A_0|^2|B_0|^2)$, $R^{\varepsilon}_{\mathrm{L}}(\mathcal{N}_{AB})=\min_{\mathcal{N}'_{AB}}R_{\mathrm{L}}(\mathcal{N}'_{AB})$, with $||\mathcal{N}_{AB}-\mathcal{N}'_{AB}||_{\diamond}\leqslant 2\varepsilon$, and $R_{\mathrm{L}}(\mathcal{N}_{AB})=\log_2(1+R(\mathcal{N}_{AB}))$.

4.3. Quantum coherence

As discussed previously in section 3.2, correlated catalysis does not help in coherence broadcasting under covariant operations (Lostaglio and Müller 2019, Marvian and Spekkens 2019). Due to these limitations on coherence transformation under covariant operations, Takagi and Shiraishi (2022)

investigated a different kind of catalyst known as marginal catalysts (Lostaglio *et al* 2015b, Wilming *et al* 2017) that could help in enhancing covariant transformations. Marginally catalytic covariant transformations are defined as follows: Suppose ρ is transformed into ρ' via a marginal catalytic conversion. Then there exists a covariant operation Λ, a constant N and a finite set of states $\otimes_{j=1}^{N}\tau_{C_j}$ such that (Takagi and Shiraishi 2022)

$$\Lambda\left(\rho\otimes_{j=1}^{N}\tau_{C_j}\right)=\rho'\otimes\tau_{C_1\cdots C_N}\text{ and} \tag{158}$$

$$\mathrm{Tr}_{\bar{C}_j}\tau_{C_1\cdots C_N}=\tau_{C_j}\text{ for all }j, \tag{159}$$

where $\bar{C}_j$ represents all the systems except C_j. Although in the final state there are correlations among different catalyst systems, the marginal state of each catalyst system remains unchanged. Therefore, the change in the global state of the catalyst produces an embezzling-like phenomenon. For more details on embezzling see section 7. With this, Takagi and Shiraishi (2022) showed that marginal catalysts can provide enormous power in state transformations under covariant transformations. Indeed for any state ρ, ρ' and $\varepsilon>0$, ρ can be transformed into a state ρ'_ε that is arbitrarily close to ρ' in trace distance, i.e. $||\rho'-\rho'_\varepsilon||_1\leqslant 2\varepsilon$ by a marginal-catalytic covariant transformation. Hence, marginal catalysts exceptionally simplify coherence transformations in this theory. Similar results for qubit states have been presented earlier by Ding *et al* (2021).

Fang *et al* (2018) studied one-shot probabilistic coherence distillation with catalytic assistance. The idea is to transform a coherent state ρ close to a maximally coherent state $|\Psi_N\rangle=\sum_{i=0}^{N-1}|i\rangle/\sqrt{N}$ of dimension N with the help of a catalyst η. In such a scenario, Fang *et al* (2018) showed that the maximum success probability of coherence distillation is given by the maximum value of p, such that

$$\Lambda\left(\rho\otimes\eta\right)=\left(p|0\rangle\langle 0|\otimes\sigma+(1-p)\,|1\rangle\langle 1|\frac{\mathbb{1}}{m}\right)\otimes\eta, \tag{160}$$

$$F(\sigma,\Psi_m)\geqslant 1-\varepsilon. \tag{161}$$

The result holds if Λ is an incoherent operation, but it can also be applied to other classes of operations studied in the resource theory of coherence (Streltsov *et al* 2017a, Fang *et al* 2018). Note that the catalyst state remains unchanged regardless of the outcome. However, this constraint was not enforced in the previous probabilistic transformation protocol (Bu *et al* 2016). With the following example, a significant improvement has been shown in probabilistic coherence distillation. Consider the initial state

$$\rho=\frac{1}{2}\left(|u_1\rangle\langle u_1|+|u_2\rangle\langle u_2|\right) \tag{162}$$

with

$$|u_1\rangle=\frac{1}{2}\left(|00\rangle-|01\rangle-|10\rangle+|11\rangle\right), \tag{163}$$

$$|u_2\rangle=\frac{1}{5\sqrt{2}}\left(2|00\rangle+6|01\rangle-3|10\rangle+|11\rangle\right). \tag{164}$$

The probability of distilling the state $|\Psi_2\rangle$ (with an error $\varepsilon\leqslant 0.01$) from ρ via dephasing-covariant incoherent operations (Marvian and Spekkens 2016, Chitambar and Gour 2016a, 2016b) can be increased significantly with the help of a catalyst in the state $|\Psi_2\rangle$ (Fang *et al* 2018). In contrast to the deterministic-exact catalytic transformation where maximally coherent states are useless (Du *et al* 2015a), this result shows that a maximally coherent state can be used as a catalyst in stochastic-approximate transformations (Kondra *et al* 2021b).

Recently, Chen *et al* (2019) studied the role of catalysis in one-shot coherence distillation under incoherent operations. In a one-shot catalytic coherence distillation protocol, the distillable coherence of ρ can be defined as follows:

$$C^{\varepsilon}_{\mathcal{O},\mathrm{c}}(\rho)=\max_{\Lambda\in\mathcal{O}}\max_{\dim\eta<\infty}\left\{\log_2 N\,:\,P\left(\Lambda\left(\rho\otimes\sigma\right),\Psi_N\otimes\sigma\right)\leqslant\varepsilon\right\} \tag{165}$$

where ρ is the initial state, $\mathcal{O}$ corresponds to different classes of incoherent operations, and P is the purified distance given as $P(\rho,\sigma)=\sqrt{1-F(\rho,\sigma)}$. This definition has been adapted from the recently introduced one-shot coherence distillation protocol (Regula *et al* 2018, Zhao *et al* 2019). Note that there is no further restriction on the catalyst σ except that the dimension is finite. In this setup, one can extract as much coherence as one wishes from any initial state ρ with some arbitrary small error ε. As there is no restriction on the catalyst, in principle the catalyst can change slightly in this procedure keeping the error within ε. Since the dimension of the catalyst could be arbitrarily large, a small change in coherence in it can provide a large value of $C^{\varepsilon}_{\mathcal{O},\mathrm{c}}(\rho)$. This phenomenon is known as embezzling and is discussed in more detail in section 7. As the previous case trivialises the transformation, in the next case Chen *et al* (2019) studied coherence distillation when the dimension of the catalyst is not more than M. In this case, they provide a lower bound on the amount of coherence that can be distilled. In fact, for an integer M and a positive number ε with the constraint $\varepsilon^2\log_2(M-1)/4\geqslant 1$, the distillable coherence of a state ρ can be lower bounded as

$$C_\varepsilon(\rho)\geqslant\frac{\varepsilon^2\left[\log_2(M-1)+1\right]-2}{2}. \tag{166}$$

This result holds for incoherent operations, and can also be extended to other operations investigated in the resource theory of coherence (Chen *et al* 2019). However, this bound appears to be loose as it is independent of the coherence of the initial state ρ, and a better bound may be possible. To avoid such embezzling, they proposed a perfect catalysis protocol with pure catalyst states where the catalyst is returned exactly, i.e. $\mathrm{Tr}_S\,\Lambda(\rho\otimes\eta)=\eta$ and the error is allowed only in the final state, i.e. $P(\mathrm{Tr}_C\,\Lambda(\rho\otimes\eta),\Psi_N)\leqslant\varepsilon$. As shown in Chen *et al* (2019), a catalyst in a pure state does not provide any advantage over standard one-shot coherence distillation where no catalyst is present to help. It is an open question whether the result is also true for a catalyst in a mixed state.

Recently, Char *et al* (2023) explored the approximate catalytic state transformations in coherence theory. Their results provide a connection between asymptotic transformation and

single-copy transformation with a catalyst. In fact, the following has been shown: if there exists an asymptotic incoherent operation transforming a state ρ into σ with unit rate, then ρ can be transformed into σ via incoherent operations with approximate catalysis (Char *et al* 2023). An analogous result in entanglement theory has been reported earlier by Kondra *et al* (2021a), see also theorem 8. Furthermore, Char *et al* (2023) showed that the relative entropy of coherence is a monotone under incoherent operations with approximate catalysis. A direct consequence of these results gives a necessary and sufficient condition for pure state transformations under incoherent operations with approximate catalysis. Particularly, the transformation from $|\psi\rangle$ to $|\phi\rangle$ is possible if and only if

$$S(\Delta[\psi]) \geqslant S(\Delta[\phi]). \tag{167}$$

The above result is analogous to the result in entanglement theory describing catalytic transformations for pure bipartite states (Kondra *et al* 2021a), see also theorem 6. Char *et al* (2023) prove equation (167) making use of the methods proposed by Shiraishi and Sagawa (2021) in quantum thermodynamics, see section 4.1.

4.4. Other quantum resource theories

One of the ways to quantify the amount of resource present in a state is the minimum amount of noise or randomness required to make it a free state (Groisman *et al* 2005). Recently, this idea has been put forward for general resource theories with the additional freedom of allowing free states to help in the process as a catalyst (Anshu *et al* 2018). In such a scenario, we want to quantify the resource content of a state ρ, or in other words, the minimum amount of noise needed to make it a free state. To achieve this, we add an extra system in a free state ρ_f to it and apply free operations on the joint state $\rho \otimes \rho_f$, such that the final state $\Lambda_f(\rho \otimes \rho_f)$ becomes arbitrarily close to a free state of the form $\sigma_f \otimes \rho_f$, where σ_f is a free state. The last point ensures that the additional state ρ_f remains almost the same in the procedure and, thus, serves as a catalyst. However, note that there is no additional constraint on the catalyst, like equation (121), which ensures that the catalyst remains exactly the same. In any case, as free states (which are not expensive) are being used as catalysts, slight changes in them do not prove to be costly. Equipped with this, Anshu *et al* (2018) showed that the asymptotic randomness rate of catalytic transformation, or the resource content of a state ρ, is given by the regularised relative entropy of the state

$$E^{\infty}(\rho) = \lim_{n\to\infty} \frac{E(\rho^{\otimes n})}{n}, \tag{168}$$

where $E(\rho) = \inf_{\sigma\in\mathcal{F}} S(\rho||\sigma)$ and $\mathcal{F}$ represents the set of free states. Furthermore, the resource content of a state in single copy regimes has also been discussed in Anshu *et al* (2018). As the above result has been derived in a general resource theoretic framework, it is applicable to resource theories of entanglement, coherence, purity, and asymmetry (Anshu *et al* 2018).

While the preceding discussion shows when an approximate catalytic transformation is possible, it does not provide any information about the catalyst's resource content. Rubboli and Tomamichel (2022) recently provided some positive answers in this direction for any general resource theory where certain monotones are additive on the tensor product. Assume that a quantity $\mathcal{D}$ completely characterises approximate catalytic state transformations in a resource theory, i.e. a transformation from ρ to σ is possible under approximate catalysis iff $\mathcal{D}(\rho) \geqslant \mathcal{D}(\sigma)$, where $\mathcal{D}(\rho) = \min_{\rho'\in\mathcal{F}} S(\rho||\rho')$. Keeping this in mind, we can consider $\mathcal{D}$ as a relevant resource measure to quantify the catalyst's resourcefulness. In fact, the order of the resource content of a catalyst has been provided for a given error in Rubboli and Tomamichel (2022).

Theorem 10 (see theorem 1 in Rubboli and Tomamichel (2022)). *Consider two quantum states ρ and σ, and $\alpha \in [1/2, 1)$ such that $\mathcal{D}_\alpha(\rho) < \mathcal{D}_\alpha(\sigma)$ and $\mathcal{D}_\alpha$ is additive for the state σ. Then, for any given error ε in an approximate catalytic transformation of ρ into σ with catalyst η, we have*

$$\mathcal{D}(\eta) = \Theta\left(\log\frac{1}{\varepsilon}\right). \tag{169}$$

Here, $\mathcal{D}_\alpha(\rho) = \min_{\rho'\in\mathcal{F}} S_\alpha(\rho||\rho')$ and $S_\alpha(\cdot||\cdot)$ is defined earlier in equation (85). Therefore, the catalyst's resourcefulness grows as the error reduces, and in the limit of zero error, the catalyst's resource content is unbounded. In Rubboli and Tomamichel (2022), the authors also derived a quantitative upper bound on the resource content of a catalyst. Furthermore, in Rubboli and Tomamichel (2022), it has been discussed that the additivity assumption, combined with $\mathcal{D}_\alpha(\rho) < \mathcal{D}_\alpha(\sigma)$ for some $\alpha \in [1/2, 1)$, implies that the approximate catalytic transformation of ρ into σ is only possible if a correlation between the system and the catalyst is allowed. Note that this kind of result has been discussed independently in the context of the resource theory of entanglement (Datta *et al* 2022).

5. Applications of quantum catalysis

5.1. Quantum state merging

Catalysis has proven useful for quantum state merging, an important protocol in quantum information theory. Before we review catalytic quantum state merging as discussed by Kondra *et al* (2021a), we consider the original quantum state merging introduced in Horodecki *et al* (2005, 2006). In the standard quantum state merging protocol, Alice (A), Bob (B), and a reference system (R) have access to many copies of a pure state $|\psi\rangle^{ABR}$. By performing LOCC operations between Alice and Bob, Alice aims to transfer her part of the state to Bob while preserving correlation with the reference system. For this, Alice and Bob can share additional singlets. In the asymptotic limit, where many copies of the overall state $|\psi\rangle^{ABR}$ are available, the minimal singlet rate for this procedure is

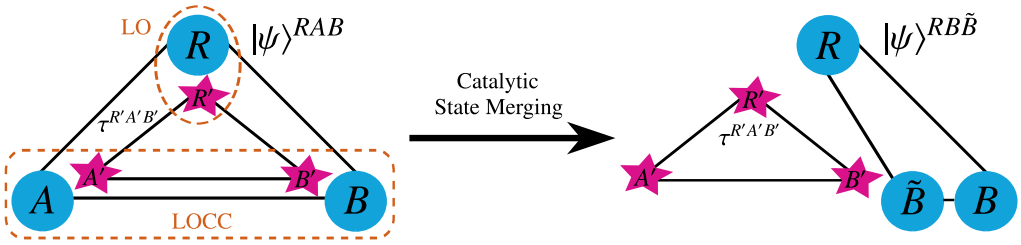


Figure 2. Catalytic quantum state merging. Alice, Bob, and a reference system share a single copy of $|\psi\rangle$. Alice aims to send her part of the state to Bob by using LOCC and approximate catalysis with catalyst states τ. Additionally, local operations (LOs) are allowed on the reference system. The process is completely characterised by the quantum conditional entropy $S(A|B)$. Reprinted figure with permission from Kondra *et al* (2021a) Copyright (2021) by the American Physical Society.

given by the quantum conditional entropy[5] (Horodecki *et al* 2005, 2006):

$$S(A|B) = S\left(\psi^{AB}\right) - S\left(\psi^{B}\right). \tag{170}$$

Depending on the value of $S(A|B)$ two situations can arise. When $S(A|B) > 0$, Alice and Bob need to share additional singlets at a rate $S(A|B)$ to accomplish the state merging protocol. However, for $S(A|B) \leqslant 0$, Alice and Bob can achieve state merging and gain extra singlets at a rate $|S(A|B)|$ to their disposal.

Catalytic quantum state merging (Kondra *et al* 2021a) allows Alice and Bob to achieve merging for a single copy of the state $|\psi\rangle^{ABR}$ by using approximate catalysis, see section 4. In this case, Alice, Bob, and the reference system share only one copy of the state $|\psi\rangle^{ABR}$. Additionally, they share a tripartite catalyst state $\tau^{A'B'R'}$, see also figure 2. In contrast to standard quantum state merging, in the catalytic case local unitaries are allowed on the reference system. However, no communication between the reference system and the other parties is required (Kondra *et al* 2021a). Similar to standard quantum state merging, the performance of this procedure can be quantified via the conditional entropy given in equation (170). For $S(A|B) > 0$, Alice and Bob can achieve catalytic state merging if they additionally have access to a pure entangled state with entanglement entropy $S(A|B)$. For $S(A|B) \leqslant 0$ Alice can merge her part of the state with Bob without any additional entanglement. Apart from merging the state, Alice and Bob will gain a pure entangled state having entanglement entropy $|S(A|B)|$. The optimality of both scenarios is discussed in Kondra *et al* (2021a).

The standard quantum state merging protocol has also been studied in the context of incoherent operations, known as incoherent quantum state merging (Streltsov *et al* 2016). In incoherent quantum state merging the main idea remains the same with some subtle changes. In comparison to LOCC in the standard state merging protocol (Horodecki *et al* 2005, 2006), here, the only change is that Bob's operations are restricted to local incoherent operations. These kinds of operations between Alice and Bob are denoted by local quantum-incoherent operations and classical communications (LQICC) (Chitambar *et al* 2016, Streltsov *et al* 2017b). Furthermore, Alice and Bob can have access to additional singlets at a rate E and only Bob has access to maximally coherent states at a rate C. Then the question is what are the achievable pairs of (E, C) such that the merging is accomplished in the asymptotic limit. In fact, for a tripartite pure state $|\psi\rangle^{ABR}$, an achievable pair of (E, C) should satisfy $E + C \geqslant S(A|B)_{\psi_{\mathrm{D}}^{AB}}$, where $\psi_{\mathrm{D}}^{AB} = \Delta(\psi^{AB})$ and $\psi^{AB} = \mathrm{Tr}R|\psi\rangle\langle\psi|^{ABR}$. As shown in Streltsov *et al* (2016), the bound can be saturated with $(E_0, 0)$, where $E_0 = S(\psi_{\mathrm{D}}^{AB}) - S(\psi_{\mathrm{D}}^{B})$. Recently, this result has been translated to single copy scenarios with the help of approximate catalysis (Char *et al* 2023) and named as catalytic incoherent state merging. As a matter of fact, it has been shown that under catalytic LQICC incoherent state merging of a single copy of $|\psi\rangle^{ABR}$ is possible when Alice and Bob have access to an additional pure state of entanglement entropy E_0 (Char *et al* 2023). Note that no additional coherence is needed on Bob's side. Furthermore, optimality has been shown by verifying that E_0 is the minimum amount of entanglement required to achieve the state merging protocol. Therefore, catalysis helps us translate the asymptotic state merging protocol into the single copy scenario.

5.2. Quantum teleportation

The role of approximate entanglement catalysis for quantum teleportation has been studied by Lipka-Bartosik and Skrzypczyk (2021b). Before presenting the result, we first briefly recapitulate the standard teleportation protocol (Bennett *et al* 1993, Popescu 1994). In general, there are two distant parties Alice and Bob sharing a quantum state ρ^{AB}. Additionally, Alice holds a particle $\tilde{A}$ in an unknown state $|\psi\rangle^{\tilde{A}}$. The goal of teleportation is to transfer Alice's state $|\psi\rangle^{\tilde{A}}$ to Bob using only LOCC. After performing an LOCC protocol Λ, the final state of Bob can be expressed as

$$\eta^{B} = \mathrm{Tr}_{A\tilde{A}}\,\Lambda\left[\rho^{AB} \otimes \psi^{\tilde{A}}\right]. \tag{171}$$

The merit of the above protocol can be quantified by the teleportation fidelity (Popescu 1994, Horodecki *et al* 1999) expressed as

[5] In section 5, the von Neumann entropy is defined with a logarithm to the base 2, i.e. $S(\rho) = -\mathrm{Tr}[\rho \log_2 \rho]$.

$$F_{\mathrm{s}}\left(\rho^{AB}\right)=\max_{\Lambda}\int \mathrm{d}\psi\,\langle\psi|\,\mathrm{Tr}_{A\tilde{A}}\,\Lambda\left[\rho^{AB}\otimes\psi^{\tilde{A}}\right]|\psi\rangle, \tag{172}$$

where the maximisation is performed over all possible LOCC operations and the integral is taken over uniform distribution $\mathrm{d}\psi$ with respect to all pure input states $|\psi\rangle$. The fidelity of teleportation can also be expressed as

$$F_{\mathrm{s}}\left(\rho^{AB}\right)=\frac{f\left(\rho^{AB}\right)d_{\tilde{A}}+1}{d_{\tilde{A}}+1}, \tag{173}$$

where $d_{\tilde{A}}$ is the dimension of the input state $|\psi\rangle^{\tilde{A}}$ and $f(\rho^{AB})=\max_{\Lambda}\langle\Phi^{+}|\Lambda(\rho^{AB})|\Phi^{+}\rangle$ is the singlet fraction maximised over all LOCC protocols (Horodecki *et al* 1999).

Recently, the standard teleportation protocol has been studied in a catalytic scenario to see whether there is an advantage in teleportation fidelity (Lipka-Bartosik and Skrzypczyk 2021b). In contrast to the standard teleportation protocol, here in the catalytic teleportation protocol Alice and Bob share an additional entangled state $\tau^{A'B'}$ such that it remains the same after the process. To be precise, the catalytic teleportation fidelity can be expressed as (Lipka-Bartosik and Skrzypczyk 2021b)

$$F_{\mathrm{c}}\left(\rho^{AB}\right)=\max_{\Lambda,\tau^{A'B'}}\int \mathrm{d}\psi\,\langle\psi|\,\mathrm{Tr}_{A\tilde{A}A'B'}\,\Lambda\left[\rho^{AB}\otimes\tau^{A'B'}\otimes\psi^{\tilde{A}}\right]|\psi\rangle \tag{174}$$

with the constraint $\mathrm{Tr}_{A\tilde{A}B}\,\Lambda[\rho^{AB}\otimes\tau^{A'B'}\otimes\psi^{\tilde{A}}]=\tau^{A'B'}$ which ensures that the catalyst remains unchanged in the process. Furthermore, in Lipka-Bartosik and Skrzypczyk (2021b) an achievable lower bound of $F_{\mathrm{c}}(\rho^{AB})$ has been derived:

$$F_{\mathrm{c}}\left(\rho^{AB}\right)\geqslant\frac{f_{\mathrm{r}}\left(\rho^{AB}\right)d_{\tilde{A}}+1}{d_{\tilde{A}}+1}, \tag{175}$$

where $f_{\mathrm{r}}(\rho^{AB})=\lim_{n\to\infty}f_n(\rho^{\otimes n})/n$ is the regularised version of the singlet fraction and $f_n(\sigma)=\max_{\Lambda}\sum_{i=1}^{n}\langle\Phi^{+}|\,\mathrm{Tr}_{\bar{i}}\,\Lambda(\sigma)|\Phi^{+}\rangle$, where $\mathrm{Tr}_{\bar{i}}$ represents partial trace over systems 1 to n except for i. As $f_{\mathrm{r}}(\rho^{AB})\geqslant f(\rho^{AB})$ for all states ρ^{AB}, we have $F_{\mathrm{c}}(\rho^{AB})\geqslant F_{\mathrm{s}}(\rho^{AB})$ (Lipka-Bartosik and Skrzypczyk 2021b). In addition, when ρ^{AB} is pure, Lipka-Bartosik and Skrzypczyk (2021b) showed that the catalytic teleportation outperforms the standard teleportation protocol for a large variety of generic quantum states. Furthermore, they studied the advantage when the dimension of the catalyst is small.

5.3. Assisted distillation of entanglement and coherence

Catalytic assisted entanglement distillation has been introduced and studied by Kondra *et al* (2021a). In the standard assisted entanglement distillation scenario, Alice, Bob and Charlie share many copies of a tripartite state $|\psi\rangle^{ABC}$, and their aim is to extract singlets between Alice and Bob by performing LOCC (DiVincenzo *et al* 1999, Smolin *et al* 2005). In Smolin *et al* (2005), the singlet rate has been given in the standard asymptotic scenario as $\min\{S(\psi^{A}),S(\psi^{B})\}$. In catalytic assisted entanglement distillation (Kondra *et al* 2021a) the parties have access to one copy of state $|\psi\rangle^{ABC}$, and can make use of tripartite entangled catalysts. By using approximate catalysis as discussed in section 4, a single copy of $|\psi\rangle^{ABC}$ can be converted into a pure entangled state shared by Alice and Bob with entanglement entropy given by $\min\{S(\psi^{A}),S(\psi^{B})\}$. The optimality of this procedure and more details are discussed in Kondra *et al* (2021a).

The role of catalysis has also been studied for assisted coherence distillation (Chitambar *et al* 2016) in Char *et al* (2023). In assisted coherence distillation Alice and Bob share many copies of a state ρ^{AB} and their target is to maximise the coherence on Bob's side by performing LQICC (Chitambar *et al* 2016). The quantifier of the performance $C_{\mathrm{d}}^{A|B}(\rho^{AB})$ represents the optimal rate at which maximally coherent states per copy of ρ^{AB} can be obtained on Bob's side. If the initial state is pure then it has been shown that (Chitambar *et al* 2016)

$$C_{\mathrm{d}}^{A|B}\left(|\psi^{AB}\rangle\right)=S\left(\Delta\left[\psi^{B}\right]\right). \tag{176}$$

With the help of an additional system as a catalyst, the above protocol can be translated to the single copy scenario where a single copy of state $|\psi\rangle^{AB}$ is used to maximise coherence on Bob's side by performing catalytic LQICC (Char *et al* 2023). Under catalytic LQICC, the optimal coherence achieved on Bob's side is $S(\Delta[\psi^{B}])$ (Char *et al* 2023), which is the same as the optimal rate at which maximally coherent states per copy of $|\psi\rangle^{AB}$ are obtained on Bob's side in the standard assisted coherence distillation protocol (Chitambar *et al* 2016).

5.4. Catalysis of noisy quantum channels

The role of entanglement catalysis for quantum communication via noisy quantum channels has been studied by Datta *et al* (2022). For this, Datta *et al* (2022) introduced and studied catalytic quantum capacity, capturing the ability of a quantum channel to transmit qubits in the presence of entangled catalysts. In the standard asymptotic setting, where the communicating parties have access to many copies of a quantum channel that can be used in parallel, the performance is quantified by quantum capacity. It represents the maximum rate at which qubits can be faithfully transmitted to a distant receiver via a channel Λ (Schumacher and Nielsen 1996, Lloyd 1997, Horodecki *et al* 2000), and can be expressed as (Shor 2002, Devetak 2005)

$$Q\left(\Lambda\right)=\lim_{n\to\infty}\frac{1}{n}\max_{\rho_n}I\left(\rho_n,\Lambda^{\otimes n}\right). \tag{177}$$

Here, I represents coherent information and can be expressed as $I(\rho,\Lambda)=S(\Lambda[\rho])-S(\mathbb{1}\otimes\Lambda[|\psi_\rho\rangle\langle\psi_\rho|])$ (Schumacher and Nielsen 1996, Lloyd 1997) and $|\psi_\rho\rangle$ represents some purification of ρ.

If the communicating parties can use the quantum channel only once, they will typically not be able to transmit qubits faithfully through the channel. However, the situation changes if they have access to entangled catalysts (Datta *et al* 2022).

The figure of merit in this setup is quantified via catalytic quantum capacity (Datta *et al* 2022)

$$Q_{\mathrm{c}}(\Lambda) = \max\left\{m : \lim_{n\to\infty} \|\mu_n - |\phi^+_{2^m}\rangle\langle\phi^+_{2^m}| \otimes \tau_n\|_1 = 0\right\}, \tag{178}$$

where $\{\mu_n\}$ is a sequence of states obtained by a single use of the noisy quantum channel Λ and approximate catalysis (Datta *et al* 2022). Note that the catalytic quantum capacity is an integer, and can be zero even for channels which are not entanglement-breaking. A lower bound on the catalytic quantum capacity can be obtained as (Datta *et al* 2022)

$$Q_{\mathrm{c}}(\Lambda) \geqslant \left\lfloor \sup_{\psi} E_{\mathrm{d}}(\mathbb{1}\otimes\Lambda[\psi]) \right\rfloor, \tag{179}$$

where E_{d} is the distillable entanglement (Bennett *et al* 1996a, 1996b) and the supremum is taken over all bipartite pure states $\psi = |\psi\rangle\langle\psi|$. For achieving this bound, Alice can prepare the two-particle state $|\psi\rangle$ locally, and send one half of it to Bob via the channel Λ. In this way, Alice and Bob end up sharing the state $\mathbb{1}\otimes\Lambda[\psi]$, which can be converted into any pure state with entanglement entropy $E_{\mathrm{d}}(\mathbb{1}\otimes\Lambda[\psi])$ (Datta *et al* 2022).

As distillable entanglement is lower bounded by $E_{\mathrm{d}}(\rho^{AB}) \geqslant S(\rho^A) - S(\rho^{AB})$ (Devetak and Winter 2005), we can obtain an easily computable lower bound for the catalytic quantum capacity (Datta *et al* 2022):

$$Q_{\mathrm{c}}(\Lambda) \geqslant \left\lfloor \log_2 d - S\left(\mathbb{1}\otimes\Lambda\left[\phi^+_d\right]\right)\right\rfloor, \tag{180}$$

where d is the dimension of the Hilbert space on which Λ is acting and $|\phi^+_d\rangle = \sum_{i=0}^{d-1}|ii\rangle/\sqrt{d}$. For a general quantum channel with Kraus operators K_i , Datta *et al* (2022) further obtained the lower bound

$$Q_{\mathrm{c}}(\Lambda) \geqslant \lfloor \log_2 d - H(p)\rfloor, \tag{181}$$

where $H(p)$ is the Shannon entropy of the probability distribution

$$p_i = \mathrm{Tr}\left[(\mathbb{1}\otimes K_i)|\phi^+_d\rangle\langle\phi^+_d|\left(\mathbb{1}\otimes K_i^\dagger\right)\right]. \tag{182}$$

As an immediate result, we observe that a channel Λ of dimension $d \geqslant 4$ has a catalytic capacity of at least one if it can be decomposed into at most two Kraus operators (Datta *et al* 2022). An upper bound on the catalytic quantum capacity has also been obtained by Datta *et al* (2022):

$$Q_{\mathrm{c}}(\Lambda) \leqslant \Delta E_{\mathrm{sq}}(\Lambda), \tag{183}$$

with

$$\Delta E_{\mathrm{sq}}(\Lambda) = \sup_{\rho^{ABC}}\left\{E_{\mathrm{sq}}^{A|BC}\left(\Lambda^C\left[\rho^{ABC}\right]\right) - E_{\mathrm{sq}}^{AC|B}\left(\rho^{ABC}\right)\right\}, \tag{184}$$

where E_{sq} is the squashed entanglement (Christandl and Winter 2004). $\Delta E_{\mathrm{sq}}(\Lambda)$ can be interpreted as the amount of entanglement that the channel Λ can transmit. Generally, computation of $\Delta E_{\mathrm{sq}}(\Lambda)$ is a formidable task. However, for certain types of channels an upper bound on it can be easily computed, e.g. for multiple copies of a single-qubit Pauli channel $\Lambda_p(\rho) = \sum_i p_i\sigma_i\rho\sigma_i$. In particular, the catalytic quantum capacity of $\Lambda_p^{\otimes n}$ is bounded as follows (Datta *et al* 2022):

$$\left\lfloor nE_{\mathrm{f}}\left(\mathbb{1}\otimes\Lambda_p\left[\phi^+_2\right]\right)\right\rfloor \geqslant Q_{\mathrm{c}}\left(\Lambda_p^{\otimes n}\right) \geqslant \lfloor n - nH(p_i)\rfloor, \tag{185}$$

where E_{f} represents entanglement of formation (Bennett *et al* 1996c). The upper bound on Q_{c} can be easily evaluated noting that entanglement of formation can be computed for all two-qubit states (Bennett *et al* 1996c, Wootters 1998).

Datta *et al* (2022) also discussed the advantage of catalysis for entanglement distribution between two distance parties. Consider a scenario where Alice wants to distribute entanglement with Bob via a single qubit depolarising channel of the form

$$\Lambda_l(\rho) = \mathrm{e}^{-\alpha l}\rho + \left(1-\mathrm{e}^{-\alpha l}\right)\frac{\mathbb{1}}{2}, \tag{186}$$

where l is the length of the channel and $\alpha \geqslant 0$ represents a damping parameter. The channel is entanglement breaking when $l \geqslant \ln 3/\alpha$. Suppose there is a node in between Alice and Bob to assist them in the entanglement distribution. More precisely, the node is allowed to perform any local operations and communicate classically with Alice and Bob, see also figure 3. Nevertheless, entanglement distribution is still not possible for $l \geqslant \ln 3/\alpha$ as shown in Datta *et al* (2022). However, if we have an entangled catalyst shared between Alice and the node then entanglement distribution between Alice and Bob is possible for $l < 2\ln 3/\alpha$ (Datta *et al* 2022). To see this, let us assume that Alice prepares a maximally entangled state locally and sends one half of it to Bob through the channel. As a result, the state between the node and Alice becomes mixed, and to ensure that the resulting state between them is entangled the position (s) of the node should satisfy $s < \ln 3/\alpha$. As all two-qubit entangled states are distillable (Horodecki *et al* 1997), it is possible to create a pure entangled state between Alice and the node using the results in Kondra *et al* (2021a). After that, the particle at the node is sent to Bob via the channel, leading to an entangled state between Alice and Bob. Therefore, the introduction of an intermediate node and a suitable catalyst makes a useless depolarising channel for entanglement distribution useful.

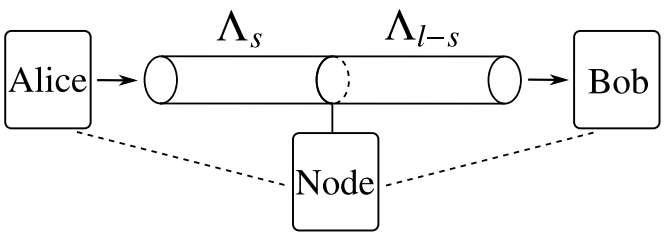


Figure 3. A schematic of entanglement distribution between Alice and Bob via a depolarising qubit channel of length l. To assist the parties in the procedure an intermediate node is considered at a distance s from Alice. Note that the node can perform local quantum operations and communicate classically (dashed lines) with the parties. Reproduced with permission from Datta *et al* (2022).

5.5. Authentication protocols based on catalysis

Catalysis has shown to be useful in authentication protocols as well (Barnum 1999). Suppose there are two parties Alice, Bob and Bob needs to authenticate himself to Alice. As we know there exist incomparable states $|\psi\rangle$ and $|\phi\rangle$ which can not be transformed into each other via LOCC. Let us say, Alice prepares locally a bipartite state $|\psi\rangle$ and sends one half of it to Bob. Additionally, Alice and Bob share a catalyst whose presence accomplishes the forbidden transformation $|\psi\rangle \nrightarrow |\phi\rangle$. Now they perform catalytic LOCC on the state $|\psi\rangle$, such that they achieved the desired final state. In the next step, Bob will send his half of the final state to Alice and Alice confirms the presence of Bob by performing a measurement in the basis $|\phi\rangle$. Suppose, there is some third-party intruder Charlie on the other side of Alice's communication channel. However, they will not be able to transform the state $|\psi\rangle$ to the desired final state $|\phi\rangle$ as Charlie does not possess the other half of the catalyst state. Note that the requirement of incomparable states is essential for this protocol. Otherwise, Alice and Charlie would have accomplished the desired transformation without the catalyst. For the error probability and security analysis we refer to (Barnum 1999).

5.6. Catalytic decoupling

Catalysis has also been useful in the decoupling of quantum information (Majenz *et al* 2017). In the standard decoupling technique, we want to approximately decouple S and E from a bipartite state ρ^{SE} by erasing a part of S (Horodecki *et al* 2005, 2006, Dupuis *et al* 2014). Precisely, the bipartite state ρ^{SE} is ε-decoupled, if there exists a unitary operation U_S such that the following holds (Majenz *et al* 2017)

$$\min_{\omega_{S_1}\otimes\omega_E} P\left(\mathrm{Tr}_{S_2}\, U_S\rho^{SE}U_S^\dagger, \omega_{S_1}\otimes\omega_E\right) \leqslant \varepsilon, \tag{187}$$

with the purified distance $P(\rho,\sigma) = (1-F(\rho,\sigma))^{1/2}$ and $S = S_1S_2$. Therefore, we successfully decouple S_1 from E at the price of losing S_2 in the process. The figure of merit is the minimum dimension of S_2 or to be precise $\log_2|S_2|$ to achieve the ε-decoupling and it is denoted by $R^\varepsilon(S;E)_\rho$. In fact, the minimum system size satisfies the following (Majenz *et al* 2017)

$$R^\varepsilon(S;E)_\rho \geqslant \frac{1}{2} I^\varepsilon_{\max}(E;S)_\rho, \tag{188}$$

where $I^\varepsilon_{\max}(E;S)_\rho$ represents smooth max-mutual information in the initial state ρ^{SE}. It is defined as

$$I^\varepsilon_{\max}(E;S)_\rho = \min_{\tilde\rho} I_{\max}(E;S)_{\tilde\rho}, \tag{189}$$

where the minimisation is taken over all bipartite states $\tilde\rho^{SE}$ such that $P(\rho^{SE},\tilde\rho^{SE}) \leqslant \varepsilon$ and

$$I_{\max}(E;S)_{\tilde\rho} = \log_2 \min_{\sigma^S}\left\{\mathrm{Tr}_{\sigma^S}\,|\sigma^S\otimes\tilde\rho^E \geqslant \tilde\rho^{SE}\right\}. \tag{190}$$

However, the bound is not achievable for all states. On the contrary, if we allow catalytic decoupling, the above bound can be achieved for any bipartite system (Majenz *et al* 2017). As a formal definition of catalytic decoupling, we say that a bipartite state ρ^{SE} is ε-decoupled with the help of a catalyst in state $\rho^{S'}$, if there exists a unitary operation $U_{\bar S}$ such that

$$\min_{\omega^{S_1S'}\otimes\omega^E} P\left(\mathrm{Tr}_{S_2}\, U_{\bar S}\rho^{\bar S E}U_{\bar S}^\dagger, \omega^{S_1S'}\otimes\omega^E\right) \leqslant \varepsilon, \tag{191}$$

with $\bar S = SS' = S_1S_2S'$, and $\rho^{\bar S E} = \rho^{SE}\otimes\rho^{S'}$. Here $\omega^{S_1S'}$ represents uncorrelated state of S_1 and S'. As mentioned in Majenz *et al* (2017), the protocol is catalytic in the sense that the share of the ancillary state $\rho^{S'}$, which becomes part of the decoupled system S_1S', remains uncorrelated with S_1 after the decoupling. However, note that other than the catalyst remaining decoupled from the system, no further restriction has been imposed on the catalyst state $\rho^{S'}$, such that it remains unaltered after the protocol. Therefore, it may not be catalytic in the sense that the state of the catalyst should remain unchanged. The minimum system size required in this scenario can be quantified as

$$R^\varepsilon_{\mathrm{c}}(S;E)_\rho = \min_{\mathcal{H}_{S'},\sigma^{S'}} R^\varepsilon(SS';E)_\rho, \tag{192}$$

where the minimum is taken over all possible Hilbert spaces $\mathcal{H}_{S'}$ and all states $\sigma^{S'}$ belong to the Hilbert space $\mathcal{H}_{S'}$. It is straightforward to check that $R^\varepsilon_{\mathrm{c}}(S;E)_\rho \geqslant R^\varepsilon(S;E)_\rho$. With this Majenz *et al* (2017) showed that the bound in equation (188) is achievable as we have

$$R^\varepsilon_{\mathrm{c}}(S;E)_\rho \lesssim \frac{1}{2} I^{\varepsilon-\delta}_{\max}(E;S)_\rho, \tag{193}$$

where $1 \geqslant \varepsilon \geqslant \delta > 0$ and '$\lesssim$' is up to terms of the order $\mathcal{O}(\log_2(1/\varepsilon))$. Furthermore, they discuss some applications of catalytic decoupling.

5.7. Catalytic cooling

Henao and Uzdin (2021) explored the advantage of catalysts to cool a system. In Henao and Uzdin (2021), the authors considered a passive state ρ_{b} as an initial state that is to be cooled. Note that a passive state cannot be cooled alone as its average energy $\langle H_{\mathrm{b}}\rangle = \mathrm{Tr}(H_{\mathrm{b}}\rho_{\mathrm{b}})$ cannot be lowered using any local unitary U_{b}, where H_{b} is the corresponding Hamiltonian (Allahverdyan *et al* 2004). More precisely a state ρ_{b} is called passive if and only if the change in average energy under any unitary operation U_{b} is greater or equal to zero, i.e. $\Delta\langle H_{\mathrm{b}}\rangle = \mathrm{Tr}[H_{\mathrm{b}}(U_{\mathrm{b}}\rho_{\mathrm{b}}U_{\mathrm{b}}^\dagger - \rho_{\mathrm{b}})] \geqslant 0$ (Allahverdyan *et al* 2004). As ρ_{b} cannot be cooled alone, we need to attach some ancillary systems to it and perform a global unitary to cool the system. If we attach a hot object which is not in a passive state, then cooling is possible. However, in Henao and Uzdin (2021), the authors considered a hot object in a passive state ρ_{r} and used a catalyst in a non-passive state ρ_{c} to cool the system in state ρ_{b}. Therefore, under a global unitary U_{brc} the initial state $\rho = \rho_{\mathrm{b}}\otimes\rho_{\mathrm{r}}\otimes\rho_{\mathrm{c}}$ transforms to $\rho' = U_{\mathrm{brc}}\rho U_{\mathrm{brc}}^\dagger$ and the process is catalytic if we have $\mathrm{Tr}_{\mathrm{br}}\,\rho' = \rho_{\mathrm{c}}$ additionally. To ensure that the system in state ρ_{b} cools we should have $\Delta\langle H_{\mathrm{b}}\rangle < 0$, and in

such a scenario, the transformation is called a catalytic cooling transformation (Henao and Uzdin 2021). Equipped with this, Henao and Uzdin (2021) showed that catalytic cooling is always possible with a sufficiently large dimensional catalyst state. More precisely, they showed the following: for a passive state $\rho_b \otimes \rho_r$ with $\rho_r \neq \mathbb{1}/d_r$ (d_r is the dimension of ρ_r), there exists a global unitary U_{brc} and a suitable catalyst state ρ_c such that catalytic cooling transformation is possible (Henao and Uzdin 2021). Furthermore, they also showed that cooling of a qubit system can be maximised with a hot qubit and a three-dimensional catalyst (Henao and Uzdin 2021).

6. Universal catalysis

So far, for the results presented above, the construction of the catalytic state for a non-trivial quantum transformation depends strongly on the choice of input and output states that are to be transformed. A natural question arises then, whether some states are more suitable than others to serve as a catalyst for a wider range of desired transformations. Perhaps one could even find a single quantum state that would be a catalyst for all allowed transformations within a particular resource theory? As we will see, such a *universal catalyst* state can be readily constructed as long as the allowed catalytic transformations induce a continuous partial order on the compact state space.

In this section, we will review the nascent theory of universal catalysis, when the catalyst state does not depend on the states undergoing the transition. The question about the existence of a universal catalysis state in the context of quantum thermodynamics was raised, and found an intriguing answer, by Lipka-Bartosik and Skrzypczyk (2021a). Soon afterwards, a similar question was posed by Kondra *et al* (2021a) for the theory of pure state quantum entanglement, and an answer was given in the subsequent work (Datta *et al* 2022). Here, we are going to focus our attention first on the results regarding quantum thermodynamics; then we will proceed to the case of quantum entanglement, leaving the general discussion about the implications of the universal catalysis to the final part of the section.

6.1. Quantum thermodynamics

For a given pair ρ_S, σ_S, of states that are block-diagonal in the energy eigenbasis of a Hamiltonian H_S of the system S, let vectors $\boldsymbol{p} = \mathrm{diag}[\rho_S]$ and $\boldsymbol{q} = \mathrm{diag}[\sigma_S]$ denote their diagonal parts. Let also $\{E_i\}$ be the eigenvalues of the Hamiltonian H_S. We set $\gamma_S = \mathrm{diag}[\boldsymbol{g}] = (g_1, g_2, \ldots, g_{d_S})$ be the system's thermal state, diagonal in the same basis, with $g_i = \mathrm{e}^{-\beta E_i}/Z_S$, $Z_S = \mathrm{Tr}\,\mathrm{e}^{-\beta H_S}$. Recall that, for a block-diagonal state in the energy eigenbasis ρ_S, such that $\boldsymbol{p} = \mathrm{diag}[\rho_S]$, with Hamiltonian H and in contact with the environment at inverse temperature β, the generalised free energies are defined as

$$F_\alpha(\boldsymbol{p}, \boldsymbol{g}) = S_\alpha(\boldsymbol{p}||\boldsymbol{g}) - Z, \tag{194}$$

where $S_\alpha(\boldsymbol{p}\,||\,\boldsymbol{g}) = \frac{\mathrm{sgn}(\alpha)}{1-\alpha}\log\left(\sum_i p_i^\alpha g_i^{1-\alpha}\right)$ is the Rényi relative entropy (Müller-Lennert *et al* 2013), and $Z = \mathrm{Tr}\,\mathrm{e}^{-\beta H}$ is the total partition function, compare also equations (84) and (85).

Now, the main result concerning the universal catalytic transformation in quantum thermodynamics can be stated as follows.

Theorem 11 (see (Lipka-Bartosik and Skrzypczyk 2021a), theorem 2). *Let ρ_S and σ_S be two states with corresponding representations $\boldsymbol{p} = diag[\rho_S]$ and $\boldsymbol{q} = diag[\sigma_S]$ which satisfy:*

$$F_\alpha(\boldsymbol{p}, \boldsymbol{g}) > F_\alpha(\boldsymbol{q}, \boldsymbol{g}) \qquad \forall \alpha \geqslant 0, \tag{195}$$

Then, for any catalyst state ω_C with $\boldsymbol{c} = diag[\omega_C]$ and sufficiently large n there exists a thermal operation $\mathcal{T}_{SC}$ such that:

$$\mathcal{T}_{SC}\left[\rho_S \otimes \omega_C^{\otimes n}\right] = \sigma'_{SC}, \tag{196}$$

and the errors on the system and the catalyst satisfy:

$$\epsilon_C := ||\,\mathrm{Tr}_S\left[\sigma'_{SC}\right] - \omega_C^{\otimes n}||_1 \leqslant \mathcal{O}\left(\mathrm{e}^{-n^\kappa}\right), \tag{197}$$

$$\epsilon_S := ||\,\mathrm{Tr}_C\left[\sigma'_{SC}\right] - \sigma_S||_1 = 0, \tag{198}$$

where $\kappa \in (0,1)$ can be chosen arbitrarily.

A remark regarding the expression 'any catalyst state' is in order here. In the proof of the above result, Lipka-Bartosik and Skrzypczyk (2021a) assume that the catalyst system of finite dimension d_C is prepared in the state ω_C that is diagonal in the eigenbasis of the system's Hamiltonian H_C. It is known that if a catalytic transformation between diagonal states ρ_S and σ_S is admissible, it is also possible to transform ρ_S into σ_S as in equation (196) with the help of a catalyst state that has a fully degenerate spectrum, compare the proof of theorem 18, Supplementary Information in Brandão *et al* (2015). Hence, without loss of generality, we can assume that the Hamiltonian H_C is trivial, $H_C \propto \mathbb{1}$. This, in turn, means that *any* quantum state ω_C can be considered a viable catalyst state, since we require only that it be diagonal in the eigenbasis of H_C. In this sense, any quantum state ω_C, as long as sufficiently many copies are available, can act as a universal catalyst.

The family of equations given in equation (195) are referred to as 'second laws of quantum thermodynamics' (Brandão *et al* 2015), as they induce a partial order on the state space, which describes allowed catalytic transformations, see also section 3.3. It is important to notice here that if one relaxes the strict condition in equation (197) on the amount of error accumulated by the transformation, in a sense that the error is allowed to vanish at the rate slower than exponential in n, the order will 'collapse' and transformations between essentially all states become possible, as in Lipka-Bartosik and Skrzypczyk (2021a), theorem 1. We will discuss this phenomenon, known in general as *catalytic embezzling* of quantum resources, thoroughly in the subsequent section 7.

An intuitive explanation of the above result is the following. Because of the law of large numbers, for an arbitrary state ω_C that is diagonal in the energy eigenbasis, as the number of copies of the state increases in the asymptotic regime ($n \to \infty$), a

subset of eigenvalues of $\omega_C^{\otimes n}$ forms a typical set: almost all probability weight is concentrated in a set that is uniformly distributed. Using a result from (Chubb *et al* 2019), see theorem 2 therein, Lipka-Bartosik and Skrzypczyk (2021a) claim that the state $\omega_C^{\otimes n}$ can be converted reversibly into multiple copies of yet another state, useful for catalytic transformation, at a rate given approximately by relative entropy. The exact quantification of the typicality property is the source of the error as in equation (197). The proof of theorem 11 utilises this kind of intuition and focuses on specifying exactly the thermal operation $\mathcal{T}_{SC}$. The actual catalytic transformation allows for a change of the catalyst. To avoid embezzling of the quantum resource, care needs to be taken in analysing the magnitude of that change, as in equation (197).

6.2. Pure state quantum entanglement

As pointed out in section 4.2, for two bipartite pure states $|\psi\rangle^{AB}$, $|\phi\rangle^{AB}$, an approximate catalytic conversion is possible if and only if $S(\psi^A) \geqslant S(\phi^A)$. This result was first proven in Kondra *et al* (2021a), where an explicit construction for the catalyst state was provided. The state of the catalyst given in Kondra *et al* (2021a) depends strongly on the states $|\psi\rangle^{AB}$, $|\phi\rangle^{AB}$, and a natural question was raised in Kondra *et al* (2021a) whether there exists a catalyst state that can catalyse all allowed transformations, even if only to some arbitrary precision. Datta *et al* (2022) give a positive answer to that question.

Theorem 12 (see Datta *et al* (2022), theorem 1). *Consider a bipartite Hilbert space of arbitrary but finite dimension. For every $\varepsilon > 0$ there exists a universal catalyst state $\tau_\varepsilon^{A'B'}$ such that for every pair of pure states $|\psi\rangle^{AB}$ and $|\phi\rangle^{AB}$ with $S(\psi^A) \geqslant S(\phi^A)$ there is an LOCC protocol Λ for which*

$$\mathrm{Tr}_{AB}\left[\Lambda\left(\psi^{AB}\otimes\tau_\varepsilon^{A'B'}\right)\right] = \tau_\varepsilon^{A'B'}, \tag{199a}$$

$$\left\|\Lambda\left(\psi^{AB}\otimes\tau_\varepsilon^{A'B'}\right) - \phi^{AB}\otimes\tau_\varepsilon^{A'B'}\right\|_1 < \varepsilon. \tag{199b}$$

As we can see, contrary to equation (197) in the previous section 6.1, equation (199*a*) assures that the universal catalyst state $\tau_\varepsilon^{A'B'}$ remains unchanged as a result of the LOCC transformation Λ. It is generally understood that the requirement in equation (199*a*) is essential if we want to talk meaningfully about universal catalysis for pure state entanglement. This is perhaps the most natural assumption to make if one wants to avoid 'quite implausible and unphysical' (Müller 2018) effects like embezzling of entanglement (see section 7). The requirement for *approximate* catalysis, as in section 4 above, is encoded in equation (199*b*). We should notice here that the amount of error allowed during the transformation does not depend on the choice of the input and output states. Hence, an arbitrary precision $\varepsilon > 0$ can be achieved for all possible catalytic transformations using only one state $\tau_\varepsilon^{A'B'}$. In this sense, one can speak of a universal catalytic state for pure-state entanglement.

The proof of theorem 12, reported in Datta *et al* (2022), makes use of the fact that the set of quantum states is compact. By discretising appropriately the set of states, one can find a catalytic state for every pair of initial and final states from the finite collection of discrete 'points' that allow a catalytic transformation as in section 4. Then the universal catalytic state is obtained by taking the tensor product of that finite number of catalysts. By controlling how fine the discretisation should be, a bound for the error of the universal catalytic transformation is achieved as in equation (199*b*).

6.3. Comparison of universal catalysis of entanglement and quantum thermodynamics

The two constructions of a universal catalytic state given above are rather different. On the one hand, the construction in Lipka-Bartosik and Skrzypczyk (2021a) for quantum thermodynamics utilises the typicality of a sufficiently large tensor power of any quantum state; on the other, the method employed in Datta *et al* (2022) for the resource theory of pure state entanglement relies on the discretisation of the compact state space. Both approaches, however, share a common assumption about what catalytic transformations are in general possible with a particular resource theory. That is the existence of a continuous function, or a family of functions, inducing a partial order of allowed transformations on the compact state space. Additionally, both results are believed to be extendable to a broader context beyond their respective resource theories: compare section III.F in Lipka-Bartosik and Skrzypczyk (2021a) and Discussion in Datta *et al* (2022). It is then of merit to ask at this point if both methods can be unified somehow and expanded to a general theory of universal catalytic states. It seems like in both cases the universality of the construction is achieved by taking a state that in some sense contains all other catalysts, at least within an assumed margin of error. Hence, the existence of a universal catalyst should be considered as another manifestation of the 'second laws' governing the allowed catalytic transformation between resourceful states.

7. Catalytic embezzling phenomena

In most parts of the previous discussion, we focused on catalytic transformations with the requirement that the state of the catalyst is left unchanged by the overall procedure. Here we will discuss phenomena which occur if a change of the catalyst state is allowed.

One of the first results in this direction has been presented by van Dam and Hayden (2003). The authors investigated exact entanglement catalysis, allowing the catalysts to change by an arbitrarily small amount, as measured by fidelity between the initial and final state of the catalyst system. In this setting, van Dam and Hayden (2003) proved that there exists a family of pure state $|\mu_n\rangle$ such that for any bipartite state $|\psi\rangle^{AB}$ there is an LOCC protocol: $|\mu_n\rangle \to |\psi\rangle^{AB}\otimes|\omega_n\rangle$ with $|\langle\mu_n|\omega_n\rangle| > 1-\varepsilon$ and vanishing $\varepsilon > 0$ as $n \to \infty$. Thus, if we consider approximate entanglement catalysis with equation (120) being the only constraint, the transformations obtained in this way allow to transform any bipartite pure state into any other bipartite pure state. In particular, for any $\varepsilon > 0$

and any two states $|\psi\rangle^{AB}$ and $|\phi\rangle^{AB}$ there exists a catalyst state τ and an LOCC protocol Λ such that equation (120) is fulfilled. This applies also to the setting where the initial state $|\psi\rangle^{AB}$ is product, and the final state $|\phi\rangle^{AB}$ corresponds to n singlets, where n is an arbitrary integer. Thus, the procedure allows to extract an unbounded number of singlets from a shared catalyst, inducing only a small change in the quantum state of the catalyst. Due to this behaviour, this phenomenon has also been termed 'embezzling entanglement' (van Dam and Hayden 2003).

The result of van Dam and Hayden (2003) indicates that perfect embezzlement with zero error is not possible in a finite-dimensional state. This result has recently been extended to infinite-dimensional systems, demonstrating that perfect embezzlement is also impossible in this case (Cleve *et al* 2017). Furthermore, in contrast to the tensor product framework, Cleve *et al* (2017) have demonstrated that perfect embezzlement is possible in a commuting operator framework.

van Dam and Hayden (2003) also proved that universal entanglement embezzling is possible, which means that for any given $\varepsilon > 0$ and given dimensions d_A and d_B there exists a catalyst achieving the conversion $|\psi\rangle^{AB} \to |\phi\rangle^{AB}$ for any initial and final state. Moreover, the conversion can be achieved by using local unitaries, and no communication between the parties is required. Additional families of states for universal entanglement embezzling have been presented by Leung and Wang (2014), and entanglement embezzling has also been used for an alternative proof for the quantum reverse Shannon theorem (Berta *et al* 2011, Bennett *et al* 2014).

A detailed study of catalytic embezzling phenomena in quantum thermodynamics has been performed by Brandão *et al* (2015). The authors introduced catalytic thermal operations (see also section 3.3), and investigated different regimes for the change of the catalyst. For the case that the catalyst is returned unchanged, the transitions are characterised by the second laws as described in section 3.3. The same holds true if the state of the catalyst changes in such a way, that the change can be corrected by investing a small amount of work (Brandão *et al* 2015). The third regime considered by Brandão *et al* (2015) requires that the state of the catalyst changes at most by $\varepsilon/\log d$ in trace norm, where d is the dimension of the catalyst. In this case, it is shown that the transitions are completely characterised by the standard second law, under the assumption that the initial and the final states are diagonal in the energy eigenbasis.

For the resource theory of purity, a careful investigation of embezzling phenomena has been performed by Ng *et al* (2015). The following condition must hold for the catalyst state, if the catalyst is to enable all state transformations via unital operations (Ng *et al* 2015):

$$\frac{1}{2}\left\|\omega-\omega'\right\|_1 \geqslant \frac{d_S-1}{1+(d_S-1)\,\frac{\log_2 d_C}{\log_2 d_S}}, \tag{200}$$

where d_S and d_C are the dimensions of the system and the catalyst, respectively. In particular, for any system states ρ and σ there exist states of the catalyst ω and ω' fulfilling equation (200) such that $\rho\otimes\omega$ can be transformed into $\sigma\otimes\omega'$ via an unital operation. Moreover, this bound is optimal: it is not possible to enable all state transformations with a catalyst violating equation (200).

For general resource theory, an interesting result regarding the resource content of a catalyst in the case of embezzlement has recently been discussed in Rubboli and Tomamichel (2022). In fact, Rubboli and Tomamichel (2022) demonstrated that the theorem 10 holds true even when the catalyst changes in the procedure, resulting in resource embezzlement. As a consequence, the results indicate that a highly resourceful catalyst is required to achieve small errors in state transformations.

In Leung *et al* (2008), catalytic embezzling phenomena have also been studied in the context of coherent state exchange. Suppose there are m parties, and they want to transform a m-partite state $|\psi\rangle$ into another m-partite state $|\phi\rangle$ by a coherent process and without any communication, i.e.

$$\alpha|0\rangle^{\otimes m}|\gamma\rangle+\beta|1\rangle^{\otimes m}|\psi\rangle \to \alpha|0\rangle^{\otimes m}|\gamma\rangle+\beta|1\rangle^{\otimes m}|\phi\rangle. \tag{201}$$

Note that in the above process the coherence remains intact. In general, the above task cannot be always accomplished for $m \geqslant 2$ (Leung *et al* 2008). However, with an additional catalyst state shared between m parties and allowed to change slightly in the process, it has been shown that coherent state exchange is always possible for any number of parties (Leung *et al* 2008). Furthermore, Leung *et al* (2008) showed that universal embezzling is possible for any m. In another recent study, universal embezzling has been explored for entangled projection games (Dinur *et al* 2013).

8. Perspectives and open problems

While significant progress has been achieved in the investigation of quantum catalysis in recent years, some fundamental problems in this research area are still open. One such open problem concerns exact entanglement catalysis, where the catalyst is left uncorrelated with the primary system at the end. To the best of our knowledge it is not known whether an entangled catalyst in a mixed state can offer an advantage over pure-state catalysts. This question is open even for transformations between bipartite pure states. In fact, the trumping conditions (Klimesh 2007, Turgut 2007) as well as the condition based on Rényi entropies (Brandão *et al* 2015) all assume that the catalyst is in a pure state. It is not known whether or not relaxing this condition will enrich the possible state transformations. Thus, a complete characterisation of exact catalytic transformations between bipartite pure states remains a pressing open question. It is worth mentioning that the above question has been solved for approximate entanglement catalysis, as theorem 6 does not require the catalyst state to be pure. Even more, it has been shown in Datta *et al* (2022) that a catalyst in a mixed state is required to achieve some transitions which are not achievable otherwise. In quantum thermodynamics, most results on catalytic state transformations are limited to specific families of states, e.g. states which are

block diagonal in the energy eigenbasis. A complete characterisation of catalytic transformations via thermal operations between general quantum states has so far remained open.

Another open question concerns the potential equivalence between asymptotic state transformations without catalysis and single-copy transformations with approximate catalysis. Isolated results demonstrating this equivalence have been presented recently, but a systematic understanding of this phenomenon is still lacking. For Gibbs-preserving operations the equivalence has been proven by Shiraishi and Sagawa (2021) showing that for any pair of asymptotically reducible states the transformation can also be achieved via approximate catalysis and vice versa. The equivalence between asymptotic reducibility and approximate catalysis has also been proven for pure-state entanglement in the bipartite setting (Kondra *et al* 2021a). For general transformations between multipartite entangled states, it has been shown that asymptotic reducibility implies that the transition is also achievable on the single-copy level via approximate catalysis (Kondra *et al* 2021a). However, it remains unclear whether catalysis can offer an advantage over asymptotic setups (without catalysis) in any of these scenarios. This leads to an intriguing question of whether bound entangled states (Horodecki *et al* 1998) can be distilled into singlets with the aid of catalysis.

In the existing literature, entanglement catalysis has primarily been explored within the confines of bipartite, or tripartite frameworks. However, an emerging and fascinating avenue for research involves the study of entanglement in many-body systems (Amico *et al* 2008, Streltsov *et al* 2020, Frérot *et al* 2023). To effectively explore catalysis in these multipartite configurations, it becomes essential to move beyond the confines of the traditional LOCC framework and embrace more advanced paradigms (Streltsov 2023). We hope that these problems will be addressed and solved in the near future, thus significantly improving our understanding of quantum catalysis and quantum systems in general.

Many of the recent results discussed in this article make use of an analogy between asymptotic and catalytic setups, as developed by Shiraishi and Sagawa (2021) for quantum thermodynamics. This technique allows to build catalytic protocols in quantum thermodynamics and other quantum resource theories, by resorting to the tools developed for the asymptotic setting. When it comes to asymptotic analysis of resource theories, an important line of research aims to build reversible resource theories from minimal assumptions, and provide quantifiers for the corresponding state transition rates. Indeed, it is believed that under few reasonable assumptions, the conversion rates between quantum states are characterised by the regularised relative entropy of the corresponding resource:

$$R(\rho \to \sigma) = \frac{E^{\infty}(\rho)}{E^{\infty}(\sigma)}. \tag{202}$$

Here, $E^{\infty}(\rho) = \lim_{n\to\infty} E(\rho^{\otimes n})/n$, and $E(\rho) = \inf_{\sigma\in\mathcal{F}} S(\rho||\sigma)$ is the relative entropy of the resource, where $\mathcal{F}$ is the set of free states of the resource theory under study.

While Brandão and Gour (2015) claimed that equation (202) holds for a general class of quantum resource theories, the proof of this result presented in Brandão and Gour (2015) relies on the generalised quantum Stein's lemma (Brandão and Plenio 2010a), which has been recently called into question (Berta *et al* 2023). It thus remains an important open problem to characterise all quantum resource theories for which equation (202) holds true. Resolving this problem is also important for the understanding of catalysis in general quantum resource theories. We also note that none of the results reviewed in this article are affected by the problem reported in Brandão and Plenio (2010a), more details are given in the appendix.

Data availability statement

No new data were created or analysed in this study.

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Appendix

Recently, Berta *et al* (2023) uncovered a gap in the proof of the main result of Brandão and Plenio (2010a), a generalisation of the quantum Stein's lemma. This puts directly into question the validity of the framework for general quantum resource theories, and the resource theory of quantum entanglement specifically, presented in Brandão and Plenio (2008, 2010a, 2010b), Brandão and Gour (2015), which tries to establish those theories as *reversible*. Briefly speaking, a reversible resource theory is such that it admits a unique entropy-like quantifier of the resource, which governs all possible asymptotic transformations between resourceful states (Chitambar and Gour 2019). As of today, it is not known whether the generalised quantum Stein's lemma holds, and hence one has to resort to other particular results, proven independently, that assure reversibility of individual resource theories. For example, it is true that the resource theory of

coherence is reversible under a relevant class of asymptotically coherence-nonincreasing operations (Berta *et al* 2023). We scrutinised all publications mentioned in this review and concluded that none of the results presented here are affected by the issue described in Berta *et al* (2023).

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